

Counterparts

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Comments welcome

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Chapter 3

Counterpart theory and the fixity of the qualitative

As you are now, so once was I
As I am now, so will you be

Tomb inscription, attributed to
Horace

3.1 Basic counterpart theory

According to what chapter 2 called the “basic idea” of modal counterpart theory, *to be possibly a certain qualitative way is to have a counterpart which is that way*. For example: to be possibly staring at a blue swan is to have a counterpart who is staring at a blue swan. To avoid the difficult issues raised by the quantification over ‘ways’, it will be helpful to consider a version of the proposal in the form of a schema:

Basic Modal Counterpart Theory (BMCT:)

$$\Diamond\phi(\alpha) := \exists x(Cx\alpha \wedge \phi(x))$$

$$\Box\phi(\alpha) := \forall x(Cx\alpha \rightarrow \phi(x))$$

Here, ‘ $\phi(\alpha)$ ’ is to be replaced with a sentence in which all vocabulary is qualitative, except for one or more occurrences of a singular term α (which could be a name or a free variable). ‘ $\phi(x)$ ’ stands for the result of replacing all the occurrences of α in $\phi(\alpha)$ with some variable x which does not already occur in $\phi(\alpha)$. ‘ $Cx\alpha$ ’ abbreviates ‘ x is a counterpart of α ’. Remember that this is functioning as a placeholder, which will be cashed out in different ways corresponding to different resolutions of the context-sensitivity of ‘possibly’ and ‘necessarily’. Finally, ‘ $:=$ ’ indicates the giving of an analysis: perhaps it

may be rendered in English as ‘for it to be the case that ... is for it to be the case that ...’.

Similarly, the “basic idea” of temporal counterpart theory is that to be *going to be* a certain qualitative way is to have a future-counterpart which is that way, and to *have been* a certain qualitative way is to have a past-counterpart which is that way. Again, it will be helpful to consider this proposal in schematic form:

Basic Temporal Counterpart Theory (BTCT):

$$\mathbf{F}\phi(\alpha) := \exists x(\mathbf{C_F}x\alpha \wedge \phi(x))$$

$$\mathbf{G}\phi(\alpha) := \forall x(\mathbf{C_F}x\alpha \rightarrow \phi(x))$$

$$\mathbf{P}\phi(\alpha) := \exists x(\mathbf{C_P}x\alpha \wedge \phi(x))$$

$$\mathbf{H}\phi(\alpha) := \forall x(\mathbf{C_P}x\alpha \rightarrow \phi(x))$$

The predicates ‘ $\mathbf{C_F}$ ’ (‘future-counterpart’) and ‘ $\mathbf{C_P}$ ’ (‘past-counterpart’) here are placeholders, which temporal counterpart theorists are free to analyse in different ways.

(One plausible constraint on such analyses is worth mentioning, however, namely that ‘ $\mathbf{C_F}xy$ ’ and ‘ $\mathbf{C_P}yx$ ’ should turn out to be equivalent: a thing’s past-counterparts are the things whose future-counterpart it is. Imposing this constraint is a natural way to validate (for the cases to which BTCT applies) the basic tense-logical “mixing” axioms, $\phi \rightarrow \mathbf{HF}\phi$ —‘What is the case has always been going to be the case’—and $\phi \rightarrow \mathbf{GP}\phi$ —‘what is the case always will have been the case’.)

The requirement that α be the only referential item in $\phi(\alpha)$ must be interpreted so as to exclude, for example, ‘Merkel is German’ as a legitimate substitution instance for ‘ $\phi(\alpha)$ ’. Modal counterpart theorists should want to be able to say that it is possible that Merkel is German and has a collection of blue swans while denying that anyone *is* German and has a collection of blue swans. The problem with the predicate ‘German’ is that it is non-qualitative: it is (roughly) equivalent to ‘from Germany’. For BMCT to be defensible, it must be restricted to a sublanguage in which the job of *referring* is done only by singular terms, and everything else is qualitative. And BTCT, too, is much more defensible when it is understood as subject to this restriction. To get an analysis of ‘It is possible that Merkel is German and has a collection of blue swans’ or ‘It will be the case that Merkel is German and not a chancellor’, we will first have to analyse away ‘is German’ in favour of something like ‘from Germany’, and then apply some (as yet unseen) generalisation of BMCT or

BTCT which can deal with cases in which multiple referential items appear within the scope of modal operators.¹

BMCT and BTCT obviously are not complete accounts of the meaning of modal and temporal operators, since they do not tell us how to make sense of the application of modal or temporal operators to sentences that contain no referential items at all, or more than one. The latter case, in which multiple referential items occur in the scope of a single modal or temporal operator, raises tricky issues best postponed until chapter 4. The former case, where a modal or temporal operator applies to a sentence built up entirely out of qualitative vocabulary, with no referential items at all, is easier to deal with.

For these sentences, as we will shortly see, acceptance of BMCT practically forces us to accept the following highly counterintuitive claim:

Modal Redundancy: When ϕ is a closed sentence built up entirely out of qualitative vocabulary, $\Box\phi$ and $\Diamond\phi$ are equivalent to ϕ .

In other words, there is no contingency at all as regards the purely qualitative facts. If it is contingently the case that ϕ , then the fact that ϕ must be a fact about at least one specific object. In slogan form: *All contingency is de re contingency*. Likewise, acceptance of BTCT practically forces us to accept

Temporal Redundancy: When ϕ is a closed sentence built up entirely out of qualitative vocabulary, $F\phi$, $P\phi$, $G\phi$ and $H\phi$ are all equivalent to ϕ .

¹Special difficulties arise when the term α in $\phi(\alpha)$ occurs in the scope of a propositional attitude verb. If we take the name 'Hesperus' to be a referential term and hence an admissible substituent for ' α ', BMCT entails the equivalence of (i) and (ii):

- (i) Necessarily $\forall x\forall y(y = \text{Hesperus} \rightarrow (x \text{ believes that Hesperus is bright} \leftrightarrow x \text{ believes that } y \text{ is bright}))$
- (ii) $\forall z(z \text{ is a counterpart of Hesperus} \rightarrow \forall x\forall y(y = z \rightarrow (x \text{ believes that } z \text{ is bright} \leftrightarrow x \text{ believes that } y \text{ is bright})))$

(ii) is an uncontroversial logical truth, but (i) is quite controversial: many theories about the behaviour of names in attitude ascriptions deny that they are "transparent" in this way (For discussion, and a defence of (i), see Dorr 2014). I count proponents of these theories as denying that 'Hesperus' is a referential singular term, in the sense relevant to what counts as a substitution instance of BMCT. Most of these proponents will *also* deny that 'Hesperus' is purely qualitative; in that case, they will not be able to use BMCT as a general analysis of modal claims involving 'Hesperus'. Nevertheless, they might endorse it as an analysis of modal claims in which 'Hesperus' occurs only outside the scope of attitude verbs, while leaving the task of analysing modal sentences that embed attitude ascriptions for another occasion.

If it is temporarily the case that ϕ , the fact that ϕ must be a fact about some particular object or objects. *All change is de re change.*

Let us see how BMCT and BTCT push us towards these surprising redundancy claims. Assume that there is at least one object that has a counterpart (perhaps itself). Let ' a ' name some object that has a counterpart (e.g. Mars), and let ' F ' be some qualitative predicate (e.g. 'red'). While BMCT does not tell us how to analyse $\Diamond\phi$ when ϕ is composed of purely qualitative vocabulary, it does yield something nearly as good, namely:

$$\Diamond(\phi \wedge (Fa \vee \neg Fa)) := \exists x(Cxa \wedge (\phi \wedge (Fx \vee \neg Fx)))$$

The right hand side of this analysis logically entails ϕ , and indeed it is equivalent to ϕ given our assumption that a has a counterpart. Thus, if we wanted to deny that $\Diamond\phi$ is equivalent to ϕ while accepting BMCT, we would have to deny that $\Diamond\phi$ is equivalent to $\Diamond(\phi \wedge (Fa \vee \neg Fa))$. But this is very implausible, since $\phi \wedge (Fa \vee \neg Fa)$ is tautologically equivalent to ϕ , and tautologically equivalent sentences can be substituted *salva veritate* in the scope of modal operators. So it looks like acceptance of BMCT commits us to regarding ϕ as equivalent to $\Diamond\phi$ whenever ϕ is purely qualitative. (Likewise, *mutatis mutandis*, for BTCT and temporal operators.)

One could resist this reasoning by rejecting the equivalence of $\Diamond\phi$ and $\Diamond(\phi \wedge (Fa \vee \neg Fa))$. While this equivalence is extremely plausible, and valid according to orthodox modal logic, there are systems of modal logic (e.g. system Q from Prior 1957) in which it fails, and indeed counterpart theory has often been linked with a rejection of such equivalences. This would make room for a view that combines BMCT with contingency in the qualitative facts, or BTCT with temporariness in the qualitative facts. But if the aim was not merely to reject *Modal Redundancy* or *Temporal Redundancy*, but to do justice to our intuitive beliefs about the extent of contingency and change, this approach is deeply inadequate. Typically, when we are pre-theoretically inclined to think that it is contingently or temporarily the case that ϕ (where ϕ is purely qualitative), we are just as strongly inclined to accept various claims to the effect that it could be or will be the case that (not- ϕ and $\psi(\alpha)$), for some referential term α . For instance, we do not merely think that it is possible that no white cubes have sides one metre long (a qualitative claim); we think that it is possible that (no white cubes have sides one metre long, and Mars is red). BMCT directly rules out the latter claim: if some white cubes have sides one metre long, then of course Mars has no red

counterparts that are such that no white cubes have sides one metre long. Thus, for those who are committed to the intuitive judgments which conflict with *Modal Redundancy* or *Temporal Redundancy*, accepting BMCT or BTCT is not a viable option.

The idea that there is no contingency, or no change, as regards purely qualitative matters looks *prima facie* bizarre. Given the discovery that a literal-minded interpretation of the “basic ideas” of modal and temporal counterpart theory (BMCT and BTCT) commit us to these strange claims, the natural reaction is to look for some way of modifying these formulations, so as to allow them to fit with some view on which modal/temporal operators need not be vacuous when applied to purely qualitative sentences. Inspiration for this project can be drawn from the versions of modal counterpart theory developed by Lewis, in Lewis 1968 and Lewis 1986b. For even when we restrict our attention to the case of singly *de re* modal claims, Lewis’s analysis differs from BMCT in a way that looks extremely relevant: unlike BMCT, Lewis’s analysis involves talk about the entities which he calls “possible worlds”, and which I am calling “Lewis-worlds”: in the final analysis (Lewis 1986b, pp. 75–6), objects any two parts of which stand in some spatiotemporal, or “analogically spatiotemporal”, relation, and which are not parts of any other such objects.

In the case of singly *de re* claims, which is our present concern, this works out as follows:

Restrictive Counterpart Theory (RCT)

$$\Diamond\phi(\alpha) := \exists w\exists x(Ww \wedge Ixw \wedge Cx\alpha \wedge [\phi(x)]^w)$$

$$\Box\phi(\alpha) := \forall w\forall x(Ww \wedge Ixw \wedge Cx\alpha \rightarrow [\phi(x)]^w)$$

Here, w and x are some new variables not already in $\phi(\alpha)$. ‘ Ww ’ means ‘ w is an accessible Lewis-world’: ‘accessible’ is a placeholder which may be filled in in different ways, depending on how we are resolving the context-sensitivity in ‘possibly’ and ‘necessarily’. ‘ Ixw ’ means ‘ x is in w ’. $\phi(x)$ is, as before, the sentence derived from $\phi(\alpha)$ by replacing each occurrence of α with one of x . $[\phi(x)]^w$ is derived from $\phi(x)$ by adding a restricting clause ‘in w ’ to every quantifier in $\phi(x)$ that is not in the scope of any other modal operator. So, for example, if $\phi(\alpha)$ is ‘Some people are taller than α ’, $[\phi(x)]^w$ is ‘Some people in w are taller than x ’; if $\phi(\alpha)$ is ‘Most geese are the same colour as some animal that belongs to α ’, $[\phi(x)]^w$ is ‘Most geese in w are the

same colour as some animal in w that belongs to x' . (A recursive definition implementing this operation for a certain formal language is given in 1968, but for our present purposes this informal presentation will do.)²

There is a natural way to generalise RCT to claims in which modal operators apply to purely qualitative sentences: what Lewis calls '*de dicto*' modal claims. And because of the role played by quantification over Lewis-worlds and the quantifier-restriction operation in RCT, this generalisation does *not* have the result that such modal operators are vacuous. For claims of this form, Lewis's analysis can be given as follows:

$$\begin{aligned} \textit{Restriction} \quad \Diamond\phi &:= \exists w(Ww \wedge [\phi]^w) \\ \Box\phi &:= \forall w(Ww \rightarrow [\phi]^w) \end{aligned}$$

Thus, for example, *Restriction* tells us that for it to be possible that some swans are blue is for there to be an accessible Lewis-world such that some swans in it are blue; for it to be possible that most swans are the same colour as some goose is for there to be an accessible Lewis-world such that most swans in it are the same colour as some goose in it; and so on.

Just like BMCT, RCT and *Restriction* must be understood as restricted to a language in which singular terms are the only carriers of reference to particular items, thus excluding non-qualitative predicates like 'is German'. However, because of the quantifier-restricting operation, Lewis needs to impose an additional requirement on the predicates that can occur in legitimate instances of RCT and *Restriction*. The point of this requirement is to rule out 'implicitly quantificational' expressions like 'invent' and 'discover' (1986b, p. 6)—expressions which under analysis reveal further quantifiers for which the restricting operation could make a difference. Before we can apply RCT or *Restriction* to claims involving predicates of this sort, we must first replace them with analyses which make the relevant quantifiers explicit, so that we can add the relevant modifiers. Thus, 'Possibly, x invents bifocals' must first be replaced with 'Possibly, x thinks of bifocals and no-one thinks of bifocals before x' ', before we can apply RCT to get 'There is a Lewis-world w , and a counterpart y of x who is in w , such that y thinks of bifocals, and no-one in w thinks of bifocals before y .'

²I have left out an important qualification: according to Lewis (1986, p. 6), it is an optional matter, to be decided on pragmatic grounds, whether a restrictor should be attached to any given quantifier in the analysis of a modal claim. In section 3.4 I will consider how this qualification affects the plausibility of Lewis's analyses of modal sentences.

Let us call a predicate ‘well-behaved’ iff it is qualitative and *not* in the relevant way ‘implicitly quantificational’, so that it can be used in legitimate instances of RCT and *Restriction*. If one takes the idea of ‘implicit quantification’ very literally, it might seem that very few predicates are well-behaved: for example, we will not be able to use ‘triangle’ in instances of these schemas, since it has an analysis involving quantification (‘figure with three angles’). But there is no need to be as demanding as that. For a wide range of predicates which are necessarily equivalent to complex formulae involving quantification, the result of substituting those complex formulae and then inserting all the relevant restrictors will make no truth-conditional difference. Given the transitivity of parthood, this will be true of any predicate that expresses an *intrinsic* property or relation. When such predicates have analyses involving quantifiers, the quantifiers in question can always be taken to be restricted to the *parts* of a given object. And since parthood is transitive, the parts of an object in a certain Lewis-world are always themselves in that Lewis-world. So, for example, the claim that there is a Lewis-world *w* that contains a white triangle—i.e. a white plane figure *x* such that exactly three angles are parts of *x*—is equivalent to the claim that there is a Lewis-world *w* that contains a white plane figure *x* such that exactly three angles *in w* are parts of *x*. Thus, we can harmlessly take the class well-behaved predicates to include all predicates expressing intrinsic (and qualitative) properties and relations. Indeed, we could even construe ‘well-behaved’ so as to apply to certain predicates expressing *extrinsic* properties and relations, so long as their analyses only involve quantification over items related to the given item by some spatiotemporal, or analogically spatiotemporal, relation. For most purposes, we will not need to worry too much about how to take the restriction to well-behaved predicates, since Lewis gives us plenty of examples of predicates (like ‘blue swan’ and ‘talking donkey’) which he is willing to use in instantiating RCT and *Restriction*.

There are some simple examples in which the analyses yielded by RCT will plausibly turn out to be equivalent to those yielded by BMCT. Consider for example the sentence ‘It is possible that Obama be staring at a blue swan’. BMCT analyses this as ‘A counterpart of Obama is staring at a blue swan’, while RCT analyses it as ‘There is a Lewis-world such that a counterpart of Obama in it is staring at a blue swan in it’. It is plausible (at least given background mereological principles accepted by Lewis) that all blue swans are parts of Lewis-worlds, and that any two things one of which is staring at the other are part of the same Lewis-world. Modulo these assumptions,

the two analyses are logically equivalent. But there is certainly no guarantee that the quantifier-restrictions introduced by RCT will always end up being redundant in this way. For example, 'It is possible that Obama be staring at every blue swan' is analysed by BMCT as 'A counterpart of Obama is staring at every blue swan', whereas RCT analyses it as 'There is a Lewis-world such that a counterpart of Obama in it is staring at every blue swan in it'.

Although the machinery of quantification over Lewis-worlds and restriction of quantifiers to those worlds makes Lewis's analysis RCT more complicated than BMCT, the added complexity may initially seem to be a price worth paying in order to avoid the highly counterintuitive claim thesis of *Modal Redundancy*, which as we have seen goes hand in hand with BMCT. However, as we will see in the next section, the machinery also leads to some puzzles, which we will need to investigate before reaching a final judgment about which of the two accounts is superior.

3.2 Some puzzles for Lewis

This section will raise a trio of puzzles about Lewis's programme for analysing modal claims. These puzzles arise already for Lewis's analysis of '*de dicto*' modal claims (*Restriction*), although similar puzzles could be raised for his analysis of singly *de re* claims (RCT). In fact, the first and second puzzles will depend only on the following very elementary (English translations of) instances of *Restriction*, which Lewis uses as examples on the first few pages of *OPW*:

- (1) It is possible that some swans are blue := there is a Lewis-world such that some swans in it are blue.
- (2) It is possible that no swans are blue := there is a Lewis-world such that no swans in it are blue.

Here the restriction to 'accessible' Lewis-worlds is omitted, as Lewis thinks it should be when we are concerned with metaphysical possibility.

I have been using the symbol ' := ' to indicate the giving of an analysis. If we have any grip at all on what it means to give an analysis of something, we know that $\lceil \phi \text{:=} \psi \rceil$ entails $\lceil \text{Necessarily, } \phi \text{ iff } \psi \rceil$. If this is given up, it becomes mysterious how the meaning of (1) and (2) go beyond the corresponding material biconditionals.³ So for example, (1) entails (1'):

³I remain neutral on the question whether $\lceil \phi \text{:=} \psi \rceil$ is *equivalent* to $\lceil \text{It is metaphysically necessary that } (\phi \leftrightarrow \psi) \rceil$. But I note that there is a reason for those, like Lewis, who seek a

- (1') Necessarily: it is possible that some swans are blue iff there is a Lewis-world such that some swans in it are blue.

Puzzle 1 depends on (1') alone. It seems obvious that it *is*, in fact, possible for it to be the case that no swans are blue. And it seems only a little less obvious that it is possible for it to be *contingently* the case that no swans are blue—i.e. that it is possible that (no swans are blue and it is possible that some swans are blue). But given (1'), this claim must be rejected. For in combination with (1') it entails that it is possible that (no swans are blue and there is a Lewis-world such that some swans in it are blue).⁴ And this cannot be possible, since it is contradictory: any blue swan that is part of a Lewis-world is a blue swan. This is surprising. Given that Lewis thought that there could have been no blue swans, could he really have been committed to denying that there could have *contingently* been no blue swans?

Puzzle 2 depends on the combination of (1') with (2). As we saw in *Puzzle 1*, (1') entails that if it is possible that no swans are blue, it is possible that it is not possible that some swans are blue. Suppose that—as Lewis believed—some Lewis-worlds don't contain blue swans, and some do. Then by (2), it is possible that no swans are blue. So by (1'), it is possible that it is not possible that some swans are blue. But it is also possible that some swans are blue, by (1). This combination of claims is inconsistent in the modal logic S5, where what is possibly necessary is simply necessary, and what is possibly impossible is simply impossible. It is thus puzzling to find it seeming to follow so directly from Lewis's commitments. For Lewis generally seems to be friendly towards S5, at least when the modal operators are interpreted as expressing metaphysical necessity and possibility and the sentences we are dealing with are "*de dicto*" (purely qualitative), as 'there are blue swans' and 'there are no blue swans' are supposed to be.

Puzzles 1 and 2 turn on the assumption that the correctness of an analysis requires the necessarily equivalence of the analysans and the analysandum. *Puzzle 3* does not require this assumption. It turns on another instance of *Restriction*:

- (3) It is possible that there are two things that are not analogically spatiotemporally related := there is a Lewis-world such that there are two things in it that are not analogically spatiotemporally related.

general analysis of modality to answer yes: it wouldn't be so exciting if one could analyse modality, counterfactuals, etc. in non-modal terms, but still had to take ':= as a primitive, manifestly non-extensional, operator.

⁴The argument here is of the evidently valid form $\Box(\phi \leftrightarrow \psi), \Diamond(\xi \wedge \phi) \vdash \Diamond(\xi \wedge \psi)$.

The right-hand side of (3) is false, since by definition, a Lewis-world is a maximal object any two parts of which are analogically spatiotemporally related. Thus, anything that has two parts which are not analogically spatiotemporally related is, ipso facto, not a Lewis-world. So if we accept (3), we have to say that it is *not* possible for there to be two things that are not analogically spatiotemporally related. This sits oddly (to say the least!) with Lewis's notorious claim that there *are* many Lewis-worlds, which of course entails that there are many things that are not analogically spatiotemporally related. Whatever *is* the case *could* be the case. This is a version of the modal axiom T, and plays an extremely basic role in our practice of reasoning with modal operators on many of their interpretations, including the metaphysical interpretation with which we are presently concerned. While no principle in logic is so uncontroversial as to be beyond all debate, the idea that some things that are necessarily the case are not the case, or that some things that are the case could not be the case, takes us deep into the alien realm of logical heresy.⁵

The obvious move for Lewis to make in response to these puzzles is to invoke the familiar idea that quantification is sometimes tacitly restricted. You can say 'There is no beer' without asserting anything false even if there is beer but none of it is in your house, or even if there is beer in your house but only in the form of a puddle of spilt beer on the floor of the fridge. Similarly, you can say 'There are no blue swans' and assert nothing false even if (as Lewis believed) there are blue swans but none of them is in the Lewis-world of which you are part.

(We need not concern ourselves with the question how the phenomenon of quantifier domain restriction fits into compositional semantics. I will write

⁵Hudson (1997) and Parsons (2009) also note the apparent failure of T induced by Lewis's analysis. Jeff Russell pointed out to me that if we interpret Lewis as committed to the rejection of T, we can still give him the related schema $\Box\phi \rightarrow \text{ACT}\phi$: what is necessarily the case is *actually* the case. Lewis could thus diagnose our temptation to think that what is necessarily the case is the case as the result of our mistaken assumption that what is actually the case is the case. This is probably the right way for the rejecter of T to go, and there is certainly textual evidence for attributing certain claims of the form ' ϕ but actually not- ϕ ' to Lewis. But the package in which neither necessity nor actuality is factive is still very alien. When people utter (3), are they right but actually wrong? We would normally take it that if David spoke truly, then David would have got a prize if everyone who actually spoke truly had got a prize; but we will not be able to say this if we think that uttering (3) involves speaking truly without actually speaking truly.

as if it involves a semantic context-sensitivity in determiners like ‘some’; but everything I say should be acceptable, *mutatis mutandis*, to those (e.g. Stanley and Szabó 2000) who locate the context-sensitivity in nouns, or in some silent syntactic elements associated with nouns.⁶)

The natural way to resolve the puzzles is to say that the analyses that generate them involve a mixture of restricted and unrestricted quantifiers. Consider (1) and (2), for example:

- (1) It is possible that some swans are blue := there is a Lewis-world such that some swans in it are blue.
- (2) It is possible that no swans are blue := there is a Lewis world such that no swans in it are blue.

The proposal is that the true readings of these claims are readings in which the quantifiers on the left (‘some’ and ‘no’) are somehow tacitly restricted, whereas the quantifiers on the right (‘there is’, ‘some’ and ‘no’) are unrestricted. We can make it clear that these are the intended readings by adding subscripts to our quantifiers: ‘u’ to indicate unrestricted quantification, and ‘r’ to indicate the relevant kind of restriction, whatever it turns out to be. Then the proposed interpretations of (1) and (2) will be represented as follows:

- (1_r) It is possible that some_r swans are blue := there is_u a Lewis world such that some_u swans in it are blue.
- (2_r) It is possible that no_r swans are blue := there is_u a Lewis world such that no_u swans in it are blue.

Of course, the proposal is not restricted to these particular analyses, but extends to all instances of *Restriction*:

$$\begin{aligned}\Diamond\phi &:= \exists w(Ww \wedge [\phi]^w) \\ \Box\phi &:= \forall w(Ww \rightarrow [\phi]^w)\end{aligned}$$

The idea is that these sentences are true only when all quantifiers on the left are all tacitly restricted in some distinctive way, while the quantifiers on the right are unrestricted. The same interpretation could be applied to instances of RCT.

⁶Even those like Bach (1994) who regard the phenomenon as pragmatic rather than semantic should be able to reconstruct my discussion to fit with their view.

This way of understanding *Restriction* makes the puzzles go away.

Puzzle 1: (1_r) does not make 'possibly some_r swans are blue' inconsistent with 'no_r swans are blue', only with 'no_u swans are blue'. Thus it is consistent with the truth of 'Possibly, it is contingently the case that there are no_r blue swans'.

Puzzle 2: for the same reason, (2_r) does not license the inference from 'Possibly, no_r swans are blue' to 'Possibly, it is not possible that some_r swans are blue'. So the inference to the S5-inconsistent 'Possibly there are no blue swans, and possibly, it is not possible for there to be blue swans' (on either interpretation) is blocked.

Puzzle 3: The intended interpretation of (3), with restricted quantifiers on the left, tells us that it is not possible that there be_r two analogically spatiotemporally unrelated objects. It follows by T that any_r two objects are analogically spatiotemporally related. This tells us something interesting about the character of the relevant form of restriction represented by 'r'. But it is perfectly consistent with Lewis's characteristic claim that there are many_u Lewis-worlds, and hence many_u analogically spatiotemporally unrelated things.

So, provided we understand *Restriction* as a schema for analysing possibility claims involving some variety of restricted quantification, trouble is averted.

However, this response raises a further question. Given that 'some' in 'Some swans are blue' admits of both unrestricted and restricted readings, it is extremely plausible that 'some' in 'It is possible that some swans are blue' also admits of both kinds of readings. So, how should we analyse 'It is possible that some swans are blue' on the unrestricted reading of its quantifier? More generally, how should we analyse claims in which unrestricted quantifiers occur in the scope of modal operators? If this is a good question, any general programme for analysing modal claims in non-modal terms had better be able to answer it.

It is hard to see how it could fail to be a good question. In general, while embedding a context-sensitive sentence inside some larger sentence often makes a difference as regards which of its readings we find most natural and salient, readings that are available when a sentence is asserted on its own do not just disappear under embedding. A semantic mechanism that made it flat-out impossible to interpret any quantifier embedded under a modal operator as unrestricted would be strange and unprecedented.

The setup in Lewis 1968 might suggest such a prohibition. There, the project of reducing modality is cast as one of finding translations from one formal language, which contains modal operators, into another, which lacks them. The output language is not just the modal-operator-free fragment of the input language, since even sentences with no modal operators are translated non-homophonically. For example, ' $\exists xFx$ ' in the input language is translated as ' $\exists x(Ix@ \wedge Fx)$ ' in the output language (' $Ix@$ ' means ' x is in the actual world'). This non-homophonic treatment of ' $\exists x$ ' suggests that the quantifier-meanings in the input language are restrictions of the quantifier-meanings in the output language. If one thought that the input language had to be capable of representing every admissible interpretation of English sentences involving a modal operator, one would be pushed to the view that quantifiers in English cannot be interpreted as unrestricted when they occur in the scope of a modal operator.

But such a prohibition is quite unmotivated. If unrestricted quantification were a rare practice engaged in only by philosophical initiates, perhaps there would be some temptation to think that this special way of talking cannot meaningfully be combined with ordinary modal operators. But in fact, ordinary people hold, and express, lots of opinions about what there is, unrestrictedly speaking. For example, they are very confident that no_u bachelors are married, and fairly confident that no_u swans are blue. Lewis, of course, thought that many ordinary negative judgments involving unrestricted quantification were *mistaken*. But if ordinary people can express unrestricted quantification when they say things like 'No swans are blue', surely they can also express unrestricted quantification when they say things like 'It is possible that no swans are white' and 'It is necessary that no bachelors are married'. Ordinary usage gives no support at all to the claim that such readings are semantically prohibited.

Moreover, it is hard to see how any prohibition of this sort could work. A restricted quantifier Q_r is presumably equivalent to the combination of an unrestricted quantifier Q_u with a predicate R —an invented one, if necessary—expressing the relevant restriction.⁷ For example, ' $\text{Most}_r Fs \text{ are } Gs$ ' will be equivalent to ' $\text{Most}_u Fs \text{ which are } R \text{ are } Gs$ '. But if ' $Q_r Fs \text{ are } G$ ' is equivalent to ' $Q_u Fs \text{ which are } R \text{ are } G$ ', 'It is possible that $Q_u Fs \text{ which are } R \text{ are } G$ ' surely should make sense, and be equivalent to 'It is possible that $Q_r Fs \text{ are } G$ '. We don't want to dismiss 'It is possible that $Q_u Fs \text{ which are } R \text{ are } G$ ' as nonsense just on the grounds that it contains an unrestricted quantifier in the scope of

⁷This certainly was Lewis's view of restricted quantification.

a modal operator. If so, the mechanism that enforces the prohibition must somehow check to make sure that every unrestricted quantifier in the scope of a modal operator ends up combining with a restrictor of the right kind—a complicated and messy task. Worse, there are sentences—for example, ‘Possibly some swans are blue’ on the restricted reading of ‘some’—whose analyses *require* unrestricted quantification. If we want ‘Possibly, possibly, some swans are blue’ to be legitimate and ‘Possibly, there is a Lewis-world such that some swans in it are blue’ to be prohibited on the unrestricted reading of ‘some’, the mechanism that enforces the prohibition will have to distinguish, using some obscure mechanism, between analysis and analysandum. Even if there were such a mechanism, it would be an arbitrary imposition, which we could easily stipulatively remove for philosophical purposes, if we found it helpful to do so.

So a general program for analysing modality *does* need an answer to the question how to analyse claims in which unrestricted quantifiers occur in the scope of modal operators, such as ‘Possibly, some_u swans are blue’. And for someone who accepts *Restriction* as an analysis of modal claims involving restricted quantification, the possible answers to this question are severely limited. In fact, for a certain class of claims, namely those derived from well-behaved predicates by applying an unrestricted *existential* quantifier, there is essentially only one possible answer:

Existential Redundancy: When F is well-behaved, $\Box\exists_u xFx$ and $\Diamond\exists_u xFx$ are both equivalent to $\exists_u xFx$.

The claim that $\exists_u x\phi(x)$ entails $\Diamond\exists_u x\phi(x)$, and that $\Box\exists_u x\phi(x)$ entails $\exists_u x\phi(x)$, should be uncontroversial, since they are consequences of T. As for the converse entailments, suppose that (like almost all philosophers other than Lewis) one took the claim that nothing_u is F to be consistent with the claim that possibly, something_u is F . Then what on earth could motivate one to accept the necessity of (4), as required by our current way of understanding *Restriction*?

$$(4) \quad \Diamond\exists_r xFx \rightarrow \exists_u xFx$$

For any kind of restriction we antecedently understand, (4) is profoundly counterintuitive. If the proposal is to be worth bothering with at all, it must be because it is an inevitable upshot of some general strategy for analysing modal claims in non-modal terms. But if we were in possession of some such strategy which allowed us to avoid the counterintuitive consequences

of *Existential Redundancy*, it would simply be perverse not to use the same trick to avoid the (equally counterintuitive) consequences of (4).

Given *Existential Redundancy*, which says that certain modal operators are redundant when applied to sentences built up in certain ways from well-behaved predicates and unrestricted quantifiers, it is natural to consider the following stronger thesis:

Generalized Redundancy: Whenever ϕ is a closed sentence built up from well-behaved predicates, unrestricted quantifiers, and truth-functional connectives, 'Possibly ϕ ' and 'Necessarily ϕ ' are both equivalent to ϕ .

One can certainly imagine views which accept *Existential Redundancy* while resisting *Generalized Redundancy*, allowing (for example) that even if some_u swans are blue, it is still possible that no_u swans should be blue. One way to formulate a view of this sort to keep the letter of *Restriction*, understood once again as an analysis of modal claims involving unrestricted quantification, while relaxing our previous stipulations about the meaning of its primitive predicate ' Ww ' (which we formerly understood to mean ' w is an accessible Lewis-world'):

$$\begin{aligned}\Diamond\phi &:= \exists w(Ww \wedge [\phi]^w) \\ \Box\phi &:= \forall w(Ww \rightarrow [\phi]^w)\end{aligned}$$

Suppose we just read ' Ww ' as ' w is an appropriate part of reality', and consider what we get when we plug in different accounts of appropriateness. One thing we can immediately see is that if we want T to be valid, the whole of reality—something such that everything whatsoever counts as being 'in' it—had better count as 'appropriate'. If it does, and if we treat the existence of such an all-encompassing entity as a logical truth, our analysis will preserve the entailments from $\Box\phi$ to ϕ , and from ϕ to $\Diamond\phi$. If not, not.⁸ Note that so long as we do hold fixed that the whole of reality is an appropriate part of reality, we will recover *Existential Redundancy*: $\Diamond\exists xFx$ is equivalent to $\exists w(Ww \wedge \exists x(Ixw \wedge Fx))$, which is equivalent to $\exists xFx$ so long as $\exists w(Ww \wedge \forall xIxw)$ can be treated as a logical truth. What else should we count as an 'appropriate part of reality'? One possible view is that the only

⁸In principle, one could preserve T without counting the whole of reality as 'appropriate', so long as one's account of appropriateness guaranteed that some part of reality which is an intrinsic duplicate of the whole of reality is appropriate. (Recall that objects that are infinite in size can have proper parts that are intrinsic duplicates of themselves.)

other appropriate parts, besides the whole of reality, are the Lewis-worlds.⁹ But this restriction is arbitrary. If there could be_u many Lewis worlds, and there could be_u exactly one, what motivation could there be for denying that there could be_u two, or any other intermediate number? It seems better to understand 'appropriate' in such a way that arbitrary fusions of appropriate parts of reality are themselves appropriate.¹⁰

However, this whole approach seems pointless. For if anything short of reality as a whole counts as "appropriate", we will no longer have an S5 modal logic, even for claims involving just unrestricted quantifiers applied to well-behaved predicates. For example, suppose that there is an appropriate part that lacks blue swans, so that it is possible that there are no_u blue swans. By *Existential Redundancy*, 'it is possible that there should be_u blue swans' is equivalent to 'there are_u blue swans'; so we are committed to the claim that it is possible that it is not possible that there should be_u blue swans. Reality could be "smaller" (it could lack blue swans), but it is necessary that it could not be bigger (necessarily, if it lacks blue swans, it couldn't contain them); once the blue swans are gone, they cannot come back. Endorsing such a claim for the sake of being able to agree with ordinary people when they claim that it is possible that no_u swans are blue seems like a bad trade. For once one accepts *Existential Redundancy*, one is already committed to rejecting a vast array of common-sense modal claims involving unrestricted quantification, for example the claim that it is possible that it is contingently the case that no_u swans are blue. It seems perverse to jettison the compelling logical principles that constitute S5 merely in order to slightly decrease the scope of this systematic disagreement as regards particular modal claims.

The 'appropriate parts of reality' approach was only one way of combining *Existential Redundancy* with the denial of *Generalized Redundancy*. Might there be some other, more attractive, way of developing such a combination? Well, I certainly can't think of one. What's more, there are some considerations which suggest that any such combination will have to be, in effect, a version of the 'appropriate parts' view. Let us assume that it is necessary that there is such a thing as "the whole of reality", something which absolutely

⁹This version of *Restriction* is equivalent to the "T-preserving analysis" discussed by Parsons (2009).

¹⁰Bricker (2001) endorses something that looks like *Restriction* with '*Ww*' interpreted to mean '*w* is a fusion of Lewis-worlds'. However, he does not intend this analysis to apply to possibility claims involving absolutely unrestricted quantifiers. I will discuss Bricker's account of the relevant kind of restriction in section 3.3 below.

everything is “in”: $\Box \exists_u x \forall_u y (Iyx)$. Then for any closed sentence ϕ built up from well-behaved predicates, unrestricted quantifiers, and truth-functional connectives, we have that $\Box(\phi \rightarrow \exists_u w[\phi]^w)$: if ϕ is true, there is bound to be at least one thing (viz. the whole of reality) such that ϕ remains true when its quantifiers are restricted to parts of that thing. So from $\Diamond\phi$, we can conclude that $\Diamond\exists_u w[\phi]^w$. If we could treat $[\phi]^w$ as the result of applying a *well-behaved* predicate to w , we could apply *Existential Redundancy* at this point, concluding that $\exists_u w[\phi]^w$. This would bring us back to the ‘appropriate parts’ account: the only way for ϕ to be possibly true is for there to be something such that ϕ is in fact true when its quantifiers are restricted to its parts. So, can we treat $[\phi]^x$ as equivalent to the result of applying a well-behaved predicate to x ? If we take the well-behaved predicates to be those which express intrinsic properties and relations, the answer is yes. Say that a formula containing free variables is “intrinsic” iff the result of turning it into a predicate, by lambda-abstracting each of its free variables, expresses an intrinsic property or relation. Conjunctions, disjunctions and negations of intrinsic formulae are themselves intrinsic. Moreover, when ψ is an intrinsic formula and v_1, v_2 are variables, $\exists_u v_1 (Iv_1 v_2 \wedge \psi)$ and $\forall_u v_1 (Iv_1 v_2 \rightarrow \psi)$ are intrinsic formulae: quantification over the parts of the given objects preserves intrinsicness. This is an extremely plausible principle about the notion of intrinsicness, and can easily be seen to be true on Lewis’s theory of intrinsic properties (as presented for example in Lewis 1983a). It follows that any formula in which x is the only free variable, in which all predicates express intrinsic properties and relations, and in which all quantifiers carry a restriction to the parts of x , is intrinsic. (If we understand ‘well-behaved’ more broadly, so as to apply got some non-intrinsic predicates provided that their application depends only on how an object is related to its Lewis-world-mates, we will very plausibly get the same result.)

The upshot of this discussion is that if we adopt *Restriction qua* analysis of modal claims involving some special kind of restricted quantification, there are no attractive alternatives to adopting *Generalized Redundancy* as our account of modal claims involving only unrestricted quantification.¹¹

¹¹Divers (1999) seems to agree that modal realists should accept *Generalized Redundancy*: ‘Whenever the possibility operator expresses a non-trivial semantic function on quantificational sentences it is, indeed, always that of altering the scope of formerly world-restricted quantifiers’ (229). The class of sentences on which Divers thinks the possibility operator operates only vacuously is, however, wider than the class of sentences built up using well-behaved predicates, unrestricted quantifiers and truth-functional operators. For Divers, the class also includes, for example, subject-predicate sentences where the subject is an object

3.3 Contingency-inducing quantification as *de re* restricted quantification

Suppose ϕ is a closed sentence built up from well-behaved predicates, truth-functional connectives, and quantifiers. According to *Generalized Redundancy*, $\Diamond\phi$ and $\Box\phi$ can differ in truth value from ϕ only when the quantifiers in ϕ are understood as tacitly restricted, somehow. But obviously not just any restriction on the quantifiers would suffice to introduce contingency. For example, if we interpret the quantifier in ‘Possibly, some swans are blue’ as tacitly restricted to *noisy* things, the sentence will be equivalent to ‘Possibly, some_u noisy swans are blue’, and hence to ‘some_u noisy swans are blue’, and hence to ‘some swans are blue’ on the same restricted interpretation. For a tacitly restricted quantifier to take us beyond the realm of the modally trivial, the restriction must not itself be expressible by some well-behaved predicate: otherwise, the claim made using the restricted quantifier could be made explicit in a language with only unrestricted quantifiers and well-behaved predicates. What, then, is the kind of restriction that makes for modal non-triviality?

One might suggest that the relevant restriction could be expressed by some predicate that is qualitative but ill-behaved. But it is hard to see how this could work. The usual examples of ill-behaved qualitative predicates are predicates for which we can provide, or at least gesture towards, analyses in terms of unrestricted quantifiers and well-behaved qualitative predicates—e.g. the analysis of ‘tallest person’ as ‘person taller than any other person’, or the analysis of ‘inventor of bifocals’ as ‘person who thought of bifocals before anyone else’. It was because we had such analyses vaguely in view that the restriction of Lewis’s schemas to sentences built up from well-behaved predicates didn’t seem like a terribly serious failure of generality. But if sentences built up from well-behaved predicates and unrestricted quantifiers are modally trivial, an ill-behaved predicate that took us beyond this realm into the realm of contingency could not be necessarily equivalent to any expression in a language with only well-behaved predicates.

What has just emerged is that, unless we posit some altogether new and mysterious category of qualitative but ill-behaved predicates, there is no way to accept *Generalized Redundancy* without accepting the following *prima facie* stronger principle, which we first encountered in section 3.1:

Modal Redundancy: When ϕ is a closed sentence built up entirely out of

that is not part of any Lewis-world. Section 5.2 will make it clear why I think this is a mistake.

qualitative vocabulary, $\Box\phi$ and $\Diamond\phi$ are equivalent to ϕ .

For unless the notion of a well-behaved predicate is restricted in some surprising way, it is extremely plausible that *every* purely qualitative proposition is necessarily equivalent to some proposition expressible in a language with only unrestricted quantifiers, truth-functional operators, and well-behaved predicates, or at least supervenient on such propositions.

I have been assuming, and will continue for now to assume, the standard view that ‘non-qualitative’ properties, relations and propositions are those that are not *de re* (‘singular’, ‘referential’, ‘haecceitistic’). Thus a non-qualitative restricted quantifier is a *de re* restricted quantifier—a quantifier restricted to items that are related in the right way to some particular thing or things. Such restrictions are common. For example, I might utter the sentence ‘there are no blue swans’ and thereby say, of a certain zoo, that there are no blue swans in it; or say, of the planet Earth, that there are no blue swans on it; or say that there are no blue swans on the same planet as *me*. These are all available readings for unembedded utterances of ‘there are no blue swans’, and surely they all remain available when this sentence is embedded under a modal operator. In saying ‘Possibly, there are no blue swans’, I might say that it is possible that there are no blue swans in a certain zoo; that it is possible there are no blue swans on Earth; or that it is possible that there are no blue swans on the same planet as *me*. A complete reductive account of modality should, in any case, tell us what it would be for each of those claims to be true. And there certainly are possible accounts of these claims which are consistent with *Generalized Redundancy*, but allow modal operators to be non-vacuous when combined with such *de re* sentences. BMCT, the simple version of counterpart theory with which we began, is one such analysis. It yields analyses such as the following:

It is possible that there are no blue swans in London Zoo := London Zoo has a counterpart in which there are no blue swans

It is possible that there are no blue swans on Earth := Earth has a counterpart on which there are no blue swans

It is possible that there are no blue swans on the same planet as *me* := one of my counterparts is such that there are no blue swans on the same planet as it

And as we saw in section 3.1, BMCT is not only consistent with *Generalized*

Redundancy: given some minimal logical assumptions, it *entails* the stronger principle of *Modal Redundancy*.

Lewis's

- (2) It is possible that no swans are blue := there is a Lewis world such that no swans in it are blue

does not look at all plausible as an account of what it is for it to be possible for there to be no blue swans *in London Zoo, or on Earth, or on the same planet as me*. Is there any way to interpret the quantifier on the left as carrying a *de re* restriction on which (2) and other instances of *Restriction* look more plausible? Yes: the obvious approach is to interpret the relevant quantifiers as carrying a *de re* restriction to things in a certain *Lewis-world*, the one of which we are parts. It will be convenient to have a name for this Lewis-world: let's call it 'Cosmo'. The suggestion, then, is that if one is determined to accept (2) on some interpretation or other, one should accept it on an interpretation that makes it equivalent to (5):

- (5) Possibly, no swans in Cosmo are blue := there is a Lewis-world such that no swans in it are blue.

More generally, the suggestion is that if we accept *Restriction* at all, we should accept it as a schema for analysing modal claims in a language each of whose quantifiers is equivalent to the result of adding the restriction 'in Cosmo' to an unrestricted quantifier.

How interested should we be in an analysis which only applies to modal claims whose quantifiers involve this particular kind of *de re* restricted quantifier, while saying nothing about any other kind of *de re* modal claim? It depends how widespread we take the practice of restricting quantifiers to things in Cosmo to be. If one thought that ordinary people believed in a plurality of Lewis-worlds, one might naturally be led to the hypothesis that our quantifiers are quite often restricted to things in Cosmo: this would perhaps help to explain why our belief in many Lewis-worlds doesn't show up more often in ordinary speech. But the idea that the folk believe in many Lewis-worlds strikes me as misguided, as it did Lewis. Honesty requires proponents of a Lewis-style ontology to recognise the radical character of their view, even if they want to go on to argue that it should be believed in spite of its radical character. Thus I see no reason to think that quantification in ordinary speech is often restricted just to the domain of things in

Cosmo. Typically, quantifiers are either unrestricted, or restricted to a much narrower domain than this one.¹²

Interesting or not, analyses like (5) are tenable (*modulo* the incredulous stare), at least on some resolution or other of the context-sensitivity of 'possible'. However, what we are ultimately looking for is a schema for analysing modal claims of all sorts. A theory which tells us what to do with sentences in which quantifiers are restricted to things in Cosmo, but says nothing about sentences in which quantifiers are restricted to things in London Zoo, on Earth, etc., is not very satisfactory. Moreover, it would be very desirable for this analysis to work in a *uniform* way, yielding an analysis of the same form for 'It is possible that no swans in *a* are blue' without having to check whether *a* names a zoo or a planet or a Lewis-world. We have already encountered an analysis that is uniform in this way, namely BMCT. BMCT can be provided to any modal claim containing exactly one singular term (and in which all other vocabulary is qualitative), no matter what sort of entity that term might designate. For example, it tells us that

- (6) Possibly, no swans in Cosmo are blue := there is a counterpart of Cosmo such that no swans in it are blue.

Prima facie, (5) and (6) look like competitors. But in fact it is not compulsory to think of them as such. We could make them consistent by giving an account of 'counterpart' according to which *the counterparts of any Lewis-world are all and only the Lewis-worlds*.¹³ In a context where 'counterpart' satisfies this constraint, BMCT entails the biconditional corresponding to (5). Indeed, it entails that this biconditional is *necessarily* true, thanks to the following instance of BMCT:

- (7) Necessarily, the counterparts of Cosmo are all and only the Lewis-worlds := every counterpart *x* of Cosmo is such that the counterparts of *x* are all and only the Lewis-worlds.

¹²Physics is the area of discourse in which it is most plausible to think that some quantifiers might be tacitly restricted to things in Cosmo. However, even here I think it is more plausible to think that absolutely unrestricted quantification is going on. Perhaps it is true that many physicists would claim to be talking only about Cosmo if we drew their attention to the fact that some deny that Cosmo is the whole of concrete reality. But in view of physicists' notorious tendency to start spouting positivistic slogans whenever a philosopher comes into the room, it would be dangerous to read much into this hedging behaviour.

¹³At at least one point Lewis suggests precisely this. At *OPW* 124, speaking of arbitrary worlds *W* and *V*, he says that 'the counterpart of *W* at *V* presumably is *V* itself'.

It is thus open to us to interpret *Restriction* not as part of a system of analyses which stands in competition with BMCT, but as nothing more than a special case of BMCT applying to sentences in which the only cases of *de re* reference occur in the form of a restrictor to things in a particular Lewis-world which attaches to every quantifier in the sentence.

Of course, this reconciliation is only successful to the extent that it is plausible to think that Lewis-worlds are one another's counterparts. Section 5.2 will consider whether this claim is well-motivated.

BMCT is not merely a useful concrete example of a theory that allows for non-trivial *de re* modal claims while being consistent with *Modal Redundancy*. Bearing in mind that 'counterpart' is a black box which we can fill in as we see fit, once we accept *Modal Redundancy* there is no real alternative to accepting BMCT for singly *de re* modal claims. Provided that we allow ourselves to engage in quantification over qualitative properties in the obvious way, if we accept *Modal Redundancy*, there is bound to be an interpretation of 'counterpart' on which BMCT can be endorsed, namely one on which ' x is a counterpart of y ' is taken to mean 'the complete qualitative profile of x is such that possibly, y instantiates it'. Under this interpretation, BMCT's analysis for ' $\Diamond\phi(\alpha)$ ' is equivalent to the following claim:

- (8) $\Diamond\phi(\alpha) :=$ there is some object x such that $\phi(x)$ and x 's complete qualitative profile is such that possibly, α instantiates it

The right-to-left direction of (8) is straightforward. Since we are requiring that all vocabulary in $\phi(\alpha)$ except for the term α is purely qualitative, the question whether an object satisfies $\phi(x)$ is determined by that object's complete qualitative profile. If some object x such that $\phi(x)$ instantiates a certain qualitative profile, then instantiation of that profile must *entail* satisfaction of ' $\phi(x)$ '; hence if it is possible that α instantiate that profile, it must be possible that α itself should satisfy ' $\phi(x)$ '. For the left-to-right direction of (8), suppose that it is possible that $\phi(\alpha)$. Then since it is necessary that α instantiate some qualitative profile or other, and qualitative properties are necessary existents, there must be some complete qualitative profile Q such that possibly ($\phi(\alpha)$ and α instantiates Q). But since the question whether $\phi(x)$ for some given x depends only on x 's qualitative character, any such Q must be such that necessarily, whenever some object x instantiates Q , $\phi(x)$. Now, if it is possible that α instantiates Q , it is of course possible that *something* instantiates Q .¹⁴ By *Modal Redundancy*, this entails that something *does* instantiate Q :

¹⁴One might want to resist the argument at this point for some special cases, where

being purely qualitative, the claim that something instantiates Q is necessary if true, and impossible if false. Thus there is some object x such that Q is x 's complete qualitative profile, such that $\phi(x)$. Since Q is a qualitative profile that α could instantiate, this establishes the right-hand side of (8).

Let me recapitulate the dialectic of the last two sections. Suppose that you think that analyses like Lewis's

- (2) It is possible that no swans are blue := there is a Lewis world such that no swans in it are blue

are true on some interpretation or other. If so, you should take the relevant interpretation to be one involving some kind of restricted quantification on the left hand side, otherwise you will run afoul of Puzzles 1–3. You will need some other account of the meaning of modal operators applied to sentences involving unrestricted quantifiers. There are essentially only two options available: the S5-rejecting, 'appropriate part of reality' analysis, and *Generalized Redundancy*, according to which sentences built out of unrestricted quantifiers and well-behaved predicates are modally trivial. The latter is more attractive. But if the standard examples of ill-behaved predicates—'tallest person', 'inventor of bifocals', etc.—are representative of the class, acceptance of *Generalized Redundancy* commits us to acceptance of *Modal Redundancy*, according to which *all* purely qualitative closed sentences are modally trivial—all contingency is *de re* contingency. But if you think this, your general account of *singly de re* modal claims must be BMCT: to be possibly a certain qualitative way is to be appropriately related to something which is that way (i.e. to have a "counterpart" which is that way). If you still really want to hold on to claims like (2), you can do so by interpreting the quantifier on the left as restricted to things in Cosmo, and accepting that the

the truth of $\phi(\alpha)$ is compatible with α 's failure to exist. If we are prepared to speak of complete qualitative profiles which entail *non-existence*, so that for each thing, necessarily, if it instantiates that qualitative profile it fails to exist, we will think that when Q is one of these special qualitative profiles, it is possible that α instantiate Q but not possible that something instantiate Q . Alternatively, we could claim that things cannot instantiate any qualitative profiles without existing: in that case, when $\phi(\alpha)$ is compatible with α 's failure to exist, we will resist the argument from 'possibly $\phi(\alpha)$ ' to 'there is some Q such that possibly ($\phi(\alpha)$ and α instantiates Q)'. However, these ways of resisting the argument are of limited interest, insofar as we still get BMCT for the wide variety of choices of $\phi(\alpha)$ which are not by anyone's lights consistent with ' α does not exist'. I will return to the question what counterpart theorists should think about possible non-existence in section 3.6.

counterparts of each Lewis-world are all and only the Lewis-worlds. But it is unclear what would motivate this claim.

People often think of counterpart theory as a view which isn't really *serious* about *de re* modality. They think of the view as driven by the idea that the *interesting* modal facts are *de dicto* modal facts—natural languages allow for *de re* modal claims, and so we need to give some theory of their truth conditions, but from the point of view of the metaphysician, this is a bit of a distraction—it would be more perspicuous if we all spoke a language in which *de re* modal claims couldn't be formulated at all. But if we develop our counterpart theory in a way that commits us to *Modal Redundancy*, this is all backwards. The theory of *de re* modality is all there *is* to the theory of modality; until we start to deal with *de re* claims, the modal facts are completely trivial and uninteresting. Lewis's way of developing counterpart theory initially seemed to give us a way of avoiding *Modal Redundancy*, and preserving an interesting role for non-*de re* modality. But, as we discovered, this was a mirage; we run into horrible trouble unless we interpret Lewis's theory as a theory of restricted quantification, and it is hard to see what the relevant restriction could be if isn't a *de re* restriction; but if it *is* a *de re* restriction, Lewis's theory does not provide us with the non-trivialising account of non-*de re* modality we thought we wanted.

There is a parallel dialectic in the temporal case. Suppose that you believe that spacetime can be partitioned into special three-dimensional regions—call them “Slices”—in such a way that each ordinary object is, in some way we need not specify at this stage, intimately associated with, or ‘in’, exactly one Slice. You seek a reductive theory of temporal operators. You might be tempted by the thought that whatever you end up saying about *de re* temporal claims, *de dicto* temporal claims can be handled by the temporal analogues of *Restriction*:

$$\begin{aligned} (Restriction_T) \quad F\phi &:= \exists s(Fs \wedge [\phi]^s) \\ P\phi &:= \exists s(Ps \wedge [\phi]^s) \end{aligned}$$

where ‘*Fs*’ and ‘*Ps*’ might be pronounced as something like ‘*s* is a futuristically-accessible Slice’ and ‘*s* is a historically-accessible Slice’. For example,

- (9) It has been the case that some swans are hungry := there is a pastly-accessible Slice some swans in which are hungry.
- (10) It will be the case that no swans are hungry := there is a futurely-

accessible Slice no swans in which are hungry.

But you should not accept such analyses on an unrestricted reading of the quantifiers on the left, since doing so would have implausible results every bit as bad as those that emerged as Puzzles 1–3. Since ‘there once were_u hungry swans’ entails ‘there are_u hungry swans’, we cannot endorse the claim that it will be the case that (there are no_u hungry swans, but there once were_u hungry swans), despite the fact that this follows (intuitively, and in standard tense logic) from the premise that there are_u hungry swans although there will not always be_u hungry swans. In fact, if we think that there will at some point be no hungry swans, we will be forced to say that it will also someday be the case that *there have never been*_u hungry swans. Moreover, even though we think that there are two things which belong to different Slices, we will have to claim that it never will be the case, and never has been the case, that there are two such things.¹⁵

If you want to accept instances of (*Restriction_T*) like (9) and (10), you had better interpret the quantifiers on the left as somehow restricted. But then you will need to give some other theory about what temporal operators mean when combined with unrestricted quantifiers. There is no tenable alternative to accepting the temporal analogue of *Generalized Redundancy*. (The temporal analogues of the ‘appropriate part of reality’ analysis are completely untenable, since the temporal analogue of the conflict with S5 which emerged in the modal case is a conflict with the mixing principles—what is the case always will have been the case; what is the case has always been going to be the case—which are absolutely basic to the logic of the temporal operators.) And just as in the modal case, unless you are willing to posit some mysterious ill-behaved yet qualitative predicate that works very differently from any ordinary examples, this means you will have to accept *Temporal Redundancy*: temporal operators are redundant when applied to purely qualitative sentences; all change is *de re* change. But if you embrace this claim, your account of *singly de re* temporal claims must be BTCT: to be going to be [to have been] a certain qualitative way is to have a future-counterpart [past-counterpart] which is that way. If you still really want to

¹⁵The badness of this result is not *quite* so obvious as the badness of violating T in the modal case. Some conceive of the notion of a ‘timeless truth’ in such a way that when it is a timeless truth that ϕ , ‘it is sometimes the case that ϕ ’ is false. I think this is a mistake, born of a confusion between infelicity due to the presence of a vacuous element and falsehood. No matter what the subject matter, it is intolerable to assert ϕ while also asserting ‘People who assert that ϕ never speak truly’.

hold on to (*Restriction_T*), you can do so by interpreting the quantifiers on the left as restricted to things in the Slice to which *we* belong, and accepting that Slices only have other Slices as future- and past-counterparts. But it remains to be seen whether this is something you should want to say.

The most interesting way to resist these arguments is to reject the step from *Generalized Redundancy* (and its temporal analogue) to *Modal Redundancy* (and *Temporal Redundancy*). We could posit some special predicate-meaning, which is not *de re*, but also not equivalent to anything expressible in terms of well-behaved predicates, and take the presence of this meaning, either as the meaning of an explicit predicate or as a tacit quantifier restriction, to be a source of contingency quite different from *de re* contingency. If we wanted, we could then endorse *Restriction* and (*Restriction_T*) on a reading where the quantifiers on the left hand side are taken to be restricted in the special way, while those on the right are unrestricted. Assuming that we want to allow for *de re* contingency [change] in any case, it is not clear what the motivation for such a posit would be; nevertheless, some philosophers seem to take it seriously. So, section 3.5 will be devoted to exploring some forms such an account might take, and showing they are deeply problematic. But first. . .

3.4 An exegetical interlude

To what extent did Lewis actually accept the views I have been offering him? Did he, for example, accept that modal operators are redundant when applied to purely qualitative sentences?

There is considerable evidence that suggests that he did. For one thing, Lewis does not seem to embrace the puzzling claims that—as we saw in section 3.2—follow from *Restriction* when all the quantifiers are taken as unrestricted. There are, moreover, several passages in which Lewis directly considers the modal status of claims involving unrestricted quantification, and seems to suggest that they are necessary when true, i.e. that *Generalized Redundancy* is true. Here is one example:

Our *contingent* knowledge that there are donkeys at *our* world requires causal acquaintance with the donkeys, or at least with what produces them. Our *necessary* knowledge that there are donkeys at *some* worlds—even talking donkeys, donkeys with dragons as worldmates, and what have you—does not require causal acquaintance either with the donkeys or with what produces them. (OPW 112)

This presupposes that it is necessary that there are donkeys at some worlds, and thus that it is necessary that there are donkeys, unrestrictedly speaking. (It also suggests, interestingly, that the true reading of 'it is contingent that there are donkeys' is equivalent to 'it is contingent that there are donkeys at our world', which seems to be a *de re* modal claim about a particular Lewis-world, or perhaps about *us*: this does something to support the attribution to Lewis of *Modal Redundancy*.) In a similar vein, Lewis says that 'reality might have been different' is not true, if we understand 'reality' in such a way as to make 'reality is the totality of everything' true (OPW 101, note 1). And in Lewis 1996, he accepts without demur the attribution to him of a theory that 'treat[s] all statements about modal reality as non-contingent: if any such statement is possibly true, then it is true *simpliciter*'.¹⁶

Also supporting the attribution of *Modal Redundancy* to Lewis are several passages in which he suggests that the accessibility relation on worlds and the counterpart relation on proper parts of worlds are just two subcases of a more general relation of one thing being a 'possibility for' another:

As other worlds are alternative possibilities for an entire world, so the parts of other worlds are alternative possibilities for lesser individuals. . . . As quantification over possible worlds is commonly restricted by accessibility relations, so quantification over possible individuals is commonly restricted by counterpart relations. (OPW 8)

But I say that *any* possible individual is a possibility, and not all possible individuals are possible worlds. Only the biggest ones are. . . . [A possible world] is a way that an entire world might possibly be. But lesser possible individuals, inhabitants of worlds, proper parts of worlds, are possibilities too. They are ways that something less than an entire world might possibly be. (OPW 230)

These claims tend to suggest that the most perspicuous and general approach to the analysis of modal claims would be expressed in terms of the generic

¹⁶Another relevant passage: 'If I had acted otherwise, for instance by taking to a life of crime, each and every good or evil that is present somewhere in the totality of worlds would still have been present, and none would have been added. It is wrong to think: then this world would have been a little worse, and the rest would have been no different, so the totality of worlds would have been a little worse.' (OPW 123) Given that Lewis endorsed the inference from 'necessarily ϕ ' to 'if it were that ψ , then it would be that ϕ ', if he had endorsed *Restriction* as an analysis of possibility claims embedding unrestricted quantifiers, he should instead have said 'if I had acted otherwise there would have been only one Lewis-world'. (See chapter 5 for more discussion of this passage.)

'possibility for' relation, and would not require singling out Lewis-worlds for any special treatment.

On the other hand, the first few pages of *OPW*, in which the role of Lewis-worlds in the analysis of modal claims is introduced, strongly suggest a view on which *Generalized Redundancy* fails. We are told that 'possibly there are blue swans iff, for some world *W*, at *W* there are blue swans', where 'at *W*' 'works mainly by restricting the domains of quantifiers in its scope, in much the same way that the restricting modifier "in Australia" does' (*OPW* 5). It would be odd to think that 'In Australia, all swans are black' expresses a truth only when we interpret the quantifier as carrying some tacit restriction: since we have gone to the trouble of making 'in Australia' explicit, there is no obvious need to posit any further implicit restriction. On Lewis's account of 'at *W*', then, it looks like it can apply restrictions *even when it is applied to a sentence that doesn't come with its own tacit restriction*, so that 'at *W*, Q_u *Fs* are *Gs*' means the same as ' Q_u *Fs* which are in *W* are *Gs*'. If this is right, 'At some Lewis-world *W*, no swans are blue', and hence also 'Possibly no swans are blue', will be true even when the quantifier is just the familiar unrestricted quantifier.

There is, moreover, some textual evidence that suggests that the restrictions apply even to unrestricted quantifiers, for example the following passage:

We wanted to ask whether two worlds could differ in their laws without differing in their distribution of local qualitative character. But if we read the 'could' as a diamond, the thesis in question turns into this: it is not the case that, possibly, two worlds differ in their laws without differing in their distribution of local qualitative character. In other words: there is no world wherein two worlds differ in their laws without differing in their distribution of local qualitative character. That's trivial—there is no world wherein two worlds do anything. At any one world *W*, there is only the one single world *W*. The sentential modal operator disastrously restricts the quantification over worlds that lies within its scope. (*OPW* 16)

Admittedly, it is not clear that this passage commits Lewis to the claim that the restrictor analysis works for modal claims expressed using the English word 'possibly': the point of the passage is to argue that certain claims in English are not well expressed in a certain *formal* language. But Lewis certainly doesn't suggest that he has up his sleeve a *better* analysis of 'pos-

sibly' in English which allows it to behave differently from 'diamond' in the relevant respect.¹⁷

Some remarks of Lewis's suggest an alternative strategy for blocking the reasoning behind the puzzles without embracing *Generalized Redundancy*. For his treatment of 'at W' includes an important qualification, which I left out in my earlier presentation of the view:

I do not suppose that [restrictive modifiers] must restrict all quantifiers in their scope, without exception. 'In Australia, there is a yacht faster than any other' would mean less than it does if the modifier restricted both quantifiers rather than just the first. . . . [A] lot is left up to the pragmatic rule that what is said should be interpreted so as to be sensible. (OPW 6).

Thus according to Lewis, sentences involving restricting modifiers, including modal operators, admit a range of readings, which differ as regards which quantifiers end up having restrictions attached to them. So, for example, 'In W there are many Lewis-worlds' and 'Possibly there are many Lewis-worlds' have readings on which they are both equivalent to 'There are many Lewis-worlds', as well as readings on which they are trivially false. And it is easy to imagine a pragmatic story about why we would tend to prefer the former readings.

The profusion of readings generated by making the quantifier-restricting effects of modal operators optional makes the puzzles from section 3.2 harder to raise, by making the application of logic within the scope of modal operators a complex affair. For example, a sentence like

- (11) Possibly (there are no blue swans and there is a Lewis world that contains at least one blue swan)

¹⁷At OPW 211, Lewis sounds like he is rejecting T, at least for some notion of possibility: 'I do not deny the existence of trans-world individuals, and yet there is a sense in which I say that they cannot possibly exist.' But this is complicated by the immediately following explanation: 'As should be expected, the sense in question involves restricted quantification.' This remark implicates that there is another sense in which 'trans-world individuals cannot possibly exist' is *false*, and that this sense involves *unrestricted* quantification. This looks like good news for the attribution of the redundancy thesis to Lewis. However, it is not entirely clear that Lewis is attributing the difference between the two readings of 'trans-world individuals cannot possibly exist' to a variation in the interpretation of the quantification introduced by 'exist'—it might be suggested that the sense in which the true reading of 'transworld individuals cannot possibly exist' 'involves restricted quantification' is just that restricted quantifiers will crop up in the *analysis* of this claim.

can be read as expressing a truth (by restricting only the first quantifier), despite the logical inconsistency of the sentence to which the operator is applied. Of course, those who accept *Generalized Redundancy* can also allow for true readings of (11), in which the first quantifier is tacitly restricted and the others are not; but on this view, the true readings of (11) correspond to readings of the embedded sentence on which it is not *really* logically inconsistent. If we want to interpret Lewis's idea of optional restriction as a genuine alternative to *Generalized Redundancy*, we cannot say this. This is a heavy cost: many uses of modal operators would be seriously compromised if we cannot reason from the logical inconsistency of a sentence on a certain resolution of its context-sensitivity to the falsehood of the result of prefixing that sentence with 'Possibly' while holding fixed the interpretation of the context-sensitive elements in the original sentence. Moreover, while the approach allows us to avoid having to make certain bad-sounding speeches, it does not straightforwardly allow us to claim validity for modal principles like S5 and T. As I understand it, the claim that T is *valid* (for metaphysical necessity, say) should commit us to the claim that for every sentence ϕ , $\ulcorner \phi \rightarrow (\text{it is metaphysically possible that } \phi) \urcorner$ is true on *all* admissible readings, provided that any ambiguity or context-sensitivity in ϕ is resolved in the same way in both occurrences. If we think that T is valid, then, there is no way to avoid claiming that the true readings of 'There are at least two Lewis-worlds, and it is not metaphysically possible that there are at least two Lewis-worlds' involve differential resolution of some context-sensitive constituent within the sentence 'there are at least two Lewis-worlds' itself. And the only plausible source of relevant context-sensitivity in *this* sentence is the one associated with quantifier domain restriction.

3.5 Alternative accounts of contingency-inducing quantification

To accept *Generalized Redundancy* without accepting *Modal Redundancy*, one must posit a way of interpreting quantifiers as carrying some restriction which is neither not *de re*, but also not equivalent to anything expressible in terms of well-behaved qualitative predicates. How could this work?

On one version of the view, the special kind of interpretation is available *only* when quantifiers are embedded under modal [temporal] operators. In this respect, it is like the bound interpretation of pronouns, which is available for pronouns in the scope of appropriate quantifiers, but corresponds to no reading that unembedded pronouns can bear. Semantically, it would be

natural to implement this view by thinking of the restrictor as involving a hidden variable, which is bound by the relevant modal [temporal] operator. This is reminiscent of the kind of story told by B-theorists. But the account of modal [temporal] operators required by this view is much more complicated than the standard B-theoretic conception of them as quantifiers over worlds [times]. For given counterpart theory, these operators are not *just* quantifiers, since they can have a non-redundant effect even when they are attached to a sentence which, like 'Obama is tall', contains no variables for them to bind.

On this view, there are two structurally different ways to get a true reading for the sentence 'It could have been the case that no swans were white'. One possibility is to posit a *de re* restriction in 'no swans'. Syntactically, this could take the form of an unpronounced constituent 'in x ', where the variable x is free and interpreted deictically. On this reading, our metaphysical account of what it is for the sentence to be true will involve quantification over counterparts of x . A quite different possibility is to posit an unpronounced constituent 'in x ', where x is *bound* by the operator 'it could have been the case that'. On this reading, the counterpart relation is irrelevant.

The main objection to this view is its needless, and arbitrary, complexity. It is one thing to suggest, as modal B-theorists do, that modal operators are *always* variable-binders. It is another thing to suggest that they *sometimes* bind hidden variables, and sometimes apply non-vacuously to closed sentences (like 'Obama is tall') by introducing quantifiers over counterparts. There is no linguistic motivation for drawing a distinction in syntactic structure between 'Possibly, Obama is not tall' and some uses of 'Possibly, no swans are white', according to which variable-binding takes place only in the latter case. Even more obviously, there is no linguistic motivation for positing two different syntactic structures for 'Possibly, no swans are white'. (How are we supposed to tell which of the two mechanisms is at work in any given case?) I am not saying, here, that our ultimate *metaphysical* account of what it is for it to be possible or necessary that P has to work in any particularly uniform way; only that metaphysical considerations should not drive us to posit differences of the sort *syntax* cares about, such as differences in variable-binding structure, unless we can motivate those differences using linguistic as opposed to metaphysical considerations.

Let us assume, then, that the distinctive kind of restricted interpretation—contingency-inducing, yet not *de re*—is available even for unembedded uses of quantifiers. I will continue to use the subscript r to indicate the relevant form of restriction. For the sake of having a concrete example, let us suppose

that no_r swans are blue, even though some_u swans are blue. Here is the question I want to press. Consider a person, Twin, who is generally similar to me as regards his intrinsic and environmental properties, but who happens to be part of a Lewis-world—Cosmo-Prime—which also contains blue swans. Suppose I say ‘some swans are blue’, and thereby say (falsely) that some_r swans are blue. When Twin says ‘some swans are blue’ under analogous circumstances, does he also say that some_r swans are blue?

There is a compelling argument that he does not. We have assumed that

(P1) No_r swans are blue.

From the truth schema (see section 1.2), we know that

(P2) The proposition that some_r swans are blue is true iff some_r swans are blue.

But it seems obvious that

(P3) When Twin says ‘some swans are blue’, everything he says is true.

When people in Cosmo-Prime see blue swans swimming around and say ‘some swans are blue!’, the belief they thereby express is (usually) paradigmatically justified by ordinary perceptual methods. Are we supposed to think of Cosmo-Prime as a place of epistemic tragedy, where following these excellent methods leads people to form systematically untrue beliefs? Surely not; hence (P3). But from (P1), (P2), and (P3), elementary logic yields

(C) When Twin says ‘some swans are blue’, he doesn’t say that some_r swans are blue.

If one follows this kind of reasoning where it leads, one will end up thinking that the practice of using ‘some swans are blue’ to say that some_r swans are blue is more or less confined to Cosmo. People in other Lewis-worlds, even ones in most respects similar to Cosmo, do not express the very same propositions as us when they say the words that in our mouths express claims about what there is_r.

(It would be a bad mistake to confuse (C) with the counterfactual claim that if I had been spatiotemporally related to blue swans but otherwise qualitatively similar to the way I actually am, *I* would not have used ‘some swans are blue’ to say that some_r swans are blue. There are lots of things that people in other Lewis-worlds never do that I would have done even if I had

been qualitatively different in all kinds of ways. For example, no-one in any other Lewis-world lives on Earth. But my living on Earth is counterfactually robust: there is a wide range of qualitative properties such that I would still live on Earth even if I had them. For example, I would still live on Earth if I were a philosopher of mind. Chapter 4 below will discuss how counterpart theorists should analyse “multiply *de re*” modal claims like ‘I could live on Earth while being a philosopher of herpetology’; but it is a constraint on the adequacy of such analyses that they be compatible with the fact that no philosophers of herpetology live on *Earth*.)

(C) would not be surprising iff we were taking the quantifiers subscripted with ‘r’ to be some kind of *de re* restricted quantifiers. On this view, to say that some_r swans are blue is to say that some swans on Earth are blue, or that some swans in Cosmo are blue, or some such thing. And of course, this isn’t the sort of thing we should expect to find people at other Lewis-worlds doing very often. For one thing, saying that some swans on Earth are blue seems to involve *referring* to Earth; and it is not usually thought to be easy to refer to objects towards which one bears no spatiotemporal or causal relations.¹⁸ When I say that some swans on Earth are blue, Twin surely does not: rather, he says that some swans on Earth-Prime are blue, where Earth-Prime is the planet Twin lives on. Similarly, when I say that some swans in Cosmo are blue, Twin under analogous circumstances says, instead, that some swans in Cosmo-Prime are blue. But if, as we are assuming, the claim that some_r swans are blue does *not* involve *de re* reference to Cosmo or any of its parts, it becomes hard to see how (C) could be true. How could the mere fact that Twin and I are parts of different Lewis-worlds be relevant to the facts about which propositions we express, other than *de re* propositions about ourselves or our surroundings? It is unmysterious how a *de re* proposition could be much easier to express for people in one Lewis-world than for people in another, very similar Lewis-world; it is very mysterious what could make for such differences in the case of a non-*de re* proposition.

The “pictorial ersatzism” Lewis argues against in *OPW* §3.3 is a view

¹⁸Perhaps there are some contexts (see Hawthorne and Manley MS, ***) in which reference is very easily achieved—maybe it is even true in some of these contexts that whenever one says that all *F*s are *G*s, and *x* is an *F*, one says that *x* is *G*. By these lax standards, Twin can easily say that some swans on Earth are blue, just by saying that some swans on every planet (unrestrictedly speaking) are blue. But it is implausible that all contexts work like this; we can run our argument within a less permissive context. And even by very permissive standards, it is still not plausible that Twin says that some swans on Earth are blue when he utters the sentence ‘some swans on Earth are blue’.

on which (P3) and (C) both fail. There is a special property “vim”, which we often use in restricting our quantifiers. Cosmo and its parts have vim; other Lewis-worlds don’t. Vim could have been distributed differently with respect to ordinary qualitative properties—for example, there could have been a Lewis-world with vim and containing blue swans. ‘Vim’ is not a well-behaved predicate: *Restriction* and RCT cannot be applied when we are dealing with sentences in which ‘vim’ occurs in the scope of a modal operator. For example, there could have been blue swans with vim even though no Lewis-worlds contain blue swans with vim. One doesn’t need to have vim to be a swan, to be a person, to be conscious, etc: thus predicates like ‘swan’, ‘person’, and ‘conscious’ can still be treated as well-behaved for the purposes of applying *Restriction*. It is reasonable for us to be highly confident that we have vim. It is also reasonable for the many, many people who lack vim to be highly confident that they have vim. But they are just wrong. The vimless part of reality is a place of pervasive epistemic tragedy, where good reasoning leads people to form systematically false beliefs about the distribution of vim.¹⁹

This is not an attractive view. Applying the epistemological standards that normally apply to the postulation of new primitive entities or properties, there seems to be no justification for postulating vim at all, or—assuming that we do postulate it—for believing anything very specific about its distribution. Of course, it is part of the view that these are the wrong standards to

¹⁹Phillip Bricker’s view (2001, 2006) is similar in many ways to the one just described. There are two unimportant differences: he calls vim ‘actuality’, and instead of saying that it is a qualitative but ill-behaved property, he says that it is non-qualitative, thereby departing from the standard usage in which ‘non-qualitative’ is equivalent to ‘*de re*’ or ‘*haecceitistic*’. There is one moderately important difference: unlike Lewis’s pictorial ersatzist, Bricker thinks it is possible that more than one Lewis-world has vim—indeed, he thinks that any fusion of Lewis-worlds could have vim. And there is one really important difference: according to Bricker, people who lack vim do not refer to vim, or at least do not do so when their ordinary qualitative properties are of such a nature that a vim-endowed person with those properties would be referring to vim. On his preferred version of the view, when we express propositions about the distribution of vim, qualitatively similar but vimless people fail to express anything at all. The vimless part of reality is, in a sense, a place of epistemic tragedy; but the tragedy is not that good reasoning leads to false beliefs, but that it leads to contentless attempts at thought. (C) is thus *true*, assuming that Twin lacks vim.

Bricker’s central argument for going down this route is his rejection of Lewis’s doctrine of the indexicality of actuality. In section 4.6, I will argue that my favourite form of modal realism—one that accepts BMCT and *Modal Redundancy*—should also jettison this part of Lewis’s view. Thus Bricker’s arguments do not provide reason for preferring his view over my favourite view.

apply, since the epistemological standards governing beliefs about vim are quite different from the standards governing beliefs about any other property. But an implausible metaphysics is not generally made better by being combined with an implausible epistemology, one tenet of which is that the metaphysics is something it is reasonable to believe. Things might be different if striking theoretical simplifications could be achieved by taking vim to be involved in the analysis of our ordinary beliefs; but it is obscure what the theoretical benefits are supposed to be.

One important motivation for wanting there to be a reading of 'no_r swans are blue' on which (C) fails is the doctrine, defended by Lewis (*OPW* 1.4), that propositions—at least in one central sense—are sets of Lewis-worlds. Anyone who accepts this will presumably think that there is a proposition, call it *p*, which is identical to the set of Lewis-worlds that contain blue swans. *p* looks like a natural candidate to be the proposition that some_r swans are blue. But if we sometimes assert *p* by saying 'some swans are blue', surely people in other Lewis-worlds can do so with equal ease. There is nothing in the nature of *p* that could make it easier to express here in Cosmo than anywhere else, given any sensible theory of what it takes to assert a set of worlds.

According to Lewis, not only some but *all* propositions can be identified with sets of Lewis-worlds. I think this is a clear mistake. Given that Earth is part of just one Lewis-world, there is just no plausible way to identify, say, the proposition *that no swans on Earth are blue* with any set of Lewis-worlds. For if this proposition is a set of worlds, which set of worlds could it be? If we help ourselves to a counterpart relation, there are a few natural candidates: the set of Lewis-worlds in which *some* counterpart of Earth lacks blue swans; the set of Lewis-worlds in which *every* counterpart of Earth lacks blue swans; the intersection of these two sets; the set of Lewis-worlds in which there is exactly one counterpart of Earth, which lacks blue swans. But any suggestion of this form is deeply problematic, for several different reasons.

- (i) 'Counterpart' is context-sensitive; we will get different sets of worlds depending on how we understand it. But while it is arguable that there is some context-sensitivity in the sentence 'No swans on Earth are blue' (e.g. because of shiftiness as regards what counts as 'blue'), it is not context-sensitive in the same way as 'counterpart'; and we could

have used a sentence ('Particle *a* is negatively charged') which is not plausibly context-sensitive at all. So the use of any particular counterpart relation as the basis for identifying a particular set of worlds as the proposition that there are no blue swans on Earth is objectionably arbitrary.

- (ii) Whichever candidate we choose, we will be committed to the claim that whenever the counterparts of *x* are exactly the counterparts of Earth, the proposition that there are no blue swans on *x* is identical to the proposition that there are no blue swans on Earth. So if Earth-Prime and Earth happen to be similar enough to share exactly the same counterparts, the proposition that no swans on Earth-Prime are blue is the proposition that no swans on Earth are blue.²⁰ But on any ordinary account of the counterpart relation, there is nothing to stop this from happening even when there *are* blue swans on Earth-Prime but not on Earth. So we get a violation of the truth schema: there are blue swans on Earth-Prime, and there are no blue swans on Earth, and yet the proposition that there are blue swans on Earth-Prime is the proposition that there are no blue swans on Earth.
- (iii) We also get trouble with propositional attitudes. If the proposition that there are blue swans on Earth-Prime is the proposition that there are blue swans on Earth, then anyone who believes or asserts that there are blue swans on Earth-Prime also believes or asserts that there are blue swans on Earth. And this is just incredible. People on Earth-Prime may very well believe, and assert, that there are blue swans on Earth-Prime; they obviously do *not* believe or assert that there are blue swans on *Earth*, a planet to which they don't bear any spatiotemporal relations, let alone the kinds of causal relations sometimes thought necessary for reference to occur. They may very well not believe or assert *anything* false.
- (iv) According to Lewis, some objects have multiple counterparts within a single Lewis-world. Suppose that Earth has two counterparts in some Lewis-world *w*, exactly one of which is inhabited by blue swans. If the

²⁰If we endorsed the claim (entertained in section 3.3) that each Lewis-world has all and only the Lewis-worlds as counterparts, then we will get an especially pernicious form of this problem: for any two Lewis-worlds *w* and *v*, the proposition that there are no blue swans in *w* will be identical to the proposition that there are no blue swans in *v*.

proposition that there are blue swans on Earth is a set of Lewis-worlds, it either does or doesn't contain w . In either case, we are in danger of having to accept something obviously false. If the proposition does contain w , it looks like 'The proposition that there are blue swans on Earth is such that possibly, it is true even though there are no blue swans on Earth' will come out true. If it doesn't, it looks like 'The proposition that there are blue swans on Earth is such that possibly, it isn't true even though there are blue swans on Earth' will come out true. At least, this is how things go if we adopt a Lewis-style counterpart theory in which modal operators introduce quantification over Lewis-worlds, and analyse 'true' in a modal context as 'contains w ', where w is a variable bound by the relevant quantifier. And if we are not planning to understand ascriptions of truth in terms of membership in this way, the identification of propositions with sets of worlds loses most of its point.²¹

Given counterpart theory, sets of Lewis-worlds are just terrible as candidates to be identified with *de re* propositions, such as the proposition that there are blue swans on Earth. If we want to identify these propositions with set-theoretic constructions of some sort, we will do much better to employ the kind of construction often used in the literature on 'structured propositions'. For example, we might take the proposition that there are blue swans on Earth to be the ordered pair ⟨Earth, the set of things on which there are blue swans⟩, or perhaps some more complicated construct that more closely mirrors the syntactic structure of the sentence 'there are blue swans on Earth'. (Even Lewis was willing to grant that such n -tuples could reasonably be identified with propositions, on one legitimate conception of propositions: see *OPW* 57–9.) So long as we have some counterpart-theoretic account of the multiply *de re* which lets us endorse claims like 'Earth's singleton set is such that necessarily, Earth is a member of it', we will then be able to endorse consequences of the truth schema like 'The proposition that there are blue swans on Earth is such that necessarily, it is true iff there are blue swans on Earth'. We can do by saying that a proposition that is an ordered pair of an object and a set is "true" iff the object belongs to the set in question, and extending this appropriately to propositions with more complicated kinds of structure.

* * *

²¹Dorr 2005 develops this point in much more detail.

So much for the claim that *all* propositions are sets of Lewis-worlds. Someone might, however, accept my arguments against this claim while insisting that it is still true that *some* propositions are sets of Lewis-worlds. It is not clear what could motivate such a view: Lewis's own argument for taking propositions to be sets of Lewis-worlds turns on systematic considerations which would have to apply across the board if they worked at all. Still, it is worth thinking through the consequences of a view which says that there are *some* propositions which can be identified with sets of Lewis-worlds, and in particular takes the set of blue-swan-containing Lewis-worlds to be the proposition expressed by 'some swans are blue' on the reading where it is contingent but not *de re*.

Proponents of such a view must find a way to resist the argument for (C) presented in section 3.5. Their response will surely turn, somehow, on a piece of ideology that the identification of propositions with sets of Lewis-worlds makes available, namely the notion of a proposition's being "true at" a Lewis-world: a proposition that is a set of Lewis-worlds is true at *w* iff *w* is one of its members. But how is the availability of this relativised truth predicate supposed to help us to resist the argument for (C), whose premises used monadic 'true', not 'true at'?

One thought is that we should be *contextualists* about monadic 'true' (at least when applied to propositions that are sets of Lewis-worlds). According to this thought, 'true' is context-sensitive: a given occasion of utterance, it is equivalent to 'true at *w*', where *w* is some conversationally relevant Lewis-world. By default, the Lewis-world relevant to an utterance is the one the speaker is part of. This default plausibly applies to a sentence like (P2), which as uttered by us will normally express the same thing as (P2'):

(P2) The proposition that some_r swans are blue is true iff some_r swans are blue.

(P2') The proposition that some_r swans are blue is true *at Cosmo* iff some_r swans are blue.

However, it is open to the contextualist to allow exceptions to the rule that the speaker's Lewis-world is always the relevant one, in cases where other Lewis-worlds have been made salient. By allowing such exceptions, the contextualist can account for the apparent truth of (P3), by claiming that in context, it is naturally interpreted as equivalent to (P3'):

(P3) When Twin says 'some swans are blue', everything he says is true.

(P3') When Twin says 'some swans are blue', everything he says is true *at Cosmo-Prime*.

Thus (P2) and (P3) are both true if we resolve their context sensitivity in the most natural way; but in that case the argument to (C) is invalid. If we resolve the context-sensitivity of 'true' uniformly, the argument is valid but unsound: if we understand 'true' as 'true at Cosmo', (P3) fails; if we understand it as 'true at Cosmo-Prime', (P2) does.

Prima facie this might look like a neat way to get around the argument for (C). Unfortunately, as we have already seen in chapter 1 (section 1.5), the claim that the truth schema holds in all contexts rules out this kind of context-sensitivity in monadic 'true'. For suppose 'true₁' and 'true₂' represented two admissible interpretations of 'true', we would have

The proposition that ϕ is true₁ iff ϕ

and

The proposition that ϕ is true₂ iff ϕ

and hence

The proposition that ϕ is true₁ iff it is true₂.

Thus, 'true' cannot be an independent source of context-sensitivity. If we want to say that 'the proposition that some_r swans are blue is true' has a hidden, truth-value-affecting contextual parameter whose value is a Lewis-world, we can do so only by saying the very same thing about 'some_r swans are blue'. But that would be to embrace the view that the relevant form of quantifier-restriction is just a kind of *de re* restriction.

The theorist who holds that some propositions are sets of Lewis-worlds is thus in the same position as the Relativists discussed in section 1.5. They have a relativised truth-predicate, to which they assign a certain theoretical importance; but they also admit a monadic truth-predicate, which conforms to the truth schema. They will thus accept (P2), which is an instance of the truth schema. Given that they want to accept (P1) while rejecting (C), they must thus simply reject (P3). Twin *does* say (and mean, and believe...) that some_r swans are blue; and what he believes is false, since no_r swans are blue. (P2'), however, is true: what Twin says is not false *at Cosmo-Prime*. *At Cosmo-Prime*, it is Twin who is right and I am who am wrong. The relation between Twin and me is like the relation between lovers and haters of Marmite, as

conceived by Relativists about ‘tasty’. The lovers believe that Marmite is tasty; the haters believe that it isn’t. Marmite either is or isn’t tasty, so either the lovers or the haters believe something false. But it’s not a big deal! For it is possible to believe something false without being “at fault”—not just in the epistemological sense that one’s belief is justified, or supported by one’s evidence, but in some deeper sense. The lovers and the haters both achieve ‘the goal of belief’, which requires only that the proposition believed be *true relative to the believer’s standards*, a status compatible with monadic falsehood.

As I discussed in chapter 1, the Relativist programme strikes me as extremely problematic even for ‘tasty’. According to Relativists, I ought to think that the things that are tasty for me are tasty, and the propositions that are true for me are true, while the propositions that are true for other people are often false; if they believe these propositions, as they ought, they will get things wrong. This seems to describe a state of extraordinary intellectual hubris: why should I be so confident that *I* am the special person truth for whom coincides with truth? The further claim that everyone else ought to (falsely) regard themselves as special in the same way does nothing to dispel the implausibility of this. However, at least in the case of ‘tasty’ there is a real philosophical itch that Relativism tries to scratch. It is certainly tempting to think those who, unlike me, dislike Marmite, *disagree* with me about whether Marmite is tasty, and thus believe (falsely) that it isn’t tasty. By reflecting on the difficulty of convincing the Marmite-haters of this, I could easily end up thinking that there is no good reasoning that could convince them that Marmite is tasty, and indeed, that good reasoning requires them to stick as firmly to their false belief as it requires me to stick to my true one. I can imagine getting this far and still feeling that in some sense we have no *deep* disagreement—and obscure as it is, the “no fault” idea perhaps does something to articulate this elusive notion of shallowness.

By contrast, when the difference between the relevant others and us is just that we are in different Lewis-worlds, there is nothing even *prima facie* appealing about Relativism. Why should I think, just on the grounds that Twin is not spatiotemporally related to me, that he will end up forming lots of false beliefs, for example the belief that some_r swans are blue, just by reasoning as he should? It is not at all plausible that, just because I think there are no swans in my Lewis-world are blue while Twin thinks that some swans in his Lewis-world are blue, Twin and I are committed to *disagreeing* about some proposition (viz. that some_r swans are blue). This idea is no more plausible than the idea that there is a proposition which

you and I are committed to disagreeing about whenever I think that there is beer in my house and you think that there is no beer in yours. One would only be tempted to think that Twin believes propositions inconsistent with propositions I believe if one confused this claim with some counterfactual claim to the effect that if things had been thus and so, I would have believed some propositions inconsistent with propositions I actually believe. But that would be a bad confusion: it would be like inferring the false conclusion that there is a city that Twin hates and I like from the (perhaps true) premise that there is a city I actually like but would have hated if I had instantiated Twin's qualitative profile.²²

The case for (C) stands. When people in other Lewis-worlds restrict their quantifiers in the way that leads to contingency, they generally do not express the same propositions that we express when we restrict our quantifiers in an analogous way. This is straightforwardly explained by the hypothesis that contingency-inducing quantification is *de re* restricted quantification. I grant that other explanations are conceivable. One could, for instance, propose that vim comes in a multitude of different flavours: our contingency-inducing quantifiers are restricted to things with our flavour of vim, while Twin's contingency-inducing quantifiers are restricted to things with his different flavour of vim.²³ But what is the point? Given the strength of the independent reasons for thinking that *de re* restricted quantification is pervasive, it is superfluous to postulate a new realm of properties, different in character both from ordinary qualitative properties and from *de re* properties, merely in order to provide an *additional* source of contingency in quantified claims. The best way to be a modal realist is to endorse BMCT, and learn to love the idea that all contingency is *de re* contingency.

²²Ironically, one of MacFarlane's favourite ways of defending Relativism in other domains is to appeal to something very like the view I have just been criticising: 'Consider Jane (who inhabits this world, the actual world) and June, her counterpart in another possible world. Jane asserts that Mars has two moons, and June denies this very proposition. Do they disagree? Not in any real way.' (MacFarlane 2007, p. 10)

²³Parsons (2002) considers a view that works rather like this in the temporal case. On the version of the A-theory he sets out, each instant of time has its own distinctive, intrinsic, monadic property, a 'degree of futurity or pastness'. These properties could have been distributed differently from how they in fact are distributed; for example, any given instant could have instantiated the particular property we call 'presentness', and indeed, each instant that does not instantiate this property either did once instantiate it, or will eventually instantiate it. Uses of the word 'present' at a given instant pick out whichever degree of futurity or pastness characterises that instant.

3.6 Necessary and eternal existence

Unrestrictedly speaking, everything is identical to something. *A fortiori*, every counterpart of anything is identical to something. Thus BMCT entails the very controversial claim that, unrestrictedly speaking, each thing is necessarily identical to something:

(NE) $\forall x \Box \exists y (y = x)$.

If we follow the logician's custom of understanding 'exists' as synonymous with 'is identical to something', we get the headline-grabbing claim that *each thing necessarily exists*. This is not a peculiarity of BMCT: since every counterpart of anything that is in a Lewis-world is identical to something in that Lewis-world, Lewis's analysis RCT also entails NE. Most philosophers have found this claim extremely implausible. According to them, Prince Charles, for example, could have failed to be identical to anything at all: Queen Elizabeth and Prince Philip could have failed to meet; if so, Prince Charles would never have been conceived; and in that case, there would have been nothing identical to him. This is one of the standard objections to counterpart theory.

A parallel objection could be raised to temporal counterpart theory. Just as BMCT entails NE, BTCT entails that each thing has always been, and always will be, identical to something (unrestrictedly speaking):

(EE) $\forall x H \exists y (y = x)$

$\forall x G \exists y (y = x)$

This is deeply controversial too. It is rejected, influentially, by 'presentists' in the tradition of Prior (1957, 1967, 1968b). However, temporal counterpart theorists who endorse EE have some influential allies, namely those philosophers—temporal B-theorists, for the most part—who take expressions like 'always' to be quantifiers that bind syntactically real variables ranging over times (see section 2.1). Those who think this must perforce take tense operators to be vacuous when applied to sentences that contain no variables for them to bind. And when the quantifier is read as unrestricted, it is hard to see how ' $\exists y (y = x)$ ' could contain any such variables. In the modal case, by contrast, NE is decidedly a minority view, although its popularity has grown somewhat in recent years thanks to the efforts of Zalta and Linsky (1994) and especially Timothy Williamson (1998, 2002, 2010, 2013).

Lewis's reaction to this objection to counterpart theory is quite concessive. He clearly thinks that there is a natural reading of 'Humphrey could have failed to exist' on which it is true iff there are Lewis-worlds that contain no counterparts of Humphrey. But he seems to despair of the task of providing a systematic semantic account of modality in English which explains how the sentence comes to have this reading:

What is the correct counterpart-theoretic interpretation of the modal formulas of the standard language of quantified modal logic? —Who cares? We can make them mean whatever we like. We are their master. We needn't be faithful to the meanings we learned at mother's knee—because we didn't. If this language of boxes and diamonds proves to be a clumsy instrument for talking about matters of essence and potentiality, let it go hang. Use the resources of modal realism *directly* to say what it would mean for Humphrey to be essentially human, or to exist contingently. (OPW 12–13)

I find this passage rather mystifying. True, I didn't learn about '□' and '◇' at mother's knee: but I *did* learn the words 'necessarily' and 'possibly' somewhere thereabouts, and later I was introduced to '□' and '◇' as synonyms for 'necessarily' and 'possibly'. True, the resources of English for talking about modal matters are rich. As well as 'necessarily' and 'possibly', we have more adverbs like 'essentially' and 'contingently', as well modal auxiliaries like 'can' and 'could', adjectives like 'possible' and 'potential', etc. We should be open to learning that not everything that can be said using these resources can be said using 'necessarily' and 'possibly'. But Lewis is not even sketching an approach to semantics of these aspects of English. Rather, it sounds like he is saying that we should be content with one-off accounts of particular sentences and give up on systematic semantics.

In any case, in the light of what we have learned in this chapter, the suggestion that the existence of Lewis-worlds containing no counterparts of Humphrey would be enough to vindicate a true reading of 'Humphrey could have failed to be identical to anything' in which the quantifier is *unrestricted* looks hopeless. That suggestion is inextricably tied to a conception of modality in which quantification over Lewis-worlds play a central role, and in which that role involves providing restrictions for quantifiers, even ones that are originally unrestricted. I have argued that this conception should be replaced by one conforming to *Modal Redundancy*. And once we make this replacement, the fact that there are Lewis-worlds containing no counter-

parts of Humphrey will look completely irrelevant to the truth of 'Humphrey could have failed to be identical to anything, unrestrictedly speaking', even if we are willing to entertain unorthodox ideas about the relation between this sentence and the language of boxes and diamonds.

I think a better option for modal counterpart theorists is to stand their ground and insist that it is just *true*, speaking unrestrictedly, that everything is necessarily identical to something. When unrestricted quantification is in play, modal realists are used to disagreeing with ordinary opinion. Nothing is more familiar to them than claiming, of some intuitively true sentence involving quantifiers, that it is true only when the quantifiers are taken to be somehow restricted. As we have seen, this is bound to happen with sentences like 'It is contingent whether some swans are blue'. Against the backdrop of this pattern of agreement and disagreement with pre-theoretical opinion, it is hard to see that counterpart theorists will incur any *further* costs if they claim, analogously, that 'It is contingent whether something is identical to Humphrey' is false when its quantifier is taken as unrestricted. So long as they agree that the sentence can be made true by interpreting its quantifier as restricted, the dialectical situation is just the familiar one.

And it is easy for counterpart theorists to come up with *restricted* readings of 'something' on which 'It is contingent whether something is identical to Humphrey' is true. For example, 'something' could carry a qualitative restriction, say to things that are *alive*. Then the sentence will be true so long as some but not all of Humphrey's counterparts are alive. Or 'something' could carry a *de re* restriction, say to things on Earth. While we have not yet considered how to analyse multiply *de re* modal claims like 'Possibly, nothing on Earth is identical to Humphrey'—this is a task for chapter 4—there is no reason at all to worry that they will turn out to be false. Or 'something' could carry a *de re* restriction to things in Cosmo. So long as our account of multiply *de re* modality can account for the truth of 'Possibly, nothing on Earth is identical to Humphrey', it will presumably be able to account in a parallel way for the truth of 'Possibly, nothing in Cosmo is identical to Humphrey.' None of these restricted readings of 'It is contingent whether Humphrey is identical to something' seems like a very plausible candidate to be what people actually have in mind when they utter sentences like 'It is contingent whether Humphrey exists'. The word 'exists' does not take so easily to contextual domain restriction as quantifiers proper: for example, I cannot felicitously say 'John does not exist' if all I mean is that John is not at my party. But I was not arguing that counterpart theorists should think of

themselves as agreeing with what the folk mean when they say that some things exist contingently; I was merely claiming that the disagreement, while real, is part of a familiar general pattern.

Lewis suggests that, in many ordinary contexts, 'counterpart' expresses a relation that human beings never bear to things that are not human, so that 'Each human being is necessarily a human being' is true. Unlike NE, this claim is not forced on us by the abstract form of the counterpart-theoretic analysis. And since 'counterpart' is certainly not one of the words we learnt at mother's knee, the right way to address the question whether human beings only human counterparts is not to directly consult our "speakers intuitions", but to consider the plausibility of the modal thesis that would follow from this claim. And *prima facie*, the claim that each human being is necessarily human looks rather implausible. Consider Prince Charles, for example. If we thought that it was necessary that he is a human being—an ordinary, flesh-and-blood human being, that is—we would have to say that he *would* have been human even if Queen Elizabeth and Prince Philip had never met (since obviously they *could* have failed to meet). That sounds implausible. Admittedly, our intuition that Charles would not have been human if Elizabeth and Philip had never met is intimately bound up with our temptation to think that there would have been nothing identical to Charles if Elizabeth and Philip had never met. So it is not clear how strongly counterpart theorists who resist that latter temptation should be swayed by the intuition. It is open to them to insist that Charles would still have been a flesh-and-blood human being even if Elizabeth and Philip had never met—perhaps a flesh-and-blood human being living on some planet that is not Earth, in some Lewis-world that is not Cosmo. But this radical claim is certainly not forced upon counterpart theorists. They could also say that Charles would under those circumstances have *failed* to be a flesh-and-blood human being, and indeed that he would have failed to be a physical thing with a mass, a shape, a location in space, etc. They could say, in other words, that Charles could have been *non-concrete*—that is, that some of his counterparts are non-concrete.

The idea that many concrete objects could have failed to be concrete plays an important role in Williamson's recent work defending NE. Williamson's idea is that ordinary claims about the possibility of things failing to exist are fairly reliable so long as we interpret 'exists' to mean something in the vicinity of 'is concrete', rather than 'is identical to something, unrestrictedly

speaking'. The exact meaning of 'concrete' is not so important: the general idea is that non-concrete things are things to which most ordinary, positive, non-modal predicates do not apply. I am inclined to think that proponents of BMCT should gratefully avail themselves of Williamson's suggestion. And as regards *temporal* counterpart theory, this seems absolutely crucial. The suggestion that if Elizabeth and Philip had never met, Charles would have been a flesh-and-blood human being living on a different planet was hard to take seriously; the analogous suggestion that *before* Elizabeth and Philip had ever met, Charles already was a flesh-and-blood human being (living on a different planet?) is just embarrassing.²⁴ But this is the sort of thing we would have to say if we claimed that it has always been the case that Charles is a human being, given that it has not always been the case that Elizabeth and Philip have met.

While "non-concrete" things are boring in many respects, they can have interesting modal and/or temporal properties. Charles has not always been a prince, but even when he was not a prince he was a *future* prince, and even when he has ceased to be a prince he will always be a *former* prince. By the lights of BTCT, this means that all Charles's past-counterparts, even the non-concrete ones, have princes among their future-counterparts, and all Charles's future-counterparts have princes among their past-counterparts. I, on the other hand, have never been a future prince, and will never be a past prince. My temporal counterparts, including the non-concrete ones, do not have princes among their temporal counterparts. Non-concrete objects can thus differ from one another as regards which objects they have as counterparts. This poses a challenge to the project of providing a non-circular analysis of temporal counterparthood in terms of something else: the relations we appeal to in the analysis must apply to non-concrete objects, and must make interesting distinctions between them. On Williamson's conception of non-concrete objects, this challenge is insuperable. On his picture, there are many distinctions among non-concrete objects that can *only* be captured in a language with temporal or modal operators, or their equivalents. However we should distinguish the general idea that before my birth and after my death I am still identical to something but lack mass, shape, a spatial location, a mental life, etc., from Williamson's own, particularly uncompromising, version of the view. What is needed to avoid absurdity is that a certain range of ordinary predicates (including 'overweight', 'in

²⁴I suppose that believes in the transmigration of souls might take this seriously. But they will not like the analogous claims about inanimate objects.

Oxford', and 'in pain', but not necessarily 'famous' or 'eagerly anticipated') did not apply to currently living beings before their conception, and will not apply to them after their death (and decomposition). This is compatible with the things in question having rich descriptions involving certain "more fundamental" predicates, upon which their temporal counterpart relations supervene.

The claim that ordinary objects have non-concrete objects among their temporal counterparts does not, thus, completely rule out any reductive analysis of temporal counterparthood in other terms, e.g. in terms of spacetime geometry. It does, however, rule out a number of otherwise attractive joint theories about the nature of ordinary objects and temporal counterparthood. Sider, for example, introduced temporal counterpart theory as part of the defence of the idea that ordinary objects are *stages*—three-dimensional spacetime regions, or objects which "occupy" such regions in such a way that no region has more than one "occupant" (Sider 1996). According to Sider, the temporal counterparts of one stage are other stages. And he suggests that the condition that two stages need to meet to be counterparts of one another is the same condition which suffices, according to worm theorists, for them to both to be temporal parts of an ordinary object of a certain kind, where the relevant kind is determined by context. In an ordinary context, then, 'temporal counterpart' expresses a relation that flesh-and-blood people bear only to other flesh-and-blood people: babies, children, and adults. It is not at all obvious how we could modify Sider's theory in such a way that the counterparts of a stage include some other entities to which ordinary "concrete" predicates do not apply. Chapter 6 will investigate some other accounts of the nature of ordinary objects which might be combined with temporal counterpart theory and a spacetime-based fundamental metaphysics.

Chapter 4

Counterpairings, worlds and instants

4.1 Multiply *de re* counterpart theory: two bad ideas

Let us now turn to the general case of counterpart theory, where modal or temporal operators are applied to formulae that may contain more than one referential term. To begin with, let's see what happens if we just mechanically extend BMCT and BTCT in the most obvious way:

$$\begin{aligned}\text{BMCT+ } \Diamond\phi(t_1, \dots, t_n) &:= \exists v_1 \dots \exists v_n (Cv_1 t_1 \wedge \dots \wedge Cv_n t_n \wedge \phi(x_1, \dots, x_n)) \\ \Box\phi(t_1, \dots, t_n) &:= \forall v_1 \dots \forall v_n ((Cv_1 t_1 \wedge \dots \wedge Cv_n t_n) \rightarrow \phi(x_1, \dots, x_n))\end{aligned}$$

$$\begin{aligned}\text{BTCT+ } F\phi(t_1, \dots, t_n) &:= \exists v_1 \dots \exists v_n (C_F v_1 t_1 \wedge \dots \wedge C_F v_n t_n \wedge \phi(x_1, \dots, x_n)) \\ G\phi(t_1, \dots, t_n) &:= \forall v_1 \dots \forall v_n ((C_F v_1 t_1 \wedge \dots \wedge C_F v_n t_n) \rightarrow \phi(x_1, \dots, x_n)) \\ P\phi(t_1, \dots, t_n) &:= \exists v_1 \dots \exists v_n (C_P v_1 t_1 \wedge \dots \wedge C_P v_n t_n \wedge \phi(x_1, \dots, x_n)) \\ H\phi(t_1, \dots, t_n) &:= \forall v_1 \dots \forall v_n ((C_P v_1 t_1 \wedge \dots \wedge C_P v_n t_n) \rightarrow \phi(x_1, \dots, x_n))\end{aligned}$$

Here $\phi(t_1, \dots, t_n)$ is any formula in which the syntactically simple referring terms (including variables) are $t_1 \dots t_n$, in order of first occurrence, and without repetition, and in which all other vocabulary is qualitative. ' $\phi(v_1, \dots, v_n)$ ' is the result of uniformly substituting for these terms variables $v_1 \dots v_n$ which do not already appear in $\phi(t_1, \dots, t_n)$.

BMCT+ and BTCT+ are hopeless. BMCT+ makes it far too easy for possibility claims to be true, and far too hard for necessity claims to be true; BTCT+ makes it far too easy for claims to the effect that something sometimes is the case to be true, and far too hard for claims to the effect that something is always the case to be true. Here are some examples that illustrate this point:

- (i) Consider Socrates' unit set, Singleton-Socrates. We would like to say (at least in ordinary contexts) that necessarily, nothing other than Socrates is a member of Singleton-Socrates. But according to BMCT+, this is necessary only if for any counterparts x_1 and x_2 of Socrates and Singleton-Socrates, respectively, nothing other than x_1 is a member of x_2 . It follows from this that if Singleton-Socrates has any nonempty counterparts, Socrates has at most one counterpart; for if he had several, each of them would have to be such that nothing other than it is a member of any of Singleton-Socrates's counterparts. And of course saying that Socrates has only one counterpart is not an option in any ordinary context, since we want to be able to say, for example, that Socrates could have died in battle and that Socrates could have died of old age. (Similarly, BTCT+ will not allow us to claim that Singleton-Socrates has never had any members other than Socrates.)
- (ii) Let Phil be the proposition that Socrates is a philosopher. By the truth schema, necessarily, Phil is true iff Socrates is a philosopher. But according to BMCT+, this is necessary only if for any counterparts x_1 and x_2 of Socrates and Phil, x_2 is true iff x_1 is a philosopher. It follows from this that if Phil has any true counterparts, all of Socrates' counterparts are philosophers: if it is possible for Phil to be true, it is necessary that Socrates is a philosopher. Which is absurd: of course it is possible for Phil to be true, since it is possible for Socrates to be a philosopher; but it is not necessary (by ordinary contextual standards) that Socrates is a philosopher. (Similarly, BTCT+ will not allow us to claim that it has always been the case that Phil is true iff Socrates is a philosopher.)
- (iii) Suppose we have a painted wooden statue, Athena. Some paint is part of Athena. (One does not need to deduct the weight of the paint from the reading on the scale when one is asked to determine Athena's weight.) But there is an object, Athena-Minus, that is made entirely of wood and overlaps all the wood in Athena throughout her life. It seems plausible that 'Athena-Minus could not possibly have been disjoint from Athena' is true, or at least that we can get it to be true by choosing an appropriate referent for 'Athena-Minus'. But according to BMCT+, this claim is false so long as *any* counterpart of Athena-Minus is disjoint from *at least one* counterpart of Athena, which is almost certainly true.

- (iv) Similarly, BTCT+ will entail that I have not always lived on Earth, so long as some past-counterpart of mine fails to live on some past-counterpart of Earth. And this is surely the case. Since I used to be a child and used to be a student; Earth used to be populated by dinosaurs and used to be populated by wooly mammoths, both Earth and I have multiple past-counterparts. And it is false that each of my past-counterparts lives on *all* of Earth's past-counterparts, since I have never been an inhabitant of many planets, and many of Earth's past-counterparts are planets.
- (v) Consider the claim that there is exactly one person who could have been Prince Charles's mother. It seems plausible that this is true in many contexts, even when the quantifier 'exactly one' is taken as completely unrestricted. But according to BMCT+, this is equivalent to the claim that there is exactly one person who has a counterpart that is a mother of a counterpart of Prince Charles. And this not plausible at all: in general, the mother of any counterpart of Prince Charles will be the counterpart of many different people (in other Lewis-worlds).
- (vi) (v) has a temporal analogue which is even more compelling. It is obviously true that there is only one person, unrestrictedly speaking, who ever *has* been a mother of Prince Charles. But according to BTCT+, everyone who has the mother of a past-counterpart of Charles among their past-counterparts is someone who has been a mother of Charles. And on plausible assumptions, there are infinitely many such people: Elizabeth herself, as well as all (or all but one) of her temporal counterparts who are mothers. Similarly, BTCT+ entails that everyone who has a past-counterpart who lives in some past-counterpart of North America is someone who has lived in North America; again, there are probably infinitely many such people.

Sider, the originator of temporal counterpart theory, notices this problem, and takes it to be so serious that he retreats to a (to my mind monstrous) hybrid view, on which 'person' has one use on which it applies to "stages" and a different use on which it applies to "worms", with temporal predications and operators presumably turning out ambiguous in a corresponding way. I think a much better response to the problem is to give up BTCT+ in favour of some other analysis of multiply *de re* temporal claims. Of course, if we are going to take temporal counterpart theory seriously, we will have to get used to the claim that

there are infinitely many people, unrestrictedly speaking. While this claim is counterintuitive, it is an inevitable commitment of any view that tries to reconcile ?? with a view that avoids the primordial objection to the B-theory. The claim that there are infinitely many people is *much* easier to believe than the claim that there are infinitely many people who have lived in North America, let alone the claim that there are infinitely many people who have been mothers of Prince Charles.

Lewis's counterpart theory avoids some of the problems that beset BMCT+, by requiring all of the sequences of counterparts witnessing the truth of a given possibility-claim to belong to the same (Lewis-)world. While chapter 3 argued that in the case of singly *de re* claims, we do better to stick with BMCT rather than giving any distinctive role in the analysis to the concept of a Lewis-world, perhaps quantification over Lewis-worlds can come into its own in the case of multiply *de re* claims, as a way of co-ordinating the assignments of counterparts. The suggestion is that it is possible that Athena and Athena-Minus are disjoint iff Athena and Athena-Minus have disjoint counterparts *which are in the same Lewis-world*. In general:

$$\begin{aligned}
 \text{RCT+} \quad \diamond \phi(t_1, \dots, t_n) &:= \exists w v_1 \dots v_n (Aw \wedge Iv_1 w \wedge \dots \wedge Iv_n w \\
 &\quad \wedge Cv_1 t_1 \wedge \dots \wedge Cv_n t_n \wedge \phi(v_1, \dots, v_n)) \\
 \square \phi(t_1, \dots, t_n) &:= \forall w v_1 \dots v_n ((Aw \wedge Iv_1 w \wedge \dots \wedge Iv_n w \\
 &\quad \wedge Cv_1 t_1 \wedge \dots \wedge Cv_n t_n) \rightarrow \phi(v_1, \dots, v_n))
 \end{aligned}$$

This is close to the analysis in Lewis 1968, except that Lewis would also have us add a restriction 'in *w*' to each quantifier in $\phi(v_1, \dots, v_n)$. As we saw in chapter 3, this is a bad idea if the quantifiers are supposed to be unrestricted (or qualitatively restricted), while if the quantifiers carry a *de re* restriction, we must make this explicit before applying any counterpart-theoretic analysis.

RCT+ makes it easier for multiply *de re* necessity claims to be true, and harder for multiply *de re* possibility claims to be true. Just as 'Athena and Athena-Minus could not have been disjoint' gets to be true so long as no counterpart of Athena is disjoint from any counterpart of Athena-minus in the same Lewis-world, 'Necessarily, nothing other than Socrates is a member of Singleton-Socrates' gets to be true provided that each nonempty counterpart of Singleton-Socrates which is in a Lewis-world has as its only member the unique counterpart of Socrates in that Lewis-world. Similarly, 'Necessarily, Phil is true iff Socrates is a philosopher' is true so long as each

Lewis-world is such that either all of Socrates' counterparts in it are philosophers and all of Phil's counterparts in it are true, or none of Socrates' counterparts in it are philosophers and none of Phil's counterparts in it are true. (To get non-crazy results from RCT+, we had better find a way of understanding 'in' that allows sets and propositions to be 'in' Lewis-worlds.)

However—as Hazen (1979) argued, and as Lewis came to agree—RCT+ still makes it too hard for multiply *de re* necessity claims like these to be true. The problem is that according to Lewis, it sometimes happens that an object has multiple counterparts in the same Lewis-world. If any Lewis-world contains two counterparts of Socrates, it cannot be necessary that nothing other than Socrates is a member of Singleton-Socrates. If any Lewis-world contains two counterparts of Athena, then it cannot be necessary that Athena and Athena-Minus are not disjoint (unless it somehow manages to be the case that every counterpart of Athena-Minus in these worlds overlaps *all* of the counterparts of Athena in it, which is wildly implausible).

Why not avoid these problems by saying that no object has more than one counterpart in any Lewis-world? In Lewis 1968, Lewis's reason for rejecting this suggestion arises from a certain analysis of counterparthood: '*x* is a counterpart of *y*' is supposed to mean '*x* is similar enough (in relevant respects) to *y*, and at least as similar to *y* as anything else in its Lewis-world'. If this were the only reason, we could reasonably brush the problem aside by saying 'So much the worse for that account of counterparthood!'. After all, 'counterpart' is already going to have to be highly context-sensitive; if its context-sensitivity is to match the context-sensitivity of 'necessary' and 'possible', we should not come to the table with too many preconceptions about the forms admissible interpretations of 'counterpart' can take. However, there are deeper reasons why Lewis is not in a position to stipulate that nothing has more than one counterpart in any Lewis-world. According to Lewis, some Lewis-worlds are qualitatively symmetric Lewis-worlds—worlds of two-way eternal recurrence, for example. Moreover, some of these Lewis-worlds are "strongly isolated", in the sense that their qualitative symmetries not only preserve all qualitative properties, but also preserve all qualitative relations to things in other Lewis-worlds.¹ Consider some such Lewis-world *w*. For any qualitative relation *R*, and object *x* not in *w*, if *x* bears *R* to any part *y* of *w*, *x* also bears *R* to all the other qualitative duplicates of *y* in *w*. But 'counterpart' is supposed to express a qualitative relation, at

¹For Lewis, a sufficient condition for a world to be strongly isolated is for none of its parts to bear any *perfectly natural* relation to anything that is not one of its parts.

least in many ordinary contexts: the facts about what qualitative properties a given object *could* have are fixed by the facts about what qualitative properties it *does* have.² So if an object has *any* counterpart in a strongly isolated, symmetric Lewis-world other than its own, it has many counterparts in that Lewis-world. The only way we could avoid claiming that some objects have multiple counterparts within the same Lewis-world, then, would be to say that objects in symmetric, strongly isolated Lewis-worlds are never counterparts of objects in any other Lewis-world. And that claim would have the implausible modal consequences that it is never *possible* for an object to be part of a symmetric, strongly isolated Lewis-world with certain qualitative features unless that object is *in fact* part of a symmetric, strongly isolated Lewis-world with those qualitative features.³

This reasoning is, however, quite specific to Lewis's system. It does not arise in temporal counterpart theory: since the parts of spacetime ('Slices') that might play the role of Lewis-worlds in a temporal analogue of RCT+ are linked by a rich array of qualitative relations, there is no obvious reason why we could not impose a one-counterpart-per-Slice constraint if we wanted to.⁴ And it might not arise in certain heterodox versions of modal counterpart theory, e.g. versions which replace Lewis's ontology with some other expansive vision of reality such as might be suggested by certain views in cosmology or in the foundations of quantum mechanics, or versions which replace Lewis's mereology-driven account of the structure of ordinary ob-

²Actually, it wouldn't matter much if we considered non-qualitative counterpart relations, such as those that would be required to deal with an interpretation of 'possible' as 'consistent with Bill Clinton's knowledge'. For all of the objects which might plausibly figure as *de re* elements of the analysis of possibility claims made by *us* are spatiotemporally related to us. Thus qualitative relations between them, us, and objects in other strongly isolated Lewis-worlds are still invariant under the symmetries of those other Lewis-worlds.

³Biting the bullet here might be less implausible for someone who thought of counterparthood in general as involving some external relation of the sort that cannot hold between strongly isolated objects. This view makes it harder for ordinary possibility claims to be true: for an object to be possibly *F*, there needs to be a chain of perfectly natural relations connecting it to something that is *F*. Lewis's doctrine of Humean Supervenience, by contrast, commits him to thinking that ordinary modal judgments are consistent with the hypothesis that the only perfectly natural relations instantiated in our Lewis-world are spatiotemporal in character. This hypothesis entails that our Lewis-world is strongly isolated.

⁴Sider (1996) claims that the one-counterpart-per-Slice constraint fails in cases of fission and fusion, and argues that this gives the temporal counterpart theorist a more attractive account of such cases than anything available to "worm" theorists. Section 6.2 below will consider whether it is a good idea for temporal counterpart theorists to accept Sider's claims about these cases.

jects with some more finely articulated picture.

However, even if we were to endorse the claim that objects never have multiple counterparts in the same Lewis-world (or Slice), some of the problems for BMCT+ remain as problems for RCT+. Just like BMCT+, RCT+ entails that each person (in whatever Lewis-world) who has a counterpart that is the mother of a counterpart of Prince Charles is someone who could have been Prince Charles's mother. Given that the relation of motherhood only holds within a Lewis-world, the move from BMCT+ to RCT+ makes no difference in this case. Similarly, in the temporal case, the analogue of RCT+ using Slices will share BTCT+'s implausible consequences that there are infinitely many people who have been mothers to Prince Charles, infinitely many people who have lived in North America, etc.

(Note that these last objections, which arise equally for RCT+ and BMCT+, essentially involve combining modal and temporal operators with unrestricted quantifiers. Those who embrace the "two languages" picture from Lewis 1968 might feel willing, on these grounds, to dismiss the claims in question as involving an illegitimate attempt to mix the two languages. But I have already explained, in section 3.3, why I don't think that such restrictions on intelligibility are tenable.)

Thus, RCT+ does at best a very partial job of overcoming the objections to BMCT+. Meanwhile, RCT+ introduces two very serious new problems which were not problems for BMCT+. First, so long as some Lewis-worlds contain no counterparts at all of certain objects, RCT+ will be unable to vindicate certain obviously valid principles of modal logic. Consider for example the following modal schemas:

$$(1) \quad \Diamond Fa \rightarrow \Diamond (Fa \vee Fb)$$

$$(2) \quad \Box (Fa \wedge Fb) \rightarrow \Box Fa$$

These seem as manifestly valid as anything could in the realm of modal logic. They are especially trivial instances of the closure of possibility and necessity under logical consequence. But look at the analysis of (1) given by RCT+:

If some Lewis-world contains a counterpart of a that is F , then some Lewis-world contains both a counterpart of a and a counterpart of b , such that either the former or the latter is F .

This is false if a has counterparts that are F , but all of them are in Lewis-worlds that contain no counterparts of b . Thus, given RCT+, guaranteeing

the truth of all instances of (1) would require claiming that every object has at least one counterpart in every Lewis-world. (The same is easily seen to be true for (2).)

In standard modal logic, claims like (1) and (2) would be derived from the fundamental axiom of modal logic, K:

$$(K) \quad \Box(\phi \rightarrow \psi) \rightarrow (\Box\phi \rightarrow \Box\psi)$$

together with the ‘rule of necessitation’ (which says that $\Box\phi$ is a logical truth whenever ϕ is). Unsurprisingly, then, the cases where (1) or (2) fails (given RCT+) are also cases where instances of K fail, for example

$$\Box((Fa \wedge Fb) \rightarrow Fa) \rightarrow (\Box(Fa \wedge Fb) \rightarrow \Box Fa)$$

The antecedent is a consequence of RCT+; the consequent is (2).⁵

The failure of K is not simply a random bug in Lewis’s analysis: it plays an important role in his defence of claims like ‘Necessarily, Prince Charles is human’. K plays a crucial role in an argument against this claim, as follows:

- (i) Necessarily, if Charles is human, then Elizabeth and Philip meet (premise).
- (ii) Necessarily, Charles is human (assumption for *reductio*).
- (iii) Necessarily, Elizabeth and Philip meet (by K, from (i) and (ii))
- (iv) Possibly, Elizabeth and Philip never meet (premise).
- (v) It is not necessary that Charles is human (*reductio* from (ii)–(iv)).

By rejecting K, one can accept the premises of this argument while rejecting its conclusion. However, I do not think that this is a compelling reason to reject K. While it is plausible that there are contexts relative to which (ii) is true, it is plausible that these are either contexts in which (iii) is true, or contexts in which (i) is false. Reasoning by K, and by the general principle that necessity and possibility are closed under logical consequence, is absolutely basic to the use of modal operators; if a particular account of what it takes for something to be a counterpart of something else makes this hard to sustain, then so much the worse for it.

⁵As far as I know, the failure of Lewis’s counterpart theory to validate K was first noted by Kripke in a letter to Lewis (Kripke 1969).

This problem with RCT+ could be avoided by requiring that every object have a counterpart in every Lewis-world. If the things ‘in’ a given Lewis-world are just the *parts* of that Lewis-world, this requirement would generate wildly implausible results. For example, consider a tiny Lewis-world w with no proper parts: w is in w , but nothing else is. If the parts of each Lewis-world had to include a counterpart for absolutely everything, w would have to be a counterpart of everything, making it true that everything could have been a spatiotemporally isolated mereological atom. This is surely not true in every legitimate context! And the broadenings of the ‘in’ relation which one might consider if, for example, one wanted to be able to think of *sets* as ‘in’ Lewis-worlds do not suggest any plausible way of getting it to be true that each object has a counterpart ‘in’ w .

The second of RCT+’s distinctive problems is more straightforward, and involves the T axiom. It is a straightforward consequence of RCT+ that for any two objects x and y , it is necessary that there is a Lewis-world containing both x and y . But if there are any two objects that are *not* in the same Lewis-world—and there had better be, otherwise RCT+ collapses into BMCT+—this is a violation of T. The objects in question are not worldmates, although it is necessary that they are. This is clearly unacceptable.

4.2 Three better ideas

I conclude that dragging Lewis-worlds (or something that plays the same role) into the analyses of multiply *de re* modal and temporal claims is not the right strategy for the counterpart theorist. Fortunately, there are better options. One was developed by Lewis himself, in the 1983 postscripts to Lewis 1968 and in section 4.4 of Lewis 1986b. According to this proposal, the analysis of a multiply *de re* modal claim involves a relation of counterpart-hood holding between *sequences* (n -tuples) of objects.⁶ Here is a statement of this analysis, incorporating the insight that there is no need to include any

⁶In postscript D to the reprint of Lewis 1968, Lewis presents the “counterparts of sequences” account not as a modification of the original translation from the formal modal language to the language of counterpart theory, but as an alternative to the obvious translation from English into the formal modal language. His idea is that the English sentence ‘It is possible that Dee and Dum are unrelated’ should be rendered formally as something like ‘ $\exists z(z = \langle \text{Dee}, \text{Dum} \rangle) \wedge \Diamond \exists x \exists y (z = \langle x, y \rangle \wedge \text{Unrelated}(x, y))$ ’ rather than ‘ $\Diamond \text{Unrelated}(\text{Dee}, \text{Dum})$ ’. Given that the ultimate point is to come up with analyses that can be stated in English, I take it nothing turns on this distinction.

special-purpose machinery for restricting quantifiers:

SEQUENCES

$$\begin{aligned}\Diamond\phi(t_1, \dots, t_n) &:= \\ &\quad \exists v_1 \dots \exists v_n (C\langle v_1, \dots, v_n \rangle \langle t_1, \dots, t_n \rangle \wedge \phi(v_1, \dots, v_n)) \\ \Box\phi(t_1, \dots, t_n) &:= \\ &\quad \forall v_1 \dots \forall v_n (C\langle v_1, \dots, v_n \rangle \langle t_1, \dots, t_n \rangle \rightarrow \phi(v_1, \dots, v_n))\end{aligned}$$

SEQUENCES (or at least, the necessitated biconditional corresponding to SEQUENCES) actually follows from BMCT, together with the principle that that any n things form a sequence:

$$(3) \quad \Box \forall x_1 \dots x_n \exists y (y = \langle x_1, \dots, x_n \rangle)$$

along with the following plausible principle of sequence-theoretic essentialism:

$$(4) \quad \forall x_1 \dots x_n y (y = \langle x_1, \dots, x_n \rangle \rightarrow \Box y = \langle x_1, \dots, x_n \rangle).$$

It follows from these principles that any multiply *de re* sentence $\phi(t_1, \dots, t_n)$ is necessarily equivalent to a singly *de re* sentence about the sequence $s = \langle t_1, \dots, t_n \rangle$, namely $\exists y_1 \dots y_n (s = \langle y_1, \dots, y_n \rangle \wedge \phi(y_1, \dots, y_n))$. (Deriving this also requires NE, the principle that everything is necessarily identical to something; but as we have already seen in section 3.6, this is a consequence of BMCT.) So we can derive SEQUENCES as follows:

$$\begin{aligned}\Diamond\phi(t_1, \dots, t_n) &\leftrightarrow \Diamond \exists y_1 \dots y_n (s = \langle y_1, \dots, y_n \rangle \wedge \phi(y_1, \dots, y_n)) \\ &\leftrightarrow \exists z (Cs z \wedge \exists y_1 \dots y_n (z = \langle y_1, \dots, y_n \rangle \wedge \phi(y_1, \dots, y_n))) \\ &\leftrightarrow \exists y_1, \dots, y_n (C\langle t_1, \dots, t_n \rangle \langle y_1, \dots, y_1 \rangle \wedge \phi(y_1, \dots, y_n))\end{aligned}$$

However, it is worth noting that SEQUENCES does not *require* any doctrine of sequence-theoretic essentialism. SEQUENCES even leaves us free to hold that the relation between an object x and its corresponding length-one sequence $\langle x \rangle$ is a contingent one: $\langle x, \langle x \rangle \rangle$ could have a counterpart $\langle y, z \rangle$ such that $z \neq \langle y \rangle$. If we do think this, then there will be a superficial conflict between SEQUENCES and BMCT: for a singly *de re* claim $\Diamond\phi(t)$, SEQUENCES yields the analysis $\exists y (C\langle y \rangle \langle t \rangle \wedge \phi(y))$ while BMCT yields the analysis $\exists y (Cyt \wedge \phi(y))$. But of course, SEQUENCES entails that BMCT is true on *some* interpretation of ‘counterpart’, namely one on which ‘ x is a counterpart of y ’ means ‘ $C\langle x \rangle \langle y \rangle$ ’, where C is the predicate of sequences used in SEQUENCES.

SEQUENCES is strictly more flexible than BMCT+ and RCT+. If we wanted to recover the effect of BMCT+ or RCT+ in some context, we could do so by analysing the counterpart relation on sequences in terms of a prior counterpart relation on objects, like this:

$$(5) \quad C\langle x_1, \dots, x_n \rangle \langle y_1, \dots, y_n \rangle := Cx_1y_1 \wedge \dots \wedge Cx_ny_n$$

or like this

$$(6) \quad C\langle x_1, \dots, x_n \rangle \langle y_1, \dots, y_n \rangle := Cx_1y_1 \wedge \dots \wedge Cx_ny_n, \text{ and } x_1 \dots x_n \text{ are all in the same accessible Lewis-world.}$$

But we have already seen why we should not expect to be able to do this in any ordinary context.⁷

By adopting appropriate postulates about the counterpart relation on sequences, SEQUENCES (and its temporal analogue) allows us to avoid all the problems that beset BMCT+:

- (i) Provided that every counterpart of $\langle \text{Socrates, Singleton-Socrates} \rangle$ is of the form $\langle x, \{x\} \rangle$, it is necessarily true (and always true) that Singleton-Socrates has no members other than Socrates.
- (ii) Provided that every counterpart of $\langle \text{Socrates, Phil} \rangle$ is of the form $\langle x, \text{the proposition that } x \text{ is a philosopher} \rangle$, it is necessarily true (and always true) that Phil is true iff Socrates is a philosopher.
- (iii) Provided that $\langle \text{Athena, Athena-Minus} \rangle$ has no counterpart that is an ordered pair of two disjoint items, it is impossible for Athena and Athena-Minus to be disjoint.
- (iv) Provided that $\langle \text{me, Earth} \rangle$ has no past-counterpart whose first element fails to live on its second element, it has always been the case that I have lived on Earth.

⁷An alternative idea, due to Graeme Forbes (1985), is, roughly, to adopt SEQUENCES while analysing ' $C\langle x_1, \dots, x_n \rangle \langle t_1, \dots, t_n \rangle$ ' as 'for some world w , $\bigwedge_i$ (either x_i is in w and Cx_it_i , or t_i has no counterpart in w and $x_i = t_i$)'. This analysis has the advantage of making (1) and (2) valid. However, it has implausible consequences as it stands, e.g., that if Gordon Brown is necessarily not an atomic Lewis-world, then for every x , necessarily, if x is an atomic Lewis-world, then Gordon Brown is Prime Minister. Forbes's remedy for this involves a much more radical departure from modal realist counterpart theory, in which atomic predicates get endowed with extra argument places that in effect turn them into relations to worlds.

- (v) Provided that there is no x other than Elizabeth such that $\langle x, \text{Charles} \rangle$ has a counterpart whose first element is the mother of its second element, Elizabeth is the only thing which could have been a mother of Charles. The mere fact that some object x has Elizabeth as a counterpart gives us no special reason to think that $x, \text{Charles}$ has Elizabeth, Charles, or any other ordered pair whose first element is the mother of its second element, among its counterparts. We might plausibly suppose that typically, when the elements of some ordered pair are in different Lewis-worlds, the elements of the counterparts of that pair are also in different Lewis-worlds.
- (vi) Similarly, provided that there is no x other than Elizabeth such that $\langle x, \text{Charles} \rangle$ has a past-counterpart whose first element is the mother of its second element, Elizabeth is the only thing which has ever been a mother of Charles. And provided that there are only finitely many people x such that $\langle x, \text{North America} \rangle$ has a past-counterpart whose first element lives in its second element, there are only finitely many people that have ever lived in North America. If we have some notion of an object being 'in' a Slice, such that the temporal counterparts of an objects are generally 'in' different Slices, then we might plausibly suppose that ordered pairs whose elements are 'in' different Slices only have other such ordered pairs as past- and future-counterparts.⁸

Nor is SEQUENCES subject to any of the distinctive problems which beset RCT+. True, it does not validate the modal axioms K and T all by itself. However, there are further constraints which we could impose on the counterpart relation for sequences which would have the effect of validating those axioms without having any further untoward effects. Validating T is straightforward: we need only require that every sequence has itself as a counterpart (even if its elements belong to different Lewis-worlds). It is not quite so easy to write down constraints which have the effect of validating K, although as we will see in section 4.3, it can be done. In particular cases, it is easy to see what is needed to guarantee the closure of possibility and necessity under some logically valid inference. For example, to validate the inference from 'Possibly Fa and Gb ' to 'Possibly Gb and Fa ', we can say that whenever $\langle x_1, x_2 \rangle$ is a counterpart of $\langle y_1, y_2 \rangle$, $\langle x_2, x_1 \rangle$ is a counterpart of $\langle y_2, y_1 \rangle$. Similarly, to validate the inference from 'Possibly Fa and Gb ' to

⁸Perhaps an exception should be made in worlds with time-travel: see section 6.4.

'Possibly Fa' , we can say that whenever $\langle x_1, x_2 \rangle$ is a counterpart of $\langle y_1, y_2 \rangle$, $\langle x_1 \rangle$ is a counterpart of $\langle y_1 \rangle$. These cases illustrate the following plausible more general constraint: when we have a pair of counterpart sequences, then any sequence of elements of the first has the corresponding sequence of elements of the second as a counterpart. To state this idea rigorously, it will be convenient to identify sequences of length n with functions whose domain is the first n natural numbers. Then we can put the claim like this:

Permutation: If π is a function from the natural numbers $< m$ to the natural numbers $< n$, and σ_1 and σ_2 are sequences of length n such that σ_2 is a counterpart of σ_1 , then $\sigma_2 \circ \pi$ is a counterpart of $\sigma_1 \circ \pi$.

This suggests that in taking 'counterpart' to apply to *pairs of sequences*, we are importing superfluous structure. The order of the elements of the sequences is irrelevant except insofar as it sets up a correspondence between the members of the first sequence and the members of the second sequence. Thus the representational work of the pair of sequences $\langle \langle a_1, \dots, a_n \rangle, \langle b_1, \dots, b_n \rangle \rangle$ could be done more perspicuously by the corresponding set of ordered pairs: $\{\langle a_1, b_1 \rangle, \dots, \langle a_n, b_n \rangle\}$. If we modify SEQUENCES in accordance with this idea, we will want to replace the two-place predicate of sequences C with a one-place predicate ' C ', pronounced 'is a counterpairing', and applying to sets of ordered pairs.

Insofar as it relies on quantification over *sets*, this proposal is still not ideal: we can achieve some desirable generality by eliminating quantification over sets in favour of plural quantification. (Following Boolos (1985), many philosophers, conspicuously including Lewis, hold that such quantification is not to be understood in terms of first-order quantification over sets or anything else.) Thus officially, 'counterpairing' will be a *plural* predicate, applying to ordered pairs taken collectively.⁹ It will be convenient to state the analysis in the notation of second-order logic, using ' Xx ' to abbreviate ' x

⁹Having said this, I will sometimes allow myself to be sloppy, saying things like 'counterpairings are relations'.

is one of X' , and ' Rxy ' to abbreviate ' $\langle x, y \rangle$ is one of R' '. So we have:

COUNTERPAIRINGS

$$\Diamond\phi(t_1, \dots, t_n) :=$$

$$\exists R \exists v_1 \dots \exists v_n (\mathbf{CR} \wedge R t_1 v_1 \wedge \dots \wedge R t_n v_n \wedge \phi(v_1, \dots, v_n))$$

$$\Box\phi(t_1, \dots, t_n) :=$$

$$\forall R \forall v_1 \dots \forall v_n ((\mathbf{CR} \wedge R t_1 v_1 \wedge \dots \wedge R t_n v_n) \rightarrow \phi(v_1, \dots, v_n))$$

If we prefer to work with SEQUENCES, we can recover it from COUNTERPAIRINGS by adopting the following analysis:

$$(7) \quad C\langle x_1, \dots, x_n \rangle \langle y_1, \dots, y_n \rangle := \exists R (\mathbf{CR} \wedge R x_1 y_1 \wedge \dots \wedge R x_n y_n)$$

And for the special case of singly *de re* sentences, we can of course recover BMCT by analysing ' x is a counterpart of y ' as ' $\exists R (\mathbf{CR} \wedge Rxy)$ '.

Like SEQUENCES, COUNTERPAIRINGS needs to be supplemented with further axioms if we want to recover a reasonable modal logic. Section 4.3 will consider in detail how such axioms about 'counterpairing' map on to schemas involving modal operators.

From my point of view, COUNTERPAIRINGS has one big advantage over SEQUENCES, namely the fact that it can very easily be extended to cover sentences in which *plural* terms and free variables occur in the scope of modal operators. All we need to do is to define ' RXY ' to abbreviate the following conjunction: (i) for any x that is one of X , there is some y that is one of Y such that Rxy and (ii) for any y that is one of Y , there is some x that is one of X such that Rxy . We can then use COUNTERPAIRINGS as it stands to analyse modal sentences involving plural terms: we simply allow the terms $t_1, \dots, t_n$ to be plural as well as singular, understanding that when t_i is plural, its corresponding variable v_i will also be plural.

I have focused on the modal case, but of course COUNTERPAIRINGS has a natural temporal analogue, in which 'counterpairing' will be replaced with 'future-counterpairing' or 'past-counterpairing' as appropriate.

It is worth mentioning a third proposal for dealing with multiply *de re*, which works only for *temporal* counterpart theory. Prior (1967) introduced *metrical* tense operators F_n , pronounced 'it will be the case n units of time hence that...' or when n is negative, 'it was the case $-n$ units of time hence that...'. If we can make sense of these, we can plausibly analyse the usual tense operators as the result of quantifying into ' n ': $F\phi$ and $P\phi$ are

respectively equivalent to $\exists n(n > 0 \wedge Fn\phi)$ and $\exists n(n < 0 \wedge Fn\phi)$. A temporal counterpart theorist will naturally want to analyse the metric tense operator in terms of a corresponding metric notion of temporal counterparthood, which lets us say that x is an ' n -unit temporal counterpart' of y ; for short, $Cnxy$. For the case of the singly *de re*, there are two obvious ways of implementing this, depending on whether we use universal or existential quantification:

$$\begin{aligned} Fn\phi(t) &:= \exists x(Cnxt \wedge \phi(x)) \\ Fn\phi(t) &:= \forall x(Cnxt \rightarrow \phi(x)) \end{aligned}$$

Which of these should we choose? There would be no need to choose we adopted the following 'uniqueness postulate': for every number n and every object x , necessarily, x has *exactly one* n -unit temporal counterpart. And the uniqueness postulate could be justified on the grounds that it accounts for the validity of the following schemas for Fn :

$$\begin{aligned} Fn\phi \wedge Fn\psi &\rightarrow Fn(\phi \wedge \psi) \\ Fn(\phi \vee \psi) &\rightarrow Fn\phi \vee Fn\psi \\ Fn\neg\phi &\rightarrow \neg Fn\phi \end{aligned}$$

If we do adopt the uniqueness postulate, then it is a straightforward matter to extend our analysis to the case of the multiply *de re*, as follows:

$$Fn\phi(t_1, \dots, t_m) := \exists v_1 \dots \exists v_m(Cnv_1t_1 \wedge \dots \wedge Cnv_mt_m \wedge \phi(v_1 \dots v_m))$$

or (equivalently, given the uniqueness postulate):

$$Fn\phi(t_1, \dots, t_m) := \forall v_1 \dots \forall v_m((Cnv_1t_1 \wedge \dots \wedge Cnv_mt_m) \rightarrow \phi(v_1 \dots v_m)).$$

The objections to BMCT+ all involves cases where objects have multiple counterparts, or where the same object is a counterpart of multiple items. Given the uniqueness postulate and the plausible further assumption that $Cnxy$ is equivalent to $C\neg nyx$, this never happens for the ' n -unit temporal counterpart' relation. So we can analyse Fn as above, and then follow Prior's analysis of the ordinary tense operators in terms of Fn .

This analysis need not be thought of as a competitor to COUNTERPAIRINGS. If we analyse ' R is a counterpairing' as $\exists n\forall x\forall y(Rxy \leftrightarrow Cnxy)$, we

can recover COUNTERPAIRINGS *modulo* the uniqueness postulate. However, COUNTERPAIRINGS is less committal than the approach using tense operators. First, it does not require the uniqueness postulate; second, it does not require us to make sense of the question how many units of time apart a given pair of temporal counterparts are—a question which may or may not raise special foundational difficulties, depending on how we think of the geometric structure of spacetime.

4.3 Constraints on counterpairings

Since modal operators are context-dependent, the answer to the question what it is for some ordered pairs to be a counterpairing must vary from one context to another. But it is plausible that certain schematic principles involving modal operators are valid across all contexts. These will correspond to constraints which all legitimate readings of ‘counterpairing’ must meet. Similarly, while there may be some context-sensitivity in temporal operators (see chapter 6), some schematic principles involving them are valid, and these will correspond to constraints which all legitimate readings of ‘future-counterpairing’ and ‘past-counterpairing’ must meet. Let us consider the modal case first.

The most central axiom of standard modal logic is *K*:

$$(K) \quad \Box(\phi \rightarrow \psi) \rightarrow (\Box\phi \rightarrow \Box\psi).$$

This does not follow from COUNTERPAIRINGS by itself. For example, ‘ $\Box((Fa \wedge b = b) \rightarrow Fa) \rightarrow (\Box(Fa \wedge b = b) \rightarrow \Box Fa)$ ’ is false if some counterpairings map *a* to objects that are not *F*, but none of these counterpairings have *b* in their domain.

The most straightforward way to guarantee the truth of every instance of (K) is to lay it down as an axiom, true across all contexts, that *counterpairings have universal domains*. That is: whenever *R* is a counterpairing, $\forall x \exists y Rxy$. It is easy to see that this validates (K).

We don’t, strictly speaking, need something as strong as this to validate (K). It would be enough if we stipulated that every counterpairing can be *extended* to a counterpairing with a universal domain:

$$(8) \quad CR \rightarrow \exists S(CS \wedge \forall x \forall y (Rxy \rightarrow Sxy) \wedge \forall x \exists y Sxy).$$

But if (K) is valid, we lose nothing by restricting our attention to those coun-

terpairings that do have universal domains: if any counterpairing witnesses the truth of ' $\Diamond\phi(t_1, \dots, t_n)$ ', some counterpairing whose domain is universal does. The reasoning is simple. Let U be *all the things there are*—the things such that everything is one of them. By the plural version of COUNTERPAIRINGS, it is (trivially) necessary that if $\phi(t_1, \dots, t_n)$ then $\phi(t_1, \dots, t_n) \wedge \forall x(Ux \rightarrow Ux)$. Suppose that it is possible that $\phi(t_1, \dots, t_n)$. Then by (K), it is possible that $\phi(t_1, \dots, t_n) \wedge \forall x(Ux \rightarrow Ux)$. By COUNTERPAIRINGS, this is true iff there is a counterpairing R and objects $v_1, \dots, v_n, X$ such that $Rt_1v_1 \dots$ and Rt_nv_n and RUX and $\phi(v_1, \dots, v_m)$ and $\forall x(Xx \rightarrow Xx)$. But for RUX to be true, R must have a universal domain. So if it is possible that $\phi(t_1, \dots, t_n)$, there must be a counterpairing R with universal domain, and objects $v_1, \dots, v_n$ such that $Rt_1v_1 \dots$ and Rt_nv_n and $\phi(v_1, \dots, v_n)$. Thus, even if we started with a sense of 'counterpairing' that applied to some relations that don't have universal domains, we will end up with equivalent analyses if we redefine 'counterpart' so that it only applies to relations with universal domains.

We can imagine *other* analyses of modal operators in terms of counterpairings on which the existence of counterpairings with non-universal domains makes a real difference. For example, there might be an analysis under which 'I could have failed to be identical to anything' is equivalent to 'there is a counterpairing whose domain does not include me'. But as we saw in section 3.6, there are formidable challenges facing reductionists about modality who want to recover an *intuitive* account of the possibility of nonexistence. While the idea of using counterpairings whose domain excludes x to represent possibilities of x 's nonexistence is initially attractive, it is very hard to see how such an analysis could allow us to endorse claims like 'Possibly, I am not identical to anything but still could have been a philosopher', or how its temporal analogue could allow us endorse claims like 'It will be the case that (I am not identical to anything but still used to be a philosopher)'. Constructing a theory consistent with such claims would require a rather radical departure from COUNTERPAIRINGS.

Another axiom that is valid in all standard quantified modal logics is Leibniz's Law:

$$(LL) \quad x = y \rightarrow (\phi \rightarrow \psi)$$

where ψ is derived from ϕ by replacing one or more occurrences of x with occurrences of y . An instance of LL is $x = y \rightarrow (\Box x = x \rightarrow \Box x = y)$. Since $\Box x = x$ is a theorem, this simplifies to the principle that identity is necessary:

$$(NI) \quad x = y \rightarrow \Box x = y.$$

Given COUNTERPAIRINGS, NI is equivalent to the claim that *counterpairings are functional*: that is, whenever R is a counterpairing and Rxy and Rxz , $y = z$. A straightforward induction on the complexity of formulas shows that this suffices for the full generality of LL in the language of quantified modal logic.

LL seems about as obviously valid as any schema could. And the obviousness persists when we consider particular instances of LL, including the instance required for the derivation of NI. If x is necessarily identical to x , and y is x , then how could y fail to be necessarily identical to x ? Nevertheless, LL has been controversial in the context of counterpart theory: Lewis rejected it in 1968, and many philosophers think of the failure of LL as one of the central interesting things about counterpart theory. In section 4.4 below I will consider Lewis's reasons for rejecting LL, and argue that they are not persuasive.

Along with LL, the requirement that counterpairings be functional also validates a plausible principle of plural modal logic, according to which when a thing is one of some things, it is necessarily one of them:

(NO) $Xx \rightarrow \Box Xx$.

Suppose x is one of X , and R is a counterpairing such that Rxy and RXY . Part of what it means to say that RXY is that for every x' that is one of X , there is some y' that is one of Y such that $Rx'y'$. But if counterpairings are functional, then y is the only thing such that Rxy ; thus y must be one of Y . In fact, the claim that NO is true for all X and x is *equivalent* to the claim that all counterpairings are functional. For if some counterpairing R maps some object x to two distinct objects y_1 and y_2 , we could take X to be the objects identical to x (following standard philosophical usage, I am allowing that "some things" are such that there is only one of them), and Y to be the objects identical to y_1 . Then RXY and Rxy_2 even though y_2 is not one of Y ; so it is not necessary that Xx . This is unsurprising, given that such counterpairings generate counterexamples to the necessity of identity: if one thought, concerning Russell, that Russell and he are identical but could have failed to be, one would also naturally think, concerning Russell and Whitehead, that Russell is one of them but could have failed to be.

If one rejected NE, one might be worried about NO even if one had no doubts about the necessity of identity. Consider Russell and Whitehead again. Would Russell still have been one of them if Whitehead had failed to be anything at all? Would Russell still have been one of them if *Russell*

had failed to be anything at all? The answers are not obvious. If we tried to tinker with counterpart theory to allow for contingent existence, we would face delicate and difficult questions about how our analysis should deal with occurrences of ‘is one of’ in modal contexts. But if we follow the advice of section 3.6 and don’t bother, I can’t think of any other reason to worry about the consequence that NO and NI stand or fall together.

The combination of the two constraints we have considered so far—that all counterpairings have universal domains, and that all counterpairings are functional—would allow us to simplify the formulation of COUNTERPAIRINGS by employing a function-symbol notation. Let us use ‘ $R(x)$ ’ to stand for ‘the object y such that Rxy ’, and likewise ‘ $R(X)$ ’ to stand for ‘the objects Y such that RXY ’. In this notation, COUNTERPAIRINGS looks like this:

$$\begin{aligned} \text{COUNTERPAIRINGS}^* \quad & \Diamond\phi(t_1, \dots, t_n) := \exists R(\mathbf{CR} \wedge \phi(R(t_1), \dots, R(t_n))) \\ & \Box\phi(t_1, \dots, t_n) := \forall R(\mathbf{CR} \rightarrow \phi(R(t_1), \dots, R(t_n))) \end{aligned}$$

However, this kind of notation creates the potential for scope ambiguity when dealing with iterated modalities. If we want COUNTERPAIRINGS* to be equivalent to COUNTERPAIRINGS, we must be careful to treat the complex terms $R(t_i)$ as having *widest possible scope*. We want ‘ $\Diamond\Diamond Fa$ ’ to be equivalent to ‘ $\exists R(\mathbf{CR} \wedge \exists S(\mathbf{CS} \wedge FS(R(a))))$ ’, not ‘ $\exists R(\mathbf{CR} \wedge \exists S(\mathbf{CS} \wedge FS(R)(S(a))))$ ’. One rigorous way to achieve this would be to follow Russell in taking sentences involving functional-symbol notation as abbreviations of quantified sentences in a language free of such notation, and stipulate that the Russellian translations are to be applied in such a way that the quantifier introduced by each complex term takes the widest possible scope.

There are many other formal schemas which have been thought to be logically valid for some or all readings of ‘necessarily’ and ‘possibly’. Many of these correspond naturally to constraints on counterpairings. Several of these schemas, including those we have already considered, are listed in Table 4.1, together with the constraints on counterpairings that correspond to them. Some technical remarks on these results:

- (i) Standard modal logics are closed under the rule of necessitation: if ϕ is a theorem, so is $\Box\phi$. If all contingency is *de re* contingency, any theory whose axioms do not themselves involve reference to particular individuals is guaranteed to have this feature. More generally, a theory

<i>Axiom</i>	<i>Constraint</i>
Duality $\Box\neg\phi \leftrightarrow \neg\Diamond\phi$	Automatic from COUNTERPAIRINGS
K $\Box(\phi \rightarrow \psi) \rightarrow (\Box\phi \rightarrow \Box\psi)$	Counterpairings have universal domains: $\mathbf{CR} \rightarrow \forall x\exists yRxy$
D $\Box\phi \rightarrow \Diamond\phi$	There is at least one counterpairing: $\exists R\mathbf{CR}$
T $\Box\phi \rightarrow \phi$	The identity relation is a counterpairing: $\forall x\forall y(Rxy \leftrightarrow x = y) \rightarrow \mathbf{CR}$
B $\phi \rightarrow \Box\Diamond\phi$	R^{-1} (the converse of R) is a counterpairing when R is: $(\mathbf{CR} \wedge \forall x\forall y(Sxy \leftrightarrow Ryx)) \rightarrow \mathbf{CS}$
4 $\Box\phi \rightarrow \Box\Box\phi$	$R \circ S$ (the relative product of R with S) is a counterpairing when R and S are: $(\mathbf{CR} \wedge \mathbf{CS} \wedge \forall x\forall y(Qxy \leftrightarrow \exists z(Sxz \wedge Rzy))) \rightarrow \mathbf{CQ}$
5 $\Diamond\phi \rightarrow \Box\Diamond\phi$	$R \circ S^{-1}$ is a counterpairing when R and S are: $(\mathbf{CR} \wedge \mathbf{CS} \wedge \forall x\forall y(Qxy \leftrightarrow \exists z(Szx \wedge Rzy))) \rightarrow \mathbf{CQ}$
NI $x = y \rightarrow \Box x = y$	Counterpairings are functional: $(\mathbf{CR} \wedge Rxy \wedge Rxz) \rightarrow y = z$
NO $Xx \rightarrow \Box Xx$	
ND $x \neq y \rightarrow \Box x \neq y$	Counterpairings are one-one: $(\mathbf{CR} \wedge Rxz \wedge Ryz) \rightarrow x = y$
NNO $\neg Xx \rightarrow \Box \neg Xx$	
CBF $\Box\forall x\phi \rightarrow \forall x\Box\phi$	Automatic from COUNTERPAIRINGS
CBF2 $\Box\forall X\phi \rightarrow \forall X\Box\phi$	
BF $\forall x\Box\phi \rightarrow \Box\forall x\phi$	Every counterpairing can be extended so that any given object is in its range: $\mathbf{CR} \rightarrow \forall x\exists S(\mathbf{CS} \wedge \forall z_1\forall z_2(Rz_1z_2 \rightarrow Sz_1z_2) \wedge \exists ySyx)$
BF2 $\forall X\Box\phi \rightarrow \Box\forall X\phi$	Counterpairings have universal ranges: $\mathbf{CR} \rightarrow \forall y\exists xRxy$

Table 4.1. Some modal axioms, together with constraints on counterpairings that generate them.

will be closed under necessitation whenever its theorems are indifferent to the identities of particular objects: if $\phi(t_1, \dots, t_n)$ is a theorem, $\forall v_1 \dots \forall v_n \phi(v_1, \dots, v_n)$ is a theorem too. Assuming that ‘counterpairing’ is itself a qualitative predicate, this means we will get necessitation for free, since the kinds of constraints we are currently considering are *general* truths about what is required for a plurality of ordered pairs to be a counterpairing.

- (ii) The axioms listed stand in the following logical relations: (a) T entails D; (b) Given K and D, 5 and T are jointly equivalent to 4 and B; (c) Modulo K and B, 4 is equivalent to 5, ND is equivalent to NI, NNO is equivalent to NO, and BF is equivalent to CBF. The same entailments holds among the corresponding constraints on counterpairings.
- (iii) Given COUNTERPAIRINGS (understood so as to apply to plural as well as singular terms), BF2 entails BF, and the same entailment holds between the corresponding constraints.
- (iv) Given COUNTERPAIRINGS, the constraints listed in the right-hand column entail every instance of the corresponding axiom-schemas in the left-hand column, with the proviso that for 4, B and 5, the entailments only hold on the assumption that ‘counterpairing’ is a qualitative predicate.¹⁰ In proving that these entailments hold, we need to rely on the fact that our definition of ‘RXY’ is well-behaved, in the following ways: if R is functional ($y = z$ whenever Rxy and Rxz), then it is also “functional on plural arguments”: if RXY and RXZ, then $\forall x(Yx \leftrightarrow Zx)$; similarly for the property of being one-to-one. If S is the converse of R ($\forall x \forall y(Sxy \leftrightarrow Ryx)$), then SXY iff RYX. And if Q is the relative product of R and S ($\forall x \forall y(Qxy \leftrightarrow \exists z(Rxz \wedge Szy))$), then QXY iff $\exists Z(RXZ \wedge SZY)$.¹¹

¹⁰Our analysis allows non-qualitative senses of ‘counterpairing’, corresponding to contexts where ‘possible’ means something like ‘historically possible as of 1900’ or ‘consistent with Clinton’s beliefs in 1995’. For these, something a bit more complicated is needed to validate 4, B or 5. Take for instance a notion of ‘counterpairing’ that takes one extra singular argument. For 4 ($\Box_x \phi \rightarrow \Box_x \Box_x \phi$) to be valid for the corresponding sense of necessity, we need it to be the case that whenever R is an x -counterpairing, S is a y -counterpairing, and Rxy , $S \circ R$ is an x -counterpairing.

¹¹I will prove the last of these facts, and show how to use it to establish the entailment for S4. *Right-to-left*: suppose that RXZ and SZY. Then $\forall x \in X \exists z \in Z(Rxz)$ and $\forall z \in Z \exists y \in Y(Szy)$, so $\forall x \in X \exists y \in Y \exists z(Rxz \wedge Szy)$. So $\forall x \in X \exists y \in Y(Qxy)$, since $Q = S \circ R$. By analogous reasoning, $\forall y \in Y \exists x \in X(Qxy)$; hence QXY. *Left-to-right*: Let Z comprise all and only those objects z such that for some x and y , $Xx \wedge Yy \wedge Rxz \wedge Szy$. If QXY, then $\forall x \in X \exists y \in Y Qxy$.

- (v) With the exception of NI/NO, ND/NN, and D, the modal axiom-schemas do not actually entail the corresponding constraints. Rather, the relation is the one we have already seen in the case of K : if we have a sense of ‘counterpairing’ under which all instances of the axiom-schema are true (given COUNTERPAIRINGS), we can use it to define a new sense of ‘counterpairing’ which satisfies the corresponding constraint; and the analyses we get when we use the new, defined sense of ‘counterpairing’ will be logically equivalent to the originals. In the case of K , BF and BF2, the new sense will *strengthen* the old sense just enough to satisfy the constraint—for BF2, for example, we can define a counterpairing in the new sense to be a counterpairing in the old sense whose range is universal. In the case of T, B, 4 and 5, the new sense will *weaken* the old sense just enough to satisfy the constraint. For 4, for example, the new sense is the closure of the old sense under the operation of taking relative products.¹² We can prove that the analyses we get from COUNTERPAIRINGS using the new sense of ‘counterpairing’ will be equivalent to analyses using the old sense of ‘counterpairing’.¹³

Similarly, given the temporal version of COUNTERPAIRINGS, we can write

So $\forall x \in X \exists y \in Y \exists z (Rxx \wedge Szy))$. By the definition of Z , every such z is one of Z . So $\forall x \in X \exists z \in Z (Rxz)$. And it follows from the definition of Z that $\forall z \in Z \exists x \in X (Rxz)$. Hence RXZ . Analogous reasoning shows that SZY .

We can now show that if $R \circ S$ is a counterpairing whenever R and S are, and $\Box\phi(t_1, \dots, t_n)$, then $\Box\Box\phi(t_1, \dots, t_n)$. By COUNTERPAIRINGS, if $\Box\phi(t_1, \dots, t_n)$, then $\phi(v_1, \dots, v_n)$ whenever Q is a counterpairing and $Qt_1v_1 \dots$ and Qt_nv_n . By the constraint, it follows that this is also true whenever $Q = R \circ S$ and R and S are counterpairings. By the above lemma, $Qt_i v_i \leftrightarrow \exists u (St_i u \wedge Ruv_i)$, whether t_i, v_i and u are singular or plural. So $\phi(v_1, \dots, v_n)$ whenever R and S are counterpairings and $\exists u_1 \dots \exists u_n St_1 u_1 \wedge Ru_1 v_1 \wedge \dots \wedge St_n u_n \wedge Ru_n v_n$. (Here u_i and v_i are plural variables iff t_i is a plural term.) That is: whenever S is a counterpairing and $St_1 u_1 \wedge \dots \wedge St_n u_n$, then: whenever R is a counterpairing and $Ru_1 v_1 \wedge \dots \wedge Ru_n v_n$, then: $\phi(v_1, \dots, v_n)$. So by a double application of COUNTERPAIRINGS, $\Box\Box\phi(t_1, \dots, t_n)$.

¹²This closure can be defined with a little trickery using plural quantification over ordered triples. Thus: R is a counterpairing in the extended sense $:=$ there is a number n and some ordered triples X such that (i) the first element of each of X is a number less than n ; (ii) whenever $0 \leq i < n$, the ordered pairs $\langle x, y \rangle$ such that $\langle i, x, y \rangle$ is one of X are a counterpairing in the old sense; (iii) any ordered pair $\langle x, y \rangle$ is one of R iff for some function f , $f(0) = x$ and $f(n) = y$ and whenever $0 \leq i < n$, $\langle i, f(i), f(i+1) \rangle$ is one of X .

¹³Here is the proof for B: the others are similar. Let C^*R be defined as $CR \vee CR^{-1}$; let $\Diamond^*$ and $\Box^*$ be defined as in COUNTERPAIRINGS but using C^* instead of C . It is obvious from the definition that for any formula ϕ , $\Diamond\phi$ entails $\Diamond^*\phi$ and $\Box^*\phi$ entails $\Box\phi$; what we need to show is that the converse entailments also hold. For simplicity, suppose ϕ contains just one singular term t and no plural terms. Assume $\Diamond^*\phi(t)$: that is, $\exists R \exists x ((CR \vee CR^{-1}) \wedge Rtx \wedge \phi(x))$. If CR , then evidently $\Diamond\phi(t)$; so suppose $CR^{-1} \wedge Rtx \wedge \phi(x)$. Since B is valid for $\Box$ and $\Diamond$, it

down constraints involving the predicates ‘future-counterpairing’ and ‘past-counterpairing’ that correspond to various standard candidate axioms for tense logic. Each of the candidate modal axioms in Table 4.1 corresponds to two different tense-logical axioms, one derived by replacing $\Box$ and $\Diamond$ with H and P , the other derived by replacing $\Box$ and $\Diamond$ with G and F . I will refer to these by subscripting the corresponding modal axiom with ‘F’ or ‘P’: thus K_F is the schema ‘ $G(\phi \rightarrow \psi) \rightarrow (G\phi \rightarrow G\psi)$ ’, and 4_P is the schema ‘ $H\phi \rightarrow HH\phi$ ’. (Note however that no-one is even tempted to accept the temporal versions of T, B and 5.) Some further candidate tense-logical axioms are listed in Table 4.2, together with corresponding constraints on temporal counterpairings.

Some remarks:

- (i) The relations between the columns are the same as in the modal case. In each case, the claim in the second column together with (the temporal analogue of) COUNTERPAIRINGS entails the schema in the first column. In the other direction, if we have an interpretation of ‘temporal counterpairing’, ‘future-counterpairing’ and ‘past-counterpairing’ which renders every instance of the schema true (given COUNTERPAIRINGS), we can use them to define new interpretations which yield logically equivalent analyses.
- (ii) Besides the logical relations noted in the discussion of the modal axioms, the tense-logical axioms stand in some further logical relations. Using the rule of necessitation (or ‘eternalisation’) and the minimal tense logic comprising Mixing_P , Mixing_F , K_P and K_F , the following

follows from $\phi(x)$ that $\Box\Diamond\phi(x)$: that is,

$$\forall R_1 \forall z_1 ((CR_1 \wedge R_1 y z_1) \rightarrow \exists R_2 \exists z_2 (CR_2 \wedge R_2 z_1 z_2 \wedge \phi(z_2)))$$

Putting R^{-1} for R_1 and t for z_1 , we can deduce that

$$\exists R_2 \exists z_2 (CR_2 \wedge R_2 t z_2 \wedge \phi(z_2))$$

—that is, $\Diamond\phi(t)$. Similar reasoning shows that $\Box\phi(t)$ entails $\Box^*\phi(t)$. The extension to sentences containing multiple variables is straightforward. For the extension to plural variables, we also need the fact that if $S = R^{-1}$ then SXY iff RXY ; this follows from the definition of SXY .

<i>Axiom</i>	<i>Constraint</i>
Mixing _P $\phi \rightarrow \text{HF}\phi$	R is a past-counterpairing $\rightarrow$ R^{-1} is a future-counterpairing
Mixing _F $\phi \rightarrow \text{GP}\phi$	R is a future-counterpairing $\rightarrow$ R^{-1} is a past-counterpairing
Totality _P $(\text{P}\phi \wedge \text{P}\psi) \rightarrow$ $(\text{P}(\phi \wedge \text{P}\psi) \vee \text{P}(\phi \wedge \psi) \vee \text{P}(\psi \wedge \text{P}\phi))$	If R and S are past-counterpairings, then either $R \circ S^{-1}$ or $S \circ R^{-1}$ is a past-counterpairing, or $R = S$.
Totality _F $(\text{F}\phi \wedge \text{F}\psi) \rightarrow$ $(\text{F}(\phi \wedge \text{F}\psi) \vee \text{F}(\phi \wedge \psi) \vee \text{F}(\psi \wedge \text{F}\phi))$	If R and S are future-counterpairings, then either $R \circ S^{-1}$ or $S \circ R^{-1}$ is a future-counterpairing, or $R = S$.
Density _P $\text{P}\phi \rightarrow \text{PP}\phi$	If R is a past-counterpairing, then $R = S \circ T$ for some past-counterpairings S and T
Density _F $\text{F}\phi \rightarrow \text{FF}\phi$	If R is a future-counterpairing, then $R = S \circ T$ for some future-counterpairings S and T

Table 4.2. Some tense-logical axioms, with corresponding constraints on counterpairings

pairs are interderivable:

4_P and 4_F ;	Density_P and Density_F ;
NI_P and ND_F ;	NI_F and ND_P ;
NO_P and NNO_F ;	NO_F and NNO_P ;
BF_P and CBF_F ; ¹⁴	BF_F and CBF_P ;
$\text{BF}2_P$ and $\text{CBF}2_F$;	$\text{BF}2_F$ and $\text{CBF}2_P$

Corresponding equivalences hold between the constraints in the second column. In particular, since COUNTERPAIRINGS automatically yields both directions of CBF and CBF2 without the need for any further constraints, the constraints corresponding to K and Mixing—i.e. that past- and future-counterpairings have universal domains, and are one another's converses—suffices to yield the constraints corresponding to both directions of BF2—i.e. that counterpairings have universal ranges—and hence also the weaker constraints corresponding to both directions of BF.

(iii) By contrast with the modal case, however, in the tense-logical case there

¹⁴This is to be expected given the standard Kripkean model-theory, in which BF_P is equivalent to the condition that the domain of each instant is a superset of the domain of each earlier instant, while CBF_F is equivalent to the condition that the domain of each instant is a subset of the domain of each later instant. We can give a direct proof, as follows. Let * and ** be the following schemas:

$$(*) \quad \text{F}\forall x\phi(x) \rightarrow \forall x\text{F}\phi(x)$$

$$(**) \quad \exists x\text{G}\phi(x) \rightarrow \text{G}\exists x\phi(x)$$

We establish the equivalence of BF_P and CBF_F by way of the following lemmas:

(a): (*) is equivalent to (**). By Duality.

(b): CBF_F entails (*). Note that $\forall x(\forall y\phi(y) \rightarrow \phi(x))$ is a truth of predicate logic, so we have $\text{G}\forall x(\forall y\phi(y) \rightarrow \phi(x))$, and hence by CBF_F , $\forall x\text{G}(\forall y\phi(y) \rightarrow \phi(x))$. Thus on the assumption that $\text{F}\forall y\phi(y)$, we can derive that $\forall x\text{F}\phi(x)$ (using K_F), establishing (*).

(c): (**) entails CBF_F . Suppose that $\text{G}\forall x\phi(x)$. Then we can introduce a redundant quantifier to get $\forall y\text{G}\forall x\phi(x)$. We can combine this with $\forall y\text{G}(\neg\phi(y) \vee \phi(y))$ to get $\forall y\text{G}(\forall x(\phi(x) \wedge \neg\phi(y)) \vee \phi(y))$, and hence $\forall y\exists z\text{G}(\forall x(\phi(x) \wedge \neg\phi(z)) \vee \phi(y))$. By (**), this yields $\forall y\text{G}\exists z(\forall x(\phi(x) \wedge \neg\phi(z)) \vee \phi(y))$. But $\phi(y)$ follows in predicate logic from $\exists z(\forall x(\phi(x) \wedge \neg\phi(z)) \vee \phi(y))$, so we have $\forall y\text{G}\phi(y)$, which establishes the relevant instance of CBF_F .

(d) BF_P entails (**). Suppose $\exists x\text{G}\phi(x)$. Then by Mixing_P , $\text{G}\text{P}\exists x\text{G}\phi(x)$, so by BF_P , $\text{G}\exists x\text{P}\text{G}\phi(x)$, and hence by Mixing_F , $\text{G}\exists x\phi(x)$.

(e) (*) entails BF_P . Suppose that $\forall x\text{H}\phi(x)$. Then by Mixing_F , $\text{H}\text{F}\forall x\text{H}\phi(x)$. So by (*), $\text{H}\forall x\text{F}\text{H}\phi(x)$, and by Mixing_P , $\text{H}\forall x\phi(x)$, which establishes the relevant instance of BF_P .

are further entailments between the constraints on counterpairings in the second column which do not correspond to any valid derivations among the axiom-schemas in the first column. Namely: given the constraints corresponding to Mixing_P and Mixing_F (past-counterpairings and future-counterpairings are one another's converses), D_P ('there is no first instant') and D_F ('there is no last instant') become equivalent to one another, and so do Totality_P ('time does not branch towards the past') and Totality_F ('time does not branch towards the future'). These further equivalences are surprising from the standpoint of the standard model theory for tense logic. There is no familiar structural property of models such that among models with that property, the two directions of *Totality*, or the two directions of *D*, become equivalent to one another.

4.4 The necessity of identity

Of the standard logical schemas considered in the previous section, the necessity of identity deserves extended consideration, since its failure has often been regarded as a distinctive advantage or disadvantage of counterpart theory.

Two main arguments against NI can be found in Lewis's work. The first appeals to the 1968 version of counterpart theory (roughly, our RCT+). Given that analysis, validating NI would require denying that a single object ever has multiple counterparts in one Lewis-world, which is implausible for various reasons (especially because of qualitatively symmetric Lewis-worlds, as discussed in section 4.1). This argument lapses once we take the independently motivated step of replacing RCT+ with an analysis in terms of counterparts of sequences or counterpairings.

The second argument involves putative counterexamples like (9):

(9) The plastic is identical to the dishpan, but might not have been.

According to Lewis (Lewis 1986b, p. 252–3), (9) can be used to say something true. But (9) cannot be true in any context if NI holds in every context. At least, this is so if we treat the descriptions 'the plastic' and 'the dishpan' as rigid, or as having wide scope with respect to the modal operator.¹⁵ To avoid

¹⁵Several authors have recently observed that it is very hard to get non-rigid readings for incomplete definite descriptions like these: see Rothschild 2007, Hawthorne and Manley MS.

distracting questions about scope, we will do better to consider an example with proper names instead of descriptions:

- (10) Goliath is identical to Lump1, but it could have been the case that Goliath was distinct from Lump1.

(The background story is from Gibbard 1975: molten bronze is poured into a statue-shaped mould, thereby creating a statue, Goliath, and a lump of bronze, Lump1. Later, the bronze is dissolved in a bath of acid. Thus Goliath and Lump1 are co-located throughout their histories.)

If we accept BMCT as our analysis of singly *de re* modal claims, there is something dialectically inappropriate about using claims like (10) to motivate an account of multiply *de re* modal claims on which the necessity of identity fails. For the natural way to motivate the claim that Goliath could have been distinct from Lump1 is by appeal to claims like (11):

- (11) Lump1 could have survived squashing, whereas Goliath could not have survived squashing.

But given BMCT, (11) cannot be true in any single context, if 'Lump1' and 'Goliath' refer to the same thing. If it is possible to speak truly and literally in uttering (11), doing so must require "changing the context in mid-utterance", i.e. resolving the context-sensitivity of "could have survived squashing" in a non-uniform way. However, for (10) to work as a counterexample to NI, it would have to be true on some *uniform* interpretations. For as I stressed in chapter 1, when one claims a schema to be "logically valid", one is committed only to its instances being true on all *uniform* resolutions of context-sensitivity. Even paradigmatic logical falsehoods like 'Mary is ready and Mary is not ready', or 'Bruce Wayne is identical to Batman and Bruce Wayne is strong and Batman is not strong', can be used to express truths when we resolve their context-sensitivity non-uniformly; this does not impugn the validity of the non-contradiction schema ' $\neg(\phi \wedge \neg\phi)$ '. But what principled grounds could there be for insisting that (10) has true uniform interpretations, if one is not prepared to make the same claim about (11)? Once we anchor ourselves in a single context, say by stipulating that by our current standards 'Lump1 could have survived squashing', and hence also 'If Goliath is identical to Lump1, Goliath could have survived squashing', are true, the temptation to assert (10) evaporates.¹⁶

¹⁶Similar points are made by Sider (2009, p. 28) and Russell (2013).

Granted, since (10) only a single occurrence of a modal operator, it is not obvious how we could generate a *non*-uniform interpretation of (10) on which it is true. Perhaps, if we think that (10) can be used to communicate a truth, we will have to regard it as some kind of nonliteral discourse, rather than merely non-uniform. A more adequate discussion of the options would require a deeper foundational account of the concept of uniform interpretation than I can provide here.

Thus, I am not persuaded by Lewis's case against the validity of NI. Nor do I see that Lewis has done anything to undermine the force of the standard argument for NI (Kripke 1971). If x is necessarily identical to x , and y is x , then y must also be necessarily identical to x ! There is no point in having a terminological debate about whether this counts as a genuine instance of Leibniz's Law. Those who deny the validity of the Kripkean argument while accepting the validity of ' a is red; $b = a$; so b is red' will of course want to use the label 'Leibniz's Law' to stand for something that licenses the latter argument but not the former one. It is not technically challenging for a counterpart theorist to do this. But this terminological achievement does nothing at all to undermine the impression that the argument is, in fact, obviously valid. Endorsing a counterpart-theoretic analysis of modal claims does not mean one has to throw away one's prior intuitions about the validity of arguments involving modal operators! Rather, as we have seen, such intuitions are a crucial guide in determining what it takes for the counterpart

relation to hold, or for some ordered pairs to be a counterpairing.¹⁷

For many philosophers, the really interesting thing about modal counterpart theory is its supposed ability to block the arguments that push people towards rich ontological pictures according to which statues are distinct from the lumps of bronze that coincide with them, people are distinct from their bodies, abstract objects of various sorts are distinct from any kind of set-theoretic construct, and so on. Consider a typical such argument:

- (12) a. Lump1 could have survived squashing.
- b. Goliath could not have survived squashing.
- c. Therefore, Goliath is not Lump1.

How might counterpart theory help rebut this argument? One standard thought goes like this. The expression ‘could have survived squashing’ is

¹⁷One further point. There is no problem understanding a sense of ‘counterpairing’ under which it applies to non-functions, or of sentences ‘ $\ulcorner$ schmossibly $\phi \urcorner$ ’ stipulated to be analysed in terms of such a notion. But ‘schmossibly’, so understood, is semantically anomalous, in that the truth-value of ‘ $\ulcorner$ schmossibly $\phi \urcorner$ ’ depends on merely *orthographic* features of ϕ . In general, we are free to introduce new words by stipulating them to be semantically equivalent to old words, and we expect the new words so introduced to be intersubstitutable *salva veritate* with the old words they were stipulated to be equivalent to, with the exception of quotation-like contexts. For example, I can stipulate that ‘Lavid Dewis’ is to be semantically equivalent to the name ‘David Lewis’. If we hold on to COUNTERPAIRINGS as our analysis of ‘ $\ulcorner$ schmossibly $\phi \urcorner$ ’, however, we find that ‘Schmossibly David Lewis $\neq$ David Lewis’ is analysed as ‘ $\exists R \exists x_1 (CR \wedge R(\text{David Lewis}, x_1) \wedge x_1 \neq x_1)$ ’, which is false, whereas ‘Schmossibly Lavid Dewis $\neq$ David Lewis’ is analysed as ‘ $\exists R \exists x_1 \exists x_2 (CR \wedge R(\text{David Lewis}, x_1) \wedge R(\text{Lavid Dewis}, x_2) \wedge x_1 \neq x_2)$ ’, which is true if there is a counterpairing that includes two distinct ordered pairs having Lewis as first element. So ‘schmossibly’ with those stipulated truth-conditions creates a quotation-like context, in which even expressions that are stipulated to be semantically equivalent cannot be substituted *salva veritate*. Our ordinary word ‘possibly’, by contrast, surely does not create a quotation-like context: the truth-value of ‘ $\ulcorner$ possibly $\phi \urcorner$ ’ is sensitive only to semantic features of ϕ .

While requiring counterpartings to be functional is one way to insure that the resulting operator is not quotation-like, it is not the only way. We could also explore a schema like COUNTERPAIRINGS except that the list $t_1, \dots, t_n$ is taken to stand for the possibly repeating list of *token* singular terms in ϕ , in order of occurrence. On this analysis, ‘ $\Diamond(Ft \wedge \neg Ft)$ ’ is equivalent to ‘ $\exists R \exists x_1 \exists x_2 (CR \wedge Rtx_1 \wedge Rtx_2 \wedge Fx_1 \wedge \neg Fx_2)$ ’, which can be true if R is non-functional. More interestingly, we could, following Kit Fine’s suggestion (2007), make the truth-conditions of ‘ $\ulcorner$ Possibly $\phi \urcorner$ ’ sensitive to semantic *relations* between the terms in ϕ that do not depend on their semantic *values*. It may be that whereas ‘Lavid Dewis’ bears the same semantic relations to ‘David Lewis’ that ‘David Lewis’ bears to itself, this is not generally true of pairs of coreferential names, or of pairs of distinct variables.

context-sensitive; each of the many admissible interpretations of ‘has a counterpart which survives squashing’ corresponds to an admissible interpretation of ‘could have survived squashing’. The choice between ‘Lumpl’ and ‘Goliath’ can make a difference by affecting how we find it natural to resolve this context-sensitivity. Because of this, each of the argument’s premises is true on its most natural, salient, easily accessible interpretations. But there is no single context (admissible interpretation) relative to which both premises are true: replacing ‘Lumpl’ with the coreferential name ‘Goliath’ preserves truth within any one context.

(Analogy: although Bruce Wayne is Batman, the sentences ‘Bruce Wayne is strong’ and ‘Batman is not strong’ are both most naturally interpreted as expressing truths. Wayne/Batman is very strong for a human being, but has no actual superpowers. It is plausible that referring to him as ‘Bruce Wayne’ makes a reading of ‘strong’ as ‘strong for a human being’ more salient, while referring to him as ‘Batman’ calls to mind the comparison class of other superheroes.)

Advocates of temporal counterpart theory have embraced analogous responses to arguments against the “extreme monist” position according to which objects which coincide *at a time* are always identical. A sample such argument:

- (13) a. No statue in the room has ever been ball-shaped
 b. Some statue-shaped lumps of clay in the room have been ball-shaped
 c. Therefore, some statue-shaped lumps of clay in the room are not statues

Again, the suggestion is that temporal counterpart theory will help by positing context-sensitivity in ‘has been ball-shaped’ associated with context-sensitivity in ‘past-counterpart’.

The role of counterpart theory in these strategies for defending various surprising identifications is rather limited: the very same moves are available to anyone who accepts that the relevant modal or temporal vocabulary is context-sensitive.¹⁸ In the modal case, that is just about everyone: the claim that modal vocabulary is context-sensitive (in the minimal sense explained in ??) is extremely banal, and must be recognised by any sensible semantic

¹⁸This point is also made by Hawthorne 2007.

account.¹⁹ The claim that temporal operators are context-sensitive (in the way that would be relevant to blocking argument (13)) is more controversial. But it is not obvious that temporal counterpart theorists should accept that claim, and one can certainly conceive of views that accept it without embracing a counterpart-theoretic account of the operators.

The pragmatic effect of word-choice on the resolution of context sensitivity is a gentle pressure. It will often be defeated by the countervailing presumption in favour of uniform resolution of context-sensitivity, especially in contexts where we are discussing arguments whose validity turns precisely on such uniform resolution. It would thus be a serious mistake for philosophers attracted to monism to assume that appeals to such pragmatic effects will allow them to systematically mimic the speeches made by "pluralists" who regard arguments like (12) and (13) as sound.

By contrast, some authors, including Lewis himself in Lewis 1971, envisage a kind of theory which achieves a much more extensive and systematic agreement with the "pluralists", by writing the idea that different singular terms invoke different counterpart relations much more directly and mechanically into the semantics. On one version of this approach, we claim that each singular term in the language is semantically associated with a "sort", in such a way that coreferential terms may be associated with different sorts. 'Goliath' and 'Lumpl', for example, might be associated with the sorts *work of art* and *lump of matter*. Each sort, in turn, picks out a distinctive counterpart relation; the counterpart relations that figure in the analysis of a given modal sentence depend on the sorts associated with its singular terms.²⁰ Thus 'Lumpl could have survived squashing' and 'Goliath could not have survived squashing' are, respectively, analysed as 'Lumpl has a lump-of-matter-counterpart that survives squashing' and 'Goliath has no work-of-art-counterpart that survives squashing'. Similarly, given a Lewis-style treatment of multiply *de re* sentences, 'Lumpl could have outlasted Goliath' will be analysed as 'Lumpl has a lump-of-matter-counterpart that

¹⁹Some counterpart theorists make the much more controversial claim that the expression 'it is metaphysically possible that' is context-sensitive. But it is not obvious that counterpart theorists have any more reason than anyone else for positing context-sensitivity in this particular piece of jargon. In section 5.4, I will argue that given the things philosophers do in introducing the notion of metaphysical possibility, it is not plausible for counterpart theorists or anyone else to regard 'metaphysically possible' as context-sensitive in any way that would help with arguments like (12).

²⁰Lewis expresses this formally in an interesting handwritten document 'Counterpart Theory, Mark II' from 1974.

outlasts some work-of-art-counterpart of Goliath that is in the same Lewis-world'.

Here we are clearly no longer dealing with a pragmatic story about the role of word-choices in making certain interpretations of context-sensitivity more salient than others; we have a fully-fledged semantic mechanism, according to which argument (12) is simply invalid.²¹ And that, I would say, is a problem, since the argument obviously *is* valid. Indeed, in order to achieve the desired effect of a systematic endorsement of the kinds of intuitive modal claims upon which pluralists base their case for pluralism, the proposal under discussion will need to reject a great deal of standard logic, including syllogisms like (14):

- (14) a. No statue could survive squashing.
- b. Every lump of clay could survive squashing.
- c. Therefore, no statue is a lump of clay.²²

Those who like this kind of story may regard it as a disadvantage of COUNTERPAIRINGS, relative to Lewis's original counterpart theory, that it provides no item that could be varied depending on the "sorts" associated with the original singular term: the predicate 'counterpairing' occurs only once in the analysis no matter how many terms occurred in the scope of the original modal operator. However, if we really wanted to, we could modify COUNTERPAIRINGS to provide a semantic role for an association between terms and sorts. In outline: instead of taking 'counterpairing' to apply plurally to

²¹What about sentences like 'There is a statue such that it could have survived squashing'? Presumably the proponent of the view will have a story about how the semantic assignment of a kind to a given variable in a sentence is a by the overall role of the variable in that sentence, including the predicates for which it is an argument.

***people to cite here: Ramachandran; Stalnaker; Gibbard; Gupta ***

²²It is controversial whether the notion of identity figures in (14). If we follow Russell in taking the 'is' in (14c) to be the 'is' of identity, we will think of (14) as containing a covert appeal to Leibniz's Law. On the other hand, if we follow authors like Strawson (1950a) and Graff (2001) in taking the 'is' in 'is an *F*' to be predicational, we should think of (14c) as having the simple form 'No *F* is *H*; every *G* is *H*; so no *F* is *G*', whose validity has nothing to do with identity. So proponents of sort-relativity who reject (14) are not simply restricting Leibniz's Law.

Perhaps some monists will simply accept argument (14). Rejecting Leibniz's Law for the predicate 'is a lump of clay', they will accept that some statues are *identical* to lumps of clay while denying that it follows from this that any statues *are* lumps of clay. But this takes us beyond anything that could be achieved by means of some modification of counterpart theory, and perhaps beyond the limits of intelligibility altogether.

ordered *pairs* of objects, we can take it to apply to ordered *triples* of the form $\langle \text{sort, object 1, object 2} \rangle$. Then the analysis of ‘Possibly $\phi(t_1, \dots, t_n)$ ’ will be as follows:

$$\exists R \exists v_1 \dots \exists v_n (\text{CR} \wedge R s_1 t_1 v_1 \wedge \dots \wedge R s_n t_n v_n \wedge \phi(v_1, \dots, v_n))$$

where $s_1 \dots s_n$ are the sorts associated with the original terms $t_1 \dots t_n$, and the new variables $v_1 \dots v_n$ are required to be semantically associated with the same sorts. The upshot is that a *single* counterpairing R can suffice to witness the truth of ‘Lumpl could have survived squashing but Goliath could not have’, or of ‘Lumpl could have outlasted Goliath’.

For some applications of the sort-relativity idea, we could stay with the conception of counterpairings as ordered pairs, simply rewriting COUNTERPAIRINGS in such a way as to incorporate a special treatment for some special kinds of terms. Such a move could, for example, be endorsed by a proponent of Lewis’s identification of qualitative properties with the sets of their instances.²³ The obvious Leibniz’s-Law based objection to this identification is that when something is a member of a set, it is necessary that it is a member of that set (at least if it and the set both exist), whereas it often happens that something has a qualitative property but could have failed to have that property (while still existing). The pragmatic, contextualist response is to say that ‘sets have their members essentially’ is true in some contexts and false in others; referring to something as ‘a set’ tends to put us into a context of the first sort, while referring to it as ‘a qualitative property’ tends to put us into a context of the second sort. A more mechanical response, appealing to a division of singular terms into equivalence classes, could be developed as follows. Each term in the language is semantically marked as either an *ordinary* term or a *qualitative-property* term. When $\phi(t_1, \dots, t_n, q_1, \dots, q_m)$ is a sentence with ordinary terms $t_1, \dots, t_n$ and qualitative-property terms $q_1, \dots, q_m$, $\ulcorner \Diamond \phi(t_1, \dots, t_n, q_1, \dots, q_m) \urcorner$ is analysed as follows:

$$\exists R \exists v_1 \dots \exists v_n (\text{CR} \wedge \phi(R(t_1), \dots, R(t_n), q_1, \dots, q_m))$$

The qualitative-property terms are simply left untouched by the counterpart-theoretic transformation. We require that counterpairings have univer-

²³Such an identification seems hopeless for *non*-qualitative properties: if Fred has only sons, the set of sons of Fred is identical to the set of children of Fred, but if we think that Fred could have had a daughter, we cannot say that the property of being a son of Fred is identical to the property of being a child of Fred.

sal domains and are functional, so that we can use the function-symbol notation without ambiguity. We also require that counterpairings treat set-membership as rigid: when s is any set and R any counterpairing, $R(s) = \{R(x) : x \in s\}$. To see how this works, suppose 'Set' and 'Prop' both name the entity that is both the set of all philosophers and the property of being a philosopher, but 'Set' is an ordinary term while 'Prop' is a qualitative-property term. 'Possibly Fred has Prop' is analysed as 'For some counterpairing R , $R(\text{Fred})$ has Prop', which is true so long as Fred has a philosopher among his counterparts. By contrast, 'Possibly Fred is in Set' is analysed as 'For some counterpairing R , $R(\text{Fred})$ is in $R(\text{Set})$ '; because counterpairings treat set-membership as rigid, this is true only if Fred in fact belongs to Set.

The point of all this special-purpose semantic machinery is to be able to systematically endorse ordinary claims about the modal or temporal differences between items belonging to different kinds (e.g. statues and lumps, or properties and sets), while still maintaining the identity of items of one kind with items of the other. One compelling objection to this program is just to insist that the relevant Leibniz's Law inferences *are* valid. Another compelling objection, made forcefully by Kit Fine (2003), is that the sort-relative semantics cannot in fact be set up in such a way as to do justice to the pattern of intuitive modal (or temporal) claims which motivate it. For example, insofar as we are inclined to judge 'Lumpl could have outlasted Goliath' to be true, we are also inclined to judge 'The first thing on Al's list could have outlasted the second thing on Al's list' true in a situation where Al's list reads 'Lumpl, Goliath'. We are still inclined to judge this true even if the speaker was just guessing and had no idea what was actually written on Al's list. It is very hard to imagine a mechanism which could assign different semantic sorts to the terms 'the first thing on Al's list' and 'the second thing on Al's list' in this sentence.

These are objections to the claim that a semantics based on the idea that terms are sort-relative in a way that can lead to failures of Leibniz's Law in modal or temporal environments is correct for *English*. There is no obvious reason why we could not stipulatively introduce a new language with different kinds of terms, which can behave differently under modal operators even when they are coreferential. And if we were doing the semantics for modal and temporal operators in a pre-existing typed language, there might be more to be said for allowing counterpairings to act differently on different types of terms. And this is not a wholly academic question, since

semanticists sometimes do resort to honest-to-goodness typed languages in explaining certain natural-language phenomena. For instance, semanticists who posit tacit ‘world variables’ and ‘time variables’ normally think of them as belonging to different types from ordinary pronouns.

4.5 Worlds and instants in counterpart theory

We have arrived at what I take to be the optimal forms for counterpart-theoretic analyses of modal and temporal operators. But there is a lot more than this to the task of giving a metaphysical account of modal and temporal discourse: many things we want to say that clearly have a modal or temporal subject matter resist paraphrase in terms of these standard operators. So, what should we try to analyse next? For several reasons, it makes most sense to set our sights on discourse about *possible worlds* and *instants of time*. First, it is a prominent target: explicit talk about entities called ‘times’, including instants of time, is quite common in ordinary language; and explicit talk about entities called ‘possible worlds’ is by now an entrenched part of the professional jargon of philosophers and semanticists. Second, explicit quantification over worlds and instants has been employed in many successful metaphysical/semantic analyses of modal and temporal claims of various kinds (e.g. the standard analyses of counterfactuals due to Stalnaker (1968) and Lewis (1973b)); if counterpart theorists can figure out how to extend their theory to include an account of discourse about worlds and instants, they will be free to incorporate these success stories into their view. Third, as we saw in section 2.1, there is much to be said for a theory of mood and tense in natural language that posits, at the level of syntax that serves as the input for semantics, variables ranging over times and worlds, and quantifiers binding these variables, even in very simple sentences; if a theory of this sort is correct, an account of the meaning of sentences containing such variables is prerequisite for any adequate semantics for natural language. Fourth, we will need to see how to integrate talk of worlds and instants into counterpart theory if we want to understand how counterpart theories fit in to the dichotomy between A- and B-theories introduced in chapter 1, since this dichotomy was partly explained by reference to questions about worlds and instants.

The central constraints on the project of fitting talk of instants and worlds into counterpart theory come from the “roles” for instants and worlds, which section 1.3 attempted to lay out systematically as a collection of schemas. If

we could find an interpretation of the relevant theoretical vocabulary—‘instant’ and ‘world’; ‘possible’, ‘future’ and ‘past’; ‘accessible from’, ‘before’ and ‘after’; ‘at’—under which all of the relevant schemas came out as valid, it is hard to see how we could reasonably reject these interpretations in favour of some other interpretations according to which the schemas in question are *not* valid. The schemas are much too central to our use of ‘instant’ and ‘possible world’ to be brushed aside in this way. Moreover, many applications of instants and worlds in semantical and metaphysical analyses implicitly rely on these schemata; even if we for some reason thought that the words ‘possible world’ and ‘instant’ should be understood in a way that violated the schemata, the analyses in question would then need to be redone, perhaps by replacing the vocabulary of worlds and instants with some other vocabulary which did satisfy the roles. Also, the schemas were part of the background which made T2 and M2 look like reasonable ways of capturing the weighty metaphysical issue that is at stake in debates between A- and B-theorists; if we ended up giving some eccentric, role-violating interpretation of the words ‘instant’ and ‘world’, that would tend to suggest that questions T2 and M2 were not, after all, good ways to capture the underlying metaphysical dispute.

The natural place to start is with Lewis’s proposal that the possible worlds are the Lewis-worlds. By making this identification, Lewis suggests that he can accommodate at least the following central elements of the possible world role:

- ($\Diamond$) Possibly ϕ iff ϕ at some possible world.
- ($\Box$) Necessarily ϕ iff ϕ at every possible world.

To evaluate this claim, we will need to know how Lewis proposes to interpret ‘At w , ϕ ’ (or as he sometimes likes to put it, ‘According to w , ϕ ’, or ‘ w represents that ϕ ’). His idea is that when ϕ is *de dicto* (and all its predicates are well-behaved), ‘at w , ϕ ’ is equivalent to $[\phi]^w$, the result of adding a restrictor ‘in w ’ to every quantifier in ϕ ; when ϕ involves expressions which refer *de re* to certain particular objects, the analysis of ‘at w , ϕ ’ will also involve quantification over counterparts of those objects in w . Clearly this strategy for making sense of ‘at w ’ has no hope of vindicating $\Diamond$ and $\Box$ unless we adopt the Lewis-style analysis of ‘possibly’ and ‘necessarily’ (in which ‘possibly’ and ‘necessarily’ introduce quantifiers over Lewis-worlds) which I have already argued against at length, in chapter 3 and in section 4.1.

But even if we set all the problems with this analysis aside, it is highly misleading of Lewis to suggest that it can be seen as the result of plugging a certain interpretation of ‘possible world’ and ‘at’ into $\Diamond$ and $\Box$. For even *given* Lewis’s analyses of possibility and necessity, getting $\Diamond$ and $\Box$ to be true would require interpreting ‘at’ in two quite different ways: in $\Diamond$, ‘at w ’ would need to be understood as introducing existential quantification over counterparts, while in $\Box$ it would need to be understood as introducing universal quantification. We could afford to ignore this difference only if we thought that each thing had exactly one counterpart in each Lewis-world, a claim that is extremely implausible given Lewis’s commitments.

As we have seen, the best counterpart theoretic analysis of modal operators drops Lewis’s ideas about the role of Lewis-worlds, and with it any prospect of being able to think of Lewis-worlds as the occupants of the possible world role. But this doesn’t mean we have to give up on $\Diamond$ and $\Box$! For as we have seen, provided that we require all counterpairings to have *universal domains* and to be *functional*, COUNTERPAIRINGS can be expressed in the following compact form:

$$\begin{aligned}\Diamond\phi(t_1, \dots, t_n) &:= \exists R(\mathbf{CR} \wedge \phi(R(t_1), \dots, R(t_n))) \\ \Box\phi(t_1, \dots, t_n) &:= \forall R(\mathbf{CR} \rightarrow \phi(R(t_1), \dots, R(t_n)))\end{aligned}$$

This looks to be of the right logical form to be a specification of $\Diamond$ and $\Box$. We need only understand ‘possible world’ to mean ‘counterpairing’, and adopt the following analysis of what it is for something to be the case “at R ”, where R denotes a plurality of ordered pairs which is functional and has a universal domain:

$$(AT) \quad \text{at } R: \phi(t_1, \dots, t_n) := \phi(R(t_1), \dots, R(t_n))$$

The situation is analogous in the temporal case. One’s natural first thought is that the “instant” role is played by certain special three-dimensional regions of spacetime (‘Slices’). But this turns out to be deeply problematic. Even if we only cared about the most basic part of the instant role, as given by principles like At-F and At-P

(At-F) It will be that ϕ iff at some future instant, ϕ

(At-P) It has been that ϕ iff at some past instant, ϕ .

the proposal could work only if we adopt a version of temporal counterpart

theory modelled closely on Lewis's counterpart theory, with Slices standing in for Lewis-worlds. But this is a bad idea for the reasons given in chapter 3 and in the first section of this chapter. And even waiving all those reasons, the only way to make things work with a univocal interpretation of 'at' would require the problematic assumption that each thing has exactly one counterpart 'in' each Slice. A much better idea falls right out of the form of COUNTERPAIRINGS: require future-counterpairings and past-counterpairings to be functional and universally defined; analyse 'at' using (AT); and identify future and past instants with future-counterpairings and past-counterpairings respectively.

There is a complication: counterpairings are not entities. That is, 'counterpairing' is officially a plural predicate; quantification over counterpairings is plural quantification. So claims like 'every possible world is a counterpairing' and 'every future instant is a future-counterpairing' are not even grammatically well-formed. Whether we have to worry about this depends on our purposes. For many of the applications of worlds and instants, it would be enough to be able to analyse the sentences in a *typed* language with special syntactic categories of 'instant variables' or 'world variables' alongside ordinary variables. And for this project, it doesn't matter at all if our analysis involves translating these special variables into plural variables rather than singular ones. This kind of "reconstructive" analysis would, for example, suffice for the purpose of integrating some Stalnaker- or Lewis-style analysis of counterfactual conditionals with a counterpart theoretic account of 'necessarily' and 'possibly': nothing is lost from these analyses if we replace the quantifiers over possible worlds with plural quantifiers over counterpairings and take 'closer than' to be a predicate taking plural arguments. A reconstructive analysis would also suffice for the purpose of providing a counterpart-theory-friendly version of the theory that natural languages themselves contain syntactically real 'time variables' and 'world variables' (see section 2.1). On the other hand, there are other purposes for which we would require a "direct reduction", which lets us identify certain particular entities as the worlds or instants. A direct reduction would be needed for an analysis of English discourse involving the words 'instant' and 'possible world', which are obviously in the same grammatical category as 'dog' and 'house'. It would also be required for applications that require us to speak of *sets* of worlds and instants, *functions* defined on all worlds and instants, etc.

Even if we decide we want a direct reduction as opposed to a recon-

struction, there is a natural way to hold on to the basic idea behind the identification of worlds or instants with counterpartings. Suppose that we find some domain of entities each of which corresponds, in some natural way, to some ordered pairs which are functional and have universal domain. And suppose that whenever some ordered pairs are a counterpairing, they correspond in the relevant way to one of these entities. Then we could think of these entities as being the worlds [instants], with the *possible* worlds, [future and past instants] being those corresponding to pluralities of ordered pairs which are [future- and past-] counterpartings. Let $P(x)$ be a plural term that stands for whichever plurality of ordered pairs which “correspond” to the relevant entity x . Then we can analyse ‘at x , ϕ ’ as ‘at $P(x)$, ϕ ’ in the sense of (AT), and thereby recover $\Diamond$ and $\Box$, or F and P .

What sorts of entities might do this job of “corresponding” to counterpartings? The most obvious candidates are *sets* of ordered pairs—*functions*, as standardly conceived. Suppose there is a set U of all non-sets, and let f be some function whose domain is $U \cup \{\emptyset\}$. Then given that sets are well-founded, there is a natural way to extend the action of f to *any* set. We simply take the image of a set under f to be the set of the images of its members under f . That is to say, f corresponds in a natural way to a plurality of ordered pairs $P(f)$ that has a universal domain and is functional, where the correspondence is defined as follows:

$$P(f)x_1x_2 :=$$

$$\exists g(\text{Function}(g) \wedge f \subset g \wedge x_2 = g(x_1) \wedge \forall y(\text{Set}(y) \rightarrow g(y) = \{g(z) | z \in y\}))$$

Of course, not all pluralities of ordered pairs can be generated in this way from sets of ordered pairs. But consider the plausible assumption that *sets have their members essentially*:

$$\forall x \forall y ((x \in y \rightarrow \Box x \in y) \wedge (x \notin y \rightarrow \Box x \notin y))$$

Given COUNTERPAIRINGS, this is equivalent to the following principle about counterpartings:

Rigid Sets Whenever R is a counterpairing and s is a set, $R(s) = \{R(x) | x \in s\}$

If Rigid Sets is true, then the *counterpairings* are among those pluralities of ordered pairs that *can* be generated from sets of ordered pairs. So we can take the *sets* to be the worlds [instants], understanding ‘ x is a possible

world [future instant/past instant]' as equivalent to ' $P(x)$ is a counterpairing [future-counterpairing/past-counterpairing]', and understanding ' $\text{at } f: \phi$ ' as equivalent to ' $\text{at } P(f): \phi$ ' in the sense of (AT).

This approach works on the controversial assumption that there is a set of all non-sets. But even if this assumption is false, we may be able to use a generalisation of this approach in defining what it is for a plurality of ordered pairs to be "generated" by a set. For it may be that among all the things there are, we can find some set-sized minority of 'basic' entities, such that everything else is 'built up' from them by means of relations analogous to set-membership in being modally and temporally rigid, and well-founded. If so, we can still take the occupants of the world or instant role to be sets of ordered pairs, and define ' $R = P(f)$ ' by means of a disjunctive definition with one clause for the "basic" entities and another for each "building" relation.

If there were nothing more to the world role than $\Diamond$ and $\Box$, and no more to the instant role than F and P , we would be finished. But as we saw in section 1.3, there is quite a bit more involved. First, there are principles about the interaction of 'at' with truth-functional operators:

(At- $\wedge$) At i : (ϕ and ψ) iff at i : ϕ and at i : ψ

(At- $\vee$) At i : (ϕ or ψ) iff at i : ϕ or at i : ψ

(At- $\neg$) At i : not- ϕ iff not (at i : ϕ)

So long as the pluralities corresponding to worlds and instants always have universal domains and are always functional, each of these principles is automatically satisfied given (AT).

(What about the following more controversial principles?)

(At- $=$) At i , $x = y$ iff $x = y$

(At- $\forall$) At i , everything ϕ s iff everything is such that at i , it ϕ s

(At- $\exists$) At i , something ϕ s iff something is such that at i , it ϕ s

Given that the pluralities corresponding to worlds and instants are functional, the right-to-left direction of At- $=$ is automatically satisfied; and given that they have universal domains, the left-to-right direction of At- $\forall$ and the right-to-left direction of At- $\exists$ are also satisfied. (This is related to the fact that CBF is a consequence of COUNTERPAIRINGS.) The other direction of At- $=$ would follow if we made the further postulate that the relevant pluralities

are always *one-to-one*, and the other directions of $\text{At-}\forall$ and $\text{At-}\exists$ would follow if we made the further postulate that the relevant pluralities always have universal *ranges*. In the temporal case, these further postulates are guaranteed by the fact that future- and past-counterpairings are one another's inverses, given that they are functional and have universal domains.)

Another important aspect of the world and instant roles concerns the *accessibility* and *beforeness* relations. In the case of worlds, this is given as follows:

(At- $\Diamond$) At w : possibly ϕ iff ϕ at some w accessible from w

(At- $\Box$) At w : necessarily ϕ iff ϕ at every w' accessible from w

The challenge posed by these principles is to find a meaning for 'accessible from' that makes it true. We can find one by substituting our analysis of 'possibly' or 'necessarily' (COUNTERPAIRINGS) into our analysis of 'at' (AT). In the simple case where 'counterpairing' is interpreted as qualitative, this is easily done:

$$\begin{aligned} &\text{at } R: \Diamond \phi(t_1, \dots, t_n) \\ &\leftrightarrow \text{at } R: \exists Q(\mathbf{C}Q \wedge \phi(Q(t_1), \dots, Q(t_n))) \\ &\leftrightarrow \exists Q(\mathbf{C}Q \wedge \phi(Q(R(t_1)), \dots, Q(R(t_n)))) \\ &\leftrightarrow \exists Q(\mathbf{C}Q \wedge \text{at}(Q \circ R): \phi(t_1, \dots, t_n)) \end{aligned}$$

Thus our reconstruction of worlds as pluralities of ordered pairs can accommodate At- $\Diamond$ (and by duality, At- $\Box$) provided we adopt the following analysis of ' S is accessible from R ':

$$(<) \quad R < S := \exists Q(\mathbf{C}Q \wedge S = Q \circ R)$$

If we are giving a direct reduction instead of a reconstruction, we can say that $x < y$ iff $P(x) < P(y)$ in the sense of ($<$). Note that if we also require the pluralities corresponding to worlds to be one-to-one, the right-hand side of ($<$) is equivalent to ' $\mathbf{C}(S \circ R^{-1})$ '. In the temporal case, we know that the pluralities of ordered pairs corresponding to future and past instants always *are* one to one, since this follows from functionality together with the requirement that future- and past-counterpairings are one another's inverses. So we can understand ' R is before S ' to mean ' $S \circ R^{-1}$ is a future-counterpairing', or equivalently, ' $R \circ S^{-1}$ is a past-counterpairing'.²⁴

²⁴What about non-qualitative interpretations of 'counterpairing', e.g. those associated

So far, then, things are going as well as we could wish for the proposal to identify worlds and instants with pluralities of ordered pairs, or with entities which correspond in some natural way to such pluralities. But trouble arises when we turn to the final component of the roles from section 1.3, the claim that facts about what is the case at a world [instant] are necessary [eternal]:

(Invariance) If at w , ϕ then necessarily, at w , ϕ

(Permanence) If at t , ϕ then always, at t , ϕ

As we saw in chapter 1, many of the standard ways of talking about worlds and instants presuppose Invariance and Permanence, and these principles need to be in the background if questions T2 and M2 are to have their envisaged significance in articulating the metaphysical issue in about which A-theorists and B-theorists disagree.²⁵ It is unfortunate, then, that Invariance and Permanence turn out to be *false* when ‘at’ is analysed in accordance with (AT), given the very plausible assumption that *ordered pairs are rigid*:

$$\forall x \forall y \forall z (z = \langle x, y \rangle \rightarrow \Box z = \langle x, y \rangle)$$

Given COUNTERPAIRINGS, this is equivalent to the following principle about counterpairings:

Rigid Pairs Whenever R is a counterpairing, $R(\langle x, y \rangle) = \langle R(x), R(y) \rangle$.

with doxastic possibility for a particular person? Suppose $\mathbf{C}(S)$ is analysed as $\psi(a_1, \dots, a_j, S)$, where ψ is qualitative and $a_1 \dots a_j$ are some singular or plural terms. Then, in the third and fourth lines of the above derivation, instead of ‘ $\mathbf{C}(Q)$ ’ we should have ‘ $\psi(R(a_1), \dots, R(a_j), Q)$ ’; so our analysis of ‘ S is accessible from R ’ should be ‘ $\exists Q(\psi(R(a_1), \dots, R(a_j), Q) \wedge S = Q \circ R)$ ’. If counterpairings are global permutations, then we can express this analysis in terms of $\mathbf{C}$: in that case, ‘ $\exists Q(\psi(R(a_1), \dots, R(a_j), Q) \wedge S = Q \circ R)$ ’ is equivalent to ‘ $\psi(R(a_1), \dots, R(a_j), S \circ R^{-1})$ ’, which is equivalent to ‘ $\psi(R(a_1), \dots, R(a_j), R(R^{-1}(S \circ R^{-1})))$ ’, which is equivalent to ‘at R : $\mathbf{C}(R^{-1}(S \circ R^{-1}))$ ’.

²⁵However, not all of the theoretical applications for worlds and instants require Invariance/Permanence. For example, even if facts about what is the case at a plurality of ordered pairs are contingent, we could still invoke quantification over such pluralities in a Stalnaker- or Lewis-style account of ‘If it were that ϕ it would be that ψ ’ as ‘Either there is no counterpairing at which ϕ , or some counterpairing at which $\phi \wedge \psi$ is closer than any counterpairing at which $\phi \wedge \neg\psi$ ’. (Since we are analysing ‘If it were that ϕ , it would be that ψ ’, rather than ‘At w : (if it were that ϕ , it would be that ψ)’, we use a two-place closeness relation.) If we were thinking of the relata as possible worlds (subject to Invariance), we would have to think of the extension of this relation as a contingent matter: at each world w , w is the closest world (or at least, among the closest worlds). However, if we give up on Invariance, we can think of the extension of ‘closer than’ as a necessary matter: the more similar a counterpairing is to the identity (in contextually relevant respects), the “closer” it is.

It follows from Rigid Pairs that a counterpairing must map each identity pair to another identity pair. Hence each counterpairing must map I , the plurality of all identity pairs, to some plurality J of identity pairs. (If counterpairings have to have universal ranges, $J = I$.) But the conjunction of 'at J : ϕ ' with the claim that J consists only of identity pairs entails ϕ . Thus for any counterpairing R , if 'at R : at I : ϕ ' is true, then 'at R : ϕ ' is true. So it is necessarily [eternally] true that if at I : ϕ , then ϕ . (If counterpairings have universal ranges, the converse also holds). Thus whenever it is contingently [temporarily] true that ϕ , it must also be contingently [temporarily] true that at I : ϕ .

Given Rigid Pairs, our reconstructive analysis of quantification over worlds or instants as plural quantification over ordered pairs runs aground on Invariance and Permanence. What about a direct reduction, in which worlds or instants are identified with entities which somehow 'correspond' to counterpairings? Clearly, given Rigid Pairs, this will allow us to accept Invariance and Permanence only if the "correspondence" relation which takes us from one of the relevant entities x to a plurality of ordered pairs $P(x)$ is a contingent [transient] one. For example, consider the entity i such that $P(i) = I$. If it were necessary that $P(i) = I$, the necessity of $\ulcorner$ at I : $\phi \rightarrow \phi \urcorner$ would yield the necessity of $\ulcorner$ at i : $\phi \rightarrow \phi \urcorner$, and hence the contingency of $\ulcorner$ at i : $\phi \urcorner$ for contingent ϕ . To preserve the necessity of at i : ϕ , we need it to be contingent that $P(i) = I$. Assuming that *sets* are rigid, this rules out our earlier suggestion that the entities should be taken to be sets of ordered pairs. And it is not at all clear what else they could be.

Giving up Rigid Pairs would be a heavy cost. But if we were prepared to pay this cost, could we get Invariance/Permanence to come out true when 'At w '/'At t ' are replaced with 'At R ', analysed in accordance with (AT)? Yes: it turns out that everything works out nicely if we (a) require all counterpairings not only to be functional and have universal domain, but also to be one-to-one and have universal range—i.e. to be *global permutations*, and (b) reject Rigid Pairs in favour of the following rather bizarre alternative:

Semi-Rigid Pairs Whenever R is a counterpairing, $R(\langle x, y \rangle) = \langle R(x), y \rangle$.

To see how this works out, note first that Semi-Rigid Pairs entails that whenever R is a counterpairing and S is any plurality of ordered pairs, $\langle x, y \rangle$ is one of $R(S)$ iff $x = R(z)$, for some z such that $\langle z, y \rangle$ is one of S . In other words, $\forall x \forall y R(S)xy \leftrightarrow \exists z (Rzx \wedge Szy)$. More succinctly still,

$$(15) \quad R(S) = S \circ R^{-1}$$

Now, consider what it takes for 'at R : at S : $\phi(t_1, \dots, t_n)$ ' to be true under these assumptions, when R is a counterpairing and S is any plurality of ordered pairs that is functional and has a universal domain. According to (AT), 'at R : at S : $\phi(t_1, \dots, t_n)$ ' is equivalent to 'at R : $\phi(S(t_1), \dots, S(t_n))$ ', and hence to

$$(16) \quad \phi(R(S)(R(t_1)), \dots, R(S)(R(t_n))).^{26}$$

Since R is a counterpairing, (15) tells us that $R(S) = S \circ R^{-1}$. Hence $R(S)(R(t_i)) = (S \circ R^{-1})(R(t_i)) = S(R^{-1}(R(t_i)))$. But given our assumption that counterpairings are one-to-one, $R^{-1}(R(t_i)) = t_i$. (Note that this reasoning works whether t_i is a singular or plural term.) So (16) is equivalent to ' $\phi(S(t_1), \dots, S(t_n))$ ', and hence to 'at S : $\phi(t_1, \dots, t_n)$ '. We have thus shown that if 'at S : $\phi(t_1, \dots, t_n)$ ' is true, 'at R : at S : $\phi(t_1, \dots, t_n)$ ' is true for every counterpairing R , so that 'at S : $\phi(t_1, \dots, t_n)$ ' is necessarily [always] true.

To elucidate this result, let us see how Semi-Rigid Pairs secures the non-contingency of 'At R , ϕ ' in a particular case. Consider the claim that at I , Lewis is a philosopher (where I is the plurality of all identity pairs). Since it is contingent whether Lewis is a philosopher, and since all I does is map Lewis to himself, how can it be necessary that at I , Lewis is a philosopher? The answer is that, given Semi-Rigid Pairs, the ordered pairs in I are not *necessarily* identity pairs. For example, the identity pair $\langle \text{Lewis}, \text{Lewis} \rangle$ is not necessarily an identity-pair, since whenever x is a counterpart of Lewis, $\langle x, \text{Lewis} \rangle$ is a counterpart of $\langle \text{Lewis}, \text{Lewis} \rangle$. But since all its counterparts have Lewis himself (not just one of his counterparts!) as their second member, all its counterparts have a philosopher as their second member: thus $\langle \text{Lewis}, \text{Lewis} \rangle$ is *necessarily* a pair whose second member is a philosopher. Likewise, it is not necessary that I are all identity pairs, but it is necessary that whichever of I has Lewis as a first member has a philosopher as its second member. Indeed, it would have been *very easy* for I not to have been identity pairs—some of I would have failed to be identity pairs if *anything* had been otherwise in any respect. For whenever it is true that ϕ , it is necessary that at I , ϕ ; and it is always necessary that (ϕ iff at whichever pairs are the identity pairs, ϕ); hence, whenever it is true that ϕ , it is necessary that (if not- ϕ , then I are not the identity pairs). *Every* counterpairing consists of identity pairs *at itself*: by Rigid Pairs, $R(R) = R \circ R^{-1} = I$ (since counterpairings have universal domains

²⁶Note that we get the same result if we first eliminate 'at R :' to get 'at $R(S)$: $\phi(R(t_1), \dots, R(t_n))$ ' and then eliminate 'at $R(S)$:'.

and are bijective). Thus 'comprises all and only the identity pairs' plays the role, in this theory, of the predicate 'is actualised' in the theory of possible worlds. This takes some getting used to: one must constantly fight the temptation to slip back into the picture of ordered pairs as rigid, which is deeply entrenched in our ordinary ways of thinking about ordered pairs.

Our problem, then, is that if we want to identify possible worlds or instants with pluralities or sets of ordered pairs, and retain Invariance/Permanence, we have to give up the extremely plausible view that ordered pairs are rigid in favour of the outlandish Semi-Rigid Pairs. There are several possible responses to this problem that do not require us to embrace Semi-Rigid Pairs. One response is that it just goes to show that we should not identify possible worlds with those particular pair-theoretic or set-theoretic constructs. After all, theories which identify objects of this or that kind (properties, propositions, events, processes, aspects, absences. . .) with set-theoretic constructs are not all that popular these days. And one common line of objection to many such theories involves the charge that the target entities have different modal properties from the set-theoretic constructs to which they are supposed to be identical. For example, this looks like a powerful objection to Lewis's identification of properties with sets, even when this is restricted to qualitative properties. Those who committed in any case to regarding the realm of abstract entities as much more extensive than the realm of sets will naturally react to the problem by keeping Rigid Pairs and rejecting the identification of possible worlds with sets or pluralities of pairs.

What *are* the entities that play the role of worlds or instants, if they are not sets of ordered pairs? If we are generally hostile to reductionism in ontology, there may not be much more to say: we describe the characteristic role played by the entities in question, which includes a certain modal profile, and that's that. However, it might be helpful to consider a version of the view that departs minimally from the proposals using ordered-pairs. Suppose that we expand our ontology to include several different kinds of ordered-pair-like objects, all governed by the basic structural principle that for any objects x and y , there is exactly one such object whose 'first element' is x and whose 'second element' is y , but having different modal profiles. As well as ordered pairs, there are ordered pairs*; for each ordered pair $\langle x, y \rangle$, there is a corresponding ordered pair $\langle x, y \rangle^*$. Semi-Rigid Pairs is false, but the corresponding principle about pairs* is true. We can then identify worlds or instants with pluralities or sets of pairs* rather than pairs. The natural correspondence between pairs and pairs* generates a natural

correspondence between such sets/pluralities and the pluralities of *pairs* to which our predicate ‘counterpairing’ continues to apply.

Of course lovers of set-theoretic reductionism, like Lewis, will not be satisfied with this approach. A more Lewisian strategy would be to keep the identification of possible worlds with sets or pluralities of ordered pairs, while deploying Lewis’s idea that choices between different ways of referring to the same thing(s) can affect the truth value of our modal claims. Here is a set *s* of ordered pairs, which is also a possible world. If we refer to *s* as “a set of ordered pairs”, that will tend to put us into a context in which Rigid Pairs is true. If we refer to *s* as “a possible world”, that will tend to put us into a different context, relative to which Semi-Rigid Pairs will be true, so that claims about what is the case “at” *s* become non-contingent. If we try to do both these things in the same breath, we will be misled into thinking that we are thinking with two distinct entities.

As we saw in section 4.4, the ability of such appeals to context-sensitivity to vindicate our ordinary modal discourse is limited. It is fairly common in philosophy to find speeches which simultaneously presuppose ordered-pair-theoretic essentialism and the idea that facts about what is the case at a world are non-contingent. For example, people will say things like ‘At every possible world, the elements of an ordered pair are the same elements it has at the actual world’. The only way for the present approach to save such speeches would require appealing somehow to non-uniform resolution of context-sensitivity; and this will be quite a stretch, since the claims in question don’t have the distinctive feel of discourse that involves “mid-sentence context-switching”.

This is a problem for the use of the context-sensitivity idea to sustain a direct reduction of possible worlds or instants to sets of ordered pairs. It is not, however, a problem for the use of the idea within a reconstructive analysis, in which the aim is to translate a many-sorted language with a syntactically distinct category of world variables (or instant variables). For as we saw in section 4.4, once we are dealing with languages whose terms and variables come in different kinds, it is easy to modify or extend COUNTERPAIRINGS in such a way that terms of different kinds get treated differently when they occur in the scope of modal/temporal operators. (Of course, for any such modification of COUNTERPAIRINGS, we will need a corresponding modification of (AT) to preserve principles like $At \diamond$ and $At \Box$). In particular, we can modify COUNTERPAIRINGS and (AT) in such a way that ordinary terms and variables get treated as before, while world/instant terms and variables

are treated in a way that automatically achieves the effect of Semi-Rigid Pairs, and its crucial consequence (15).

Let us sketch what it would look like to work this strategy out rigorously. We start with a language $\mathcal{L}_W$ with three sorts of terms: ordinary singular terms, ordinary plural terms, and world variables. Besides quantifiers, truth-functional connectives, predicates taking ordinary terms as arguments (including a one-place plural predicate C , 'is a counterpairing') and the ordered-pair-forming functor, $\mathcal{L}_W$ contains an 'at' operator that takes a world variable and a sentence and delivers a sentence, and two predicates P ('is a possible world') and A ('is actualised'), each of which takes one world variable as its argument. We want to translate each sentence of $\mathcal{L}_W$ into a sentence of $\mathcal{L}$, the sublanguage of $\mathcal{L}_W$ with only ordinary singular and plural terms. The translations will proceed via an intermediary language $\mathcal{L}^+$, which is like $\mathcal{L}_W$ except that world terms are allowed to occur in any syntactic position where a plural term could occur. We proceed in two stages. At the first stage, we replace each formula ϕ of $\mathcal{L}_W$ with a formula ϕ' of $\mathcal{L}^+$ in which the 'At' operator does not occur; at the second stage, we will eliminate all the world variables from ϕ' leaving us with a formula ϕ'' of $\mathcal{L}$.

To eliminate the 'At' operator at stage one, we will apply the following analysis as many times as necessary:

(17)

$$\text{at } W_0: \phi(t_1, \dots, t_n, W_1, \dots, W_k) := \phi(W_0(t_1), \dots, W_0(t_n), W_1 \circ W_0^{-1}, \dots, W_k \circ W_0^{-1})$$

Here W_0 is any world variable, and $\phi(t_1, \dots, t_n, W_1, \dots, W_k)$ stands for any formula of $\mathcal{L}^+$ with ordinary terms (singular and plural) $t_1, \dots, t_n$ and world variables $W_1, \dots, W_k$, in order of first occurrence. By repeatedly applying (17), we can turn each sentence of $\mathcal{L}_W$ into an 'at'-free sentence.

(As usual, we treat sentences containing complex terms like ' $R(t_i)$ ' and ' $W_i \circ W_0^{-1}$ ' in (broadly) Russellian fashion, as abbreviations of quantified sentences. So, for example, the result of applying (17) to $\text{at } W_0: \phi(x_0, W_1)$, i.e. $\phi(W_0(x_0), W_1 \circ W_0^{-1})$, can be taken to be an abbreviation for the following formula of $\mathcal{L}^+$:

$$\begin{aligned} & \exists x_1 \exists W_2 (W_0 \langle x, y \rangle \wedge \\ & \quad \forall x_2 \forall x_3 (W_2 \langle x_2, x_3 \rangle \leftrightarrow \exists x_4 (W_0 \langle x_4 x_2 \rangle \wedge W_1 \langle x_4 x_3 \rangle)) \wedge \phi(x_1, W_2)) \end{aligned}$$

Note that this in effect gives each complex term the widest scope it could possibly have.)

At stage two, we want to transform the ‘At’-free formula ϕ' of $\mathcal{L}^+$ into a formula ϕ'' of $\mathcal{L}_{\text{ACT}}$. We do so by uniformly replacing each world variable W with a plural variable V that does not already occur in ϕ' , while restricting the quantifier binding V to pluralities of ordered pairs which are global permutations. That is, $\exists W_i(\dots)$ will be replaced by $\exists V(\forall x\exists yV\langle x, y\rangle \wedge \forall z(V\langle x, z\rangle \rightarrow z = y) \wedge \exists y(V\langle y, x\rangle \wedge \forall z(V\langle z, x\rangle \rightarrow z = y)) \wedge \dots)$, where x, y , and z are singular variables that do not already occur in ϕ' . Finally, we need to eliminate occurrences of the predicates P (‘is a possible world’) and A (‘is actualised’), which take world arguments in $\mathcal{L}_W$ and hence do not occur in $\mathcal{L}$. We do this by replacing each formula ‘ PV ’ with ‘ CV ’ (‘ V is a counterpairing’), and replacing each formula of the form ‘ AV ’ with ‘ $\forall x(Vx \leftrightarrow \exists y(x = \langle y, y\rangle))$ ’ (where x and y are singular variables that do not already occur in ϕ_1).

Here is an example of the machinery at work. Suppose our original sentence ϕ of $\mathcal{L}_{\text{ACT}}$ is

$$(18) \quad \exists W_0(AW_0 \wedge \exists W_1(PW_1 \wedge \text{at } W_1: (\exists x(Fx \wedge \neg \text{at } W_0: Fx))))$$

We can render this in English as ‘there is a possible world at which there is an F thing that is not F at the actual world’. Eliminating the two occurrences of ‘At’ in reverse order yields first

$$(19) \quad \exists W_0(AW_0 \wedge \exists W_1(PW_1 \wedge \text{at } W_1: (\exists x(Fx \wedge \neg FW_0(x))))$$

and then

$$(20) \quad \exists W_0(AW_0 \wedge \exists W_1(PW_1 \wedge \exists x(Fx \wedge \neg F(W_0 \circ W_1^{-1})(x))))$$

(Note that we would get a logically equivalent result if we eliminated the occurrences in the opposite order.) That concludes step one. At step two, we convert this (abbreviation of a) formula of $\mathcal{L}^+$ into the following (abbreviation of a) formula of $\mathcal{L}$:

$$(21) \quad \exists X_0(\text{GlobalPermutation}(X_0) \wedge \forall x'(X_0x' \leftrightarrow \exists yx' = \langle y, y\rangle) \\ \wedge \exists X_1(\text{GlobalPermutation}(X_1) \wedge \mathbf{C}X_1 \wedge \exists x(Fx \wedge \neg F(X_0 \circ X_1^{-1})(x))))$$

Given standard plural logic and the theory of ordered pairs, this formula is equivalent to

$$(22) \quad \exists X_1(\text{GlobalPermutation}(X_1) \wedge \mathbf{C}X_1 \wedge \exists x(Fx \wedge \neg FX_1^{-1}(x))))$$

and hence to

$$(23) \quad \exists y \exists X_1 (\neg Fy \wedge \text{GlobalPermutation}(X_1) \wedge \mathbf{C}X_1 \wedge FX_1(y))$$

which is what we would have expected.

4.6 Actuality

Consider Twin, off in some other Lewis-world. He is about six feet tall. He could have been much shorter than that, say five feet tall. And if so, he would have been much shorter than he in fact is, since in fact he is about six feet tall.

‘Actually’ and ‘in fact’ are roughly synonymous. The way Twin in fact is is the way he actually is. Once we agree that Twin is about six feet tall, it would be absurd to deny that he is *in fact* about six feet tall. It would be no less absurd to deny that he is *actually* six feet tall.

When Lewis discusses actuality, he generally focuses on the adjective ‘actual’ rather than the adverb ‘actually’. He claims that only things that are in Cosmo are actual; things that are in other Lewis-worlds are non-actual. ‘Actual’, according to him, is indexical. As used by us, its extension includes only things that are parts of Cosmo; as used by someone in another Lewis-world, its extension includes only things that in that world.

I am not quite sure how ‘actual’ is supposed to relate to ‘actually’. I am not even sure that ‘is actual’, as opposed to ‘is an actual *F*’, is part of standard English. But it would certainly be strange to agree with Lewis that Twin is not actual, while accepting (as I claimed we must) that he is actually six feet tall. Maybe we should say to be actual is to be actually identical to something. If so, then for the same reason that I think we must say that Twin, being six feet tall, is actually six feet tall, I think we must say that Twin, being identical to something, is actually identical to something, and thus actual.

If we thought that Lewis-worlds were *ways things might be*, we would naturally be led to think that there must be one Lewis-world that is *the way things are*, and thus to give this Lewis-world a special role in the semantics of ‘actual’ and ‘actually’.²⁷ But once we have absorbed the lesson that the

²⁷Lewis says that he ‘would find it very peculiar to say that modality, as ordinarily understood, is quantification over parts of actuality. If I were convinced that I ought to call all the worlds actual. . . then it would become very implausible to say that what might happen is what does happen at some or another world’ (OPW 100). It is not obvious how to square this with his allowing (OPW 231–2) that actual objects can be one another’s counterparts.

concept *Lewis-world* plays no special role in the semantics for modal claims, we will see that this is a mistake. Lewis-worlds, like everything else, are counterparts, and in that sense represent ways certain specific things—the things whose counterparts they are—might be. If the Lewis-worlds are all one another's counterparts, then each Lewis-world represents a way any other given Lewis-world might be.²⁸ There is only one Lewis-world—namely Cosmo itself—that represents *the way Cosmo actually is*. But this doesn't give Cosmo a distinctive role *vis-à-vis* actuality, since every Lewis-world represents the way it itself is. 'Actually ϕ ' involves indexical reference to Cosmo only when ϕ already involves indexical reference to Cosmo. Since quantification is often tacitly restricted in a broadly "indexical" way, many claims which contain no overt indexical words are indexical all the same. There is no particular reason to think that covert indexical reference to *Cosmo* is particularly common among the folk, however, although it might be more common among certain physicists and philosophers.

So much for what actuality is *not*. But what is it? What does 'actually' *actually* mean? Hazen (1979) was the first to notice the difficulty of extending the counterpart-theoretic translation scheme of Lewis 1968 to a language containing an 'actually' operator. Fara and Williamson (2005) explain why some natural ideas, as well as some more intricate proposals due to Graeme Forbes and Murali Ramachandran ***cite***, do not succeed in endowing the 'actually' operator with anything like the right logical behaviour.

So long as we admit the existence of entities playing the standard possible-world role—either by identifying counterpartings with sets of pairs and endorsing Semi-Rigid Pairs, or by rejecting set-theoretic reductionism—this challenge is easily met in the present framework. We can simply analyse 'Actually ϕ ' as 'at $w@$: (ϕ)', where ' $w@$ ' is a name for the entity corresponding the plurality consisting of all and only the identity pairs. (In the temporal context, we could do the same for 'Now ϕ '.)

One might object to this approach on the grounds that even if there are entities playing the possible world role, their existence and their modal profiles are not *logical* matter, whereas 'actually' is a logical operator, certain principles involving which are logical truths. I am not, myself, much concerned with this kind of objection. But it is interesting to see that there is another approach to the analysis of 'Actually' to which this worry does not

²⁸Cf. OPW 230: '[A] possible world. . . is a way that an entire world might possibly be. But lesser possible individuals, inhabitants of worlds, proper parts of worlds, are possibilities too. They are ways that something less than an entire world might possibly be.'

apply. In this approach, a sentence ϕ involving multiple ‘actually’ is first translated into a sentence in the multi-sorted language $\mathcal{L}_W$ introduced in the previous section. Suppose we begin with a plural modal language with the modal operators $\Diamond$, $\Box$, and ACT. Given a formula ϕ of this language, we can transform it into a formula of $\mathcal{L}_W$ by (i) replacing each occurrence of ‘ACT’ with ‘at W_0 ’; (ii) replacing each occurrence of ‘ $\Diamond$ ’ with ‘ $\exists W_1(PW_1 \wedge \dots)$ ’; (iii) replacing each occurrence of ‘ $\Box$ ’ with ‘ $\forall W_1(PW_1 \rightarrow \dots)$ ’; and (iv) writing ‘ $\exists W_0(AW_0 \wedge \dots)$ ’ around the resulting expression. We can then apply the method from the previous section to transform this formula of $\mathcal{L}_W$ back into a formula of the modal-operator-free fragment of the original language. It is easy to see that this translation will translate logical truths in the original language into logical truths, so long as we are prepared to treat the existence and uniqueness of ordered pairs as a logical matter.

It is important to understand the difference between the sense in which this theory provides an explanation of sentences involving ‘Actually’ and the sense in which the counterpart theory of section 4.1 provides an explanation of claims of the form ‘Possibly ϕ ’. The latter can be viewed as a collection of *analyses*—English sentences of the form ‘For it to be possible that ϕ is for it to be the case that $\dots$ ’. Our “translations” of claims involving ‘actually’ are not like this. The translation of ‘Actually Mars is red’ into $\mathcal{L}_W$ is ‘ $\exists W_0(AW_0 \wedge \text{at } W_0: \text{Mars is red})$ ’. And the translation of this back into the original language is ‘ $\exists V(\forall x(Vx \leftrightarrow \exists y(x = \langle y, y \rangle)) \wedge \exists x(V\langle \text{Mars}, x \rangle \wedge x \text{ is red}))$ ’. Given the existence and uniqueness of ordered pairs, this sentence (call it ψ) is equivalent to ‘Mars is red’. Since it is not necessary that ((Actually, Mars is red) iff Mars is red), it is not necessary that ((Actually, Mars is red) iff ψ). This seems to rule out our claiming that *for it to be the case that* actually, Mars is red is *for it to be the case that* ψ .

Does this mean that those who take this approach are, after all, forced to accept “primitive modality” in the form of the ‘actually’ operator? I think not. It is a familiar point that the philosopher’s ‘actually’ is impoverished by comparison with comparable devices in natural language. Actual sentences using ‘actually’ seem to exhibit something akin to scope ambiguity: it can locally undo some but not all of the modal operators in whose scope it occurs. For example, ‘Necessarily, everyone could have had a child he doesn’t actually have’ is naturally read as inconsistent with the claim that possibly, some parent x has a child y and could not have had any child other than y . This militates in favour of treating these sentences using some formal language more flexible than the standard language of ‘actually’. The

obvious approach is a language with explicit quantification over worlds, which lets us use the indices on the world variables to keep track of the exact effect we want for a given occurrence of ‘actually’. If that is what is going on in the natural language sentences that serve to introduce us to the formal actuality operator, it is far from obvious that the standard logic of that operator is a good guide to the character of the propositions it allows us to express. It would not, in that case, be outrageous to claim that ‘Actually, Mars is red’ does *not* semantically express a necessarily true proposition. Sentences involving ‘actually’ might be semantically analogous to sentences involving free variables, in that their compositional semantic contribution cannot reasonably be captured by associating them with a single proposition.

Of course, if we embrace the existence of *entities* (e.g. sets of ordered pairs*) playing the role of possible worlds, there will be no obstacle to our introducing a rigid designator ‘@’ stipulated to refer to whichever one of these entities corresponds to the plurality of all identity pairs, and stipulating that ‘Actually’ means the same as ‘At @’. In this ontological setting, there is no obstacle to our taking the logic of ‘actually’ fully seriously as a guide to the modal profiles of the propositions expressed. My considered view is that counterpart theorists should, ultimately, want a full-fledged direct analysis of English sentences involving the words ‘possible world’ and ‘instant’, as opposed to a reconstructive analysis of sentences in a multi-sorted language. And if they have a direct analysis, they can happily accept the standard view that sentences that begin with ‘actually’ express non-contingent propositions. I am merely arguing that the case for that standard view is not strong enough to provide an *additional* reason to be dissatisfied with a reconstructive analysis.

4.7 Counterpart theories are A-theories

We are finally ready to answer the question how counterpart theories fit into chapter 1’s dichotomy between A- and B-theories. As we noticed in chapter 2, counterpart theorists are not subject to the “primordial objection” to the B-theories. How do they manage this? Do they show that there is something wrong in the conception of the B-theories presupposed by the primordial objection? Or do they manage it by being, in fact, A-theories?

Recall that chapter 1 reconstructed the dispute between temporal A- and B-theorists as turning on questions T1 and T2:

(T1) Are there temporarily true propositions?

(T2) Is the unit class of the present instant much *more natural* than the unit classes of most other instants?

and the dispute between modal A- and B-theorists as turning on questions M1 and M2:

(M1) Are there temporarily true propositions?

(M2) Is the unit class of the actual much *more natural* than the unit classes of most other possible worlds?

It is clear that temporal and modal counterpart theorists should answer 'yes' to T1 and M1 respectively. Given counterpart theory, a temporarily [contingently] true proposition is just a true proposition with false counterparts (i.e. a true proposition that is mapped to a false proposition by some counterpairing). And given COUNTERPAIRINGS, the only way to make true a sentence like

(24) The proposition that Obama is president is such that necessarily [always], it is true iff Obama is president

is to ensure that every counterpairing that maps Obama to a non-president maps the proposition that Obama is president to an untrue proposition. So if sentences like (24) are to be true (on *any* reading!), there must be temporarily [contingently] true propositions.

The status of T1 and M1 in counterpart theory is thus fairly clear. And given what we have learnt about the right way to integrate talk of worlds and instants into counterpart theory, the answers to T2 and M2 should be just as clear. Suppose that we identify worlds/instants with *sets of ordered pairs*. As such sets go, the one consisting of all and only the identity pairs is clearly a very natural one; its unit class is much more natural than the unit classes of most of the other sets of ordered pairs that are worlds or instants. This can be seen as arising from the fact that *identity* is a very natural *relation*, as it clearly is. Admittedly, the identification of worlds or instants with sets of ordered pairs faces a problem: in order to accommodate the entire world role, it requires us to give up the natural principle that ordered pairs have their elements necessarily. But if we are prepared to bite this bullet, the case that the set of identity pairs is especially natural is not affected: the mere fact that the respect in which it is distinctive is a *contingent* one does not diminish its naturalness. Meanwhile, if we find the bullet too big to bite, and consequently reject the identification of worlds or instants with sets of ordered

pairs (perhaps in favour of identifying them with sets of ordered pairs*), the case for a positive answer to M2/T2 is no less strong. Somehow or other, the entities that are worlds/instants “correspond” to pluralities of ordered pairs. And whatever our theory of this correspondence relation might be, for the purposes of assessing the naturalness of sets of worlds/instants, we should treat it as a natural relation. When *F*-ness is a much more natural property than *G*-ness, and *x* is some reasonably fundamental entity, we want the set of worlds at which *x* is *F* (that is, the set of worlds which correspond to pluralities of pairs that map *x* to an *F* thing) to be much more natural than the set of worlds at which *x* is *G* (that is, the set of worlds corresponding to pluralities which map *x* to a *G* thing). The status of the ‘correspondence’ relation itself can be factored out when we are thinking in this way, in just the same way that, when we are thinking about the naturalness of sets of ordered pairs, the status of the relation between an ordered pair and its members can be factored out. For this reason, I cannot see that the denial of the naturalness of the correspondence relation provides a tenable route towards a negative answer to M2 or T2.

I conclude that, properly developed, modal counterpart theory is a kind of modal A-theory, and temporal counterpart theory is a kind of temporal A-theory. This conclusion is especially important in the temporal case, since unlike the better-known forms of temporal A-theory, temporal counterpart theory is consistent with a full-fledged endorsement of ???. The temporal counterpart theorist has no need to posit any suspect additional structure in spacetime beyond what we are used to. Spacetime is broadly homogeneous; no slice through it enjoys any very special distinction. Nevertheless, the present instant does enjoy a very special distinction. This is consistent, because ‘the present instant’ is not a name for a slice through spacetime: rather, it is a name for a certain set of ordered pairs, or something that in some way corresponds to a plurality of ordered pairs; and it is presentness just amounts to its being a set of identity pairs, or corresponding to all and only the identity pairs.

I expect that many philosophers will resist my claim that temporal counterpart theory is a version of the A-theory. In the literature, temporal counterpart theory has normally been discussed as one of the tenets of ‘the stage view’, whose other main tenet is the ordinary objects are “stages” (things each of which occupies a single three-dimensional spacetime region). It is common to come across the idea that the difference between the stage view

and the competing 'worm view' is merely a "semantic" one, as opposed to a "metaphysical" one. For example, Sider introduces the stage view as follows:

At one level, I accept the ontology of the worm view. I believe in spacetime worms, since I believe in temporal parts and aggregates of things I believe in. I simply don't think spacetime worms are what we typically call persons, name with proper names, quantify over, etc. The metaphysical view shared by this 'stage view' and the worm view may be called 'four-dimensionalism', and may be stated roughly as the doctrine that temporally extended things divide into temporal parts. (Sider 1996, p. 433)

Here Sider is echoing Lewis, who says something similar when he is considering the analogous contrast in the modal case:

The theory that ordinary things are trans-world individuals, unified by counterpart relations among their stages, really is a different theory from mine. But the difference is limited. There is no disagreement about what there is; there is no disagreement about the analysis of modality. Rather, there is extensive *semantic* disagreement. It is a disagreement about which of the things my opponent and I both believe in are rightly called persons, or sticks, or stones. (Lewis 1986b, p. 218)

And many authors have followed Sider and Lewis in emphasising this contrast between shallow 'semantics' and deep 'metaphysics'. If you are used to thinking in this way, and you think that the A-theory/B-theory dispute is a matter of metaphysics (and if *depth* is a hallmark of metaphysics, you definitely should think this this, since the A-theory/B-theory dispute is one of the deepest in all of philosophy) you will find it hard to believe my conclusion that temporal counterpart theory is a kind of A-theory, and you will think that something must have gone wrong with my argument for that conclusion.

But the idea that any of these disputes are *merely* about 'semantics' cannot be taken seriously. Given uncontroversial principles of disquotational reasoning, any dispute about what a predicate 'F' applies to, or what it takes for a sentence 'S' to be true, will also involve a dispute about the question which things are F, or about what it takes for it to be the case that S; and the latter dispute is (in most cases) not about language at all. Those uncontroversial principles allow us to move back and forth as we please between the

material mode and the formal mode; and this is so whether the questions we are interested in are deep and interesting or shallow and boring.

What, then, can it mean to say that the stage theorist and the worm theorist have the same 'metaphysics' or 'ontology'? Presumably the claim is not just that they agree as regards the cardinality of the universe! Perhaps the thought is that they agree as regards *the fundamental structure of the world*. I am not quite sure what that means, but let us grant that it is so. What I don't see is why the A-theory/B-theory dispute needs to be understood as involving different answers to the question 'what is the fundamental structure of the world'? For what would those answers be? If we want to bring the word 'fundamental' into a characterisation of the A-theory/B-theory dispute, the natural thing to say is that A-theorists think that the fundamental facts are *changeable*, whereas B-theorists think that they are not. The dispute need not concern the question how the world *is* structured; the dispute concerns whether that structure is a changing matter.

Orthodox A-theorists, e.g. 'moving spotlight' theorists who enrich a standard spacetime-based fundamental metaphysics with something like a special 'glow of presentness', clearly do disagree with orthodox B-theorists about the fundamental structure of the world. However, we can *imagine* a strange B-theorist who, at some particular instant t , completely agrees with some moving spotlight theorist's account of the fundamental structure of the world. This B-theorist thinks that, in addition to all the structure studied by physics, there is a special extra fundamental property ('glow') that is uniquely instantiated—permanently, of course—by the slice t . (As t has been approaching, this B-theorist has been getting more and more excited.) At t , these characters agree about the specialness of t . Their disagreement is over whether this specialness is a changeable or a permanent matter. And this need not manifest itself in any *other* divergence between their accounts of fundamental structure. In particular, the A-theorist might be the kind of A-theorist who accepts biconditionals like 'it will be the case that ϕ iff at some time after the present, ϕ ' and 'it has been the case that ϕ iff at some time before the present, ϕ ' as *reductive analyses* of the tense operators, and thus sees no need to add posit any special 'tense operator structure' in addition to the glow of presentness. (Nor need she take 'at' as primitive: perhaps she analyses 'At t , ϕ ' as 'at a world just like the actual world except that t glows, ϕ '.) We can, if we want to, approach this dispute at the linguistic level, in which case we will characterise the A-theorist as claiming, and the B-theorist as denying, that 'glow' features in a certain way in the semantics of tensed

discourse. But the possibility of such semantic ascent should not cause us to dismiss the debate as just about words! For *any* philosophical dispute, it is possible to be a “reductionist” about some of the vocabulary in term of which the dispute is framed. If we think the only deep disputes are ones in which both sides agree that all the relevant vocabulary is fundamental, almost no philosophical disputes are deep!

My interest in the question what side counterpart theorists should take in the A-theory/B-theory dispute is not motivated merely by classificatory zeal. I am hoping to persuade some people who are attracted to the A-theory that they should be temporal counterpart theorists, on the grounds that this will allow them to avoid the argument from spacetime. For this dialectical purpose, what matters is not so much whether temporal counterpart theory counts as a kind of A-theory, as whether temporal counterpart theory provides whatever it is that attracts these people. So, what does motivate people to be A-theorists? Chapter 2 listed several possible factors—various fancy arguments, as well as a certain sense of rock bottom obviousness that I attempted to channel in discussing ‘the primordial objection’. None of these factors does anything to favour the more familiar varieties of A-theory over temporal counterpart theory.

Did I miss something? Well, there may be some A-theorists who find it primitively obvious that *tense operators are fundamental*—facts stated with their aid cannot be “reduced” to any facts statable without tense operators. Those who are gripped by this conviction will be hostile not only to B-theoretic reductions of tense operators, but to the many reductionistic strategies that are consistent with the A-theory as understood in chapter 1. For example, they will resist the view on which ‘It will be the case that ϕ ’ can be reduced to ‘ ϕ at some future instant’, even when this is combined with the claim that futurity is a fundamental, irreducible, monadic property of instants. While they might agree that these claims of these two forms are necessarily and always alike in truth value, it will strike them as obvious that no such necessary equivalence could satisfy whatever further features are required for a “reduction”. I find this conviction rather hard to sympathise with: it seems to me that if one is going to go around theorising about which bits of vocabulary are fundamental, one should be open to the possibility that any given ordinary expression fails to be fundamental, although one may legitimately dismiss certain particular hypotheses about *how* a given expression reduces to the fundamental as obviously false. But maybe it is

true that some A-theorists are driven by this sort of conviction.²⁹ If you are one of them, it is not open to you to accept temporal counterpart theory *qua* reduction of the tense operators. But I still want to urge you to accept temporal counterpart theory *qua* extensionally adequate account of the truth values of sentences involving temporal operators, at the actual world (and other nomologically possible worlds) if not at all metaphysically possible worlds. For by doing this, you will still be able to make your view consistent with the 'Homogeneous Manifold' theory, as this was spelled out in section 2.7. For this theory did *not* include the claim that all other vocabulary can be *reduced* to the vocabulary of spacetime physics. That would be an extremely tendentious claim, for several reasons: the most weighty involves the seeming possibility of worlds with alien physics, at which various ordinary predicates (e.g. 'person') still apply in interesting ways despite the fact that all the predicates of spacetime physics are empty. All the Homogeneous Manifold commits us to is the adequacy of the relevant vocabulary to the task of capturing the structure of *the actual world*. (Although perhaps I should have strengthened that to all nomologically possible worlds.) And the relevant notion of structure was there cashed out in terms of natural classes: the idea is that the only natural classes of spacetime points and regions (and sequences and other set-theoretic constructions thereof) are those classes that can be picked out in some reasonably simple way in the language of spacetime physics. Where there is homogeneity as regards the distribution of physical fields, there is homogeneity *simpliciter*. And *this* is consistent with the idea that there are all kinds of further facts which do not reduce to the facts statable in terms of that language, or even metaphysically supervene on those facts; what matters is just that these further facts *as a matter of fact* (and perhaps of physical law) do not draw any interesting distinctions between items which are on a par from the standpoint of spacetime physics. We could accept all this without having to regard temporal counterpart theory as a reduction, or even as delivering metaphysically necessary and sufficient conditions for the truth of tensed sentences—nomological necessity is enough.

²⁹This might help explain the appeal of speculations about two-dimensional time, which arguably (see section 1.1) arises from some thought to the effect that *nothing* we could say about how things *are*—even if this includes claims about what *times* there are, and how things are at those times, and which of those times have the monadic fundamental properties of pastness, presentness, and futurity!—could ever entail anything about how things *were* or *will be*.

(The only caveat is that for this reconciliation to work, one will have to combine one's temporal counterpart theory with a fairly *simple* analysis of 'future-counterpart' and 'past-counterpart' (or 'future-counterpairing' and 'past-counterpairing') in spacetime-theoretic terms. Otherwise, the fact that this particular relation enjoys the privilege of relating to the facts about how things will be and were, while other similarly simple relations enjoy no such privilege, will constitute a violation of the Homogeneous Manifold Theory. This might or might not be a big constraint. Chapter 6 will consider in more detail the choices one faces in giving a theory about what temporal counterparthood *is*.)

Another factor which may play an important role in motivating people to be A-theorists is the conviction that there are (or have been, or will be) temporary beings—things such that it is not always the case that they are identical to something, unrestrictedly speaking. (This is customarily conceived of as a commitment of "presentism".) As we have seen, this claim is *prima facie* inconsistent with the B-theory—given the standard instant role, the present instant is such that there is nothing that is not, at it, identical to something; if not all instants have this feature, it amounts to the sort of natural distinction for the present instant which B-theorists reject. I am not, myself, tempted to regard it as pre-theoretically obvious that there are temporary beings in the relevant precise sense; but I will not now try to address the question whether this would be a good starting point for an argument for the A-theory. Suppose that you do find it pre-theoretically obvious that there are temporary beings. Then you cannot accept temporal counterpart theory in the form in which it has been developed in the last two chapters, even as merely extensionally adequate. For if BTCT is extensionally adequate, then the claim that there is something that will fail to be anything must have the same truth value as the claim that there is something that has a future-counterpart that fails to be anything, which is a logical falsehood.

This need not, however, prevent you from accepting the basic strategy for reconciling the A-theory with the Homogeneous Manifold Theory, which we have up to now been considering as part of counterpart theory. Suppose that you merely accept ***

Many philosophers think of modal counterpart theory as claiming that there is something deeply right about the 'hyper-essentialist' idea that any object which has a property cannot exist without having that property. The coun-

terpart theorists' accounts of how sentences of the form 'possibly, *a* is *F*' can be true even when *a* is not in fact *F* are conceived of as nothing more than devices for "speaking with the vulgar". Given the counterpart theorists' semantics, we cannot say that 'objects have all their properties essentially' is literally true in English, but on this conception, counterpart theorists still think that there is something importantly right that can be gestured towards by such formulae. If you thought that counterpart theory had to involve some such commitment, you would surely not regard temporal counterpart theory as doing justice to the truth of the A-theory. For on the relevant conception of temporal counterpart theory, they are committed to there being something deeply right about the slogan 'any object which has a property *never* exists without having that property': given the semantics, the slogan is not literally true, but it still gestures towards something importantly true. A-theorists are of course unlikely to think that there is anything importantly true that can be gestured towards by the slogan: objects really do change! But temporal counterpart theory as I have developed it involves no commitment to the idea that there is any interesting truth in the vicinity of this slogan; nor does it suggest any obvious candidates for the role of such a truth.

Even those philosophers who do not think of counterpart theorists as closet hyper-essentialists often think of counterpart theory as embodying a certain attitude of "metaphysical unseriousness" about *de re* facts. The attitude in question might be expressed using the following metaphor: when God makes the world, all he has to do is decide the answers to the purely qualitative questions. Having decided which qualitative roles are to be played, He does not have to make further decisions about the identity of the particular objects which play the roles. The best-understood way to cash out such metaphors is in modal terms: necessarily, for any *de re* truth, there is some purely qualitative truth that entails it. Call this 'modal anti-haecceitism'. Modal anti-haecceitism is certainly an interesting view, but it is not a commitment of most versions of counterpart theory, including Lewis's original version. Suppose that at a certain world *w*, and at every world where the purely qualitative facts are just as they are *w*, *x* has two counterparts, only one of which is *F*. Then it is possible that (*x* is *F* and there is no purely qualitative truth *P* such that necessarily, if *P*, then *x* is *F*). And of course modal anti-haecceitism is not a commitment of the modal counterpart theory developed in the last two chapters: on that theory, there are contingent *de re* truths, and no such truth could be entailed by the purely

qualitative facts, which are necessary.

However, there is another, much less well-understood, but potentially more interesting way of cashing out metaphors about God's choices: namely, as attempts to articulate *non-factualism* about the *de re*. (For the general idea of non-factualism, see section 1.5.) Non-factualists about the *de re* might claim that while there are *de re* propositions, some of which are true and others of which are false, every PROPOSITION—every proposition which is either TRUE or FALSE—is purely qualitative. Or they might claim that whenever it is *factually* the case that *P*, the purely qualitative facts entail that *P*. The idea of the metaphor is that God only has to concern himself with settling the TRUTH-VALUES of PROPOSITIONS, or with settling what is factually the case. Small-p propositions get to be true or false without his needing to bother about them.

Non-factualism about the *de re* is an extreme view: even those who are comfortable with non-factualism as a general theoretical option might regard it as too outlandish to be taken seriously. When we are introducing non-factualist views about other domains, the relevant notion of factuality is often introduced with the help of paradigms, which are often *de re* facts: the Earth is round; this is a table; there is a tree in the Quad. A view where none of these paradigms is factual pushes our understanding of 'non-factual' very far, perhaps to the breaking point. Moreover, non-factualism about the *de re* is hard to assimilate to the best-developed non-factualist projects. Based on the example of expressivism in ethics, it is natural to think that non-factualists about a certain subject matter face the burden of telling some revealing story about the true nature of the states that we normally call *beliefs* about the relevant subject matter, where the story in question has to be told in factual language. But when *p* is a *de re* proposition, *believing p* is a *de re* property—if we are not modal anti-haecceitists, we should not expect to be able to state necessary and sufficient conditions in purely qualitative terms for, e.g., believing Obama to be tall. It is thus not clear that there is any body of theory we could develop that would flesh out non-factualism about the *de re* in anything like the way in which substantive expressivist accounts of the states normally known as 'moral beliefs' flesh out non-factualism about morality.

There is very little basis for attributing non-factualism about the *de re* to Lewis. Lewis does not seem to be hospitable to the ideology required to get non-factualist views onto the table in any domain. In Lewis's discussions of the areas where non-factualist views have been most prominent—ethics,

chance, laws of nature—he does not even mention these views as options; and in the one explicit discussion of non-factualism that I know of (Lewis 2005) is quite unsympathetic. Moreover, as we saw in chapter 3, there is a strong current in Lewis's work that pushes him towards the view that all contingency is *de re* contingency. The combination of this claim with non-factualism about the *de re* yields non-factualism about the contingent. And it would be a real stretch to hang *that* on Lewis. For Lewis, beliefs about the contingent are the paradigm case of belief—indeed, his theory of belief has a lot of trouble dealing with the apparent fact that not everyone believes every necessary proposition. If anything, one would be more tempted to think of Lewis as a non-factualist about the *non*-contingent, insofar as he is drawn to a coarse-grained theory of belief. On such a view, 'Necessarily ϕ ' entails 'Everyone believes that ϕ ' on the most basic, metaphysically perspicuous way of understanding 'believes that ϕ '. If 'Everyone believes that ϕ ' is false on some more ordinary interpretation, then the property expressed on that interpretation by 'believes that ϕ ' is not perspicuously understood as that of believing a certain proposition, just as, according to expressivists about morality, the property we normally express using 'believes that lying is wrong' is not perspicuously understood as that of believing a certain proposition.

It is more tempting to attribute non-factualism about the *de re* to those philosophers who think of themselves as counterpart theorists but do not endorse anything like Lewisian modal realism. It is characteristic of these philosophers that while they talk about "possible worlds", they do so in a way that departs from the standard possible world role laid out in section 1.3. For example, they may sometimes affirm there is exactly one possible world at which ϕ (where ϕ is purely qualitative), while also claiming that for some x , it is possible that (ϕ and x is F), and also possible that (ϕ and x is not F), which is incoherent given the standard role. In interpreting such philosophers, we face the question why they use 'world' in this non-standard way, rather than using it to talk about some entities which *do* play the standard world role. Are they just being perverse? Perhaps in some cases the answer is that they don't believe that there *are* any entities that play the standard world role; but most of the time this is unlikely to be the full story, if the philosophers in question believe in set theory, they should be able to use it to construct entities that do a better job of fitting the standard role than the ones they call 'possible worlds'. The hypothesis that the philosophers in question are non-factualists about the *de re* helps to explain why they

should care about the entities they call 'possible worlds', even when they believe in other entities that play a fuller version of the possible world role. For it is characteristic of possible worlds, on the usual view, that each of them determines a natural mapping from propositions to truth-values. A non-factualist about certain subject matters would find it natural to focus for many purposes on entities each of which determines a natural mapping from PROPOSITIONS to TRUTH-VALUES. Such entities will play the possible world role so long as we confine ourselves to factual discourse, although they may not be fine-grained enough to continue playing the role after we have expanded our language to include the relevant non-factual vocabulary. A non-factualist about *de re* propositions would thus find it natural to focus on entities that play the possible world role as conceived by the non-modal-realist counterpart theorists, which coincides with the usual possible world role for purely qualitative sentences. This helps make sense of what would otherwise seem an uninteresting and pointless departure from the accepted use of 'possible world'.

The combination of non-factualism about the *de re* with the thesis that all change is *de re* change yields non-factualism about the changeable: PROPOSITIONS never change their TRUTH-VALUES; whenever it is factually the case that *P*, it is always the case that *P*. Likewise, the combination of non-factualism about the *de re* with the thesis that all contingency is *de re* contingency yields non-factualism about the contingent.

Chapter 1, discussed non-factualism about the changeable and non-factualism about the contingent as possible strategies for resisting the argument from propositional temporalism [contingentism] to temporal [modal] elitism. It is plausible that if one recognises a category of non-factual truths, one should not take these truths into account in the same way as factual truths when one is assessing the naturalness of a given class of objects. For example, non-factualists about morality will presumably not want to say that the class of morally wrong actions is a very natural one, despite the fact that it is the extension of the very easy-to-refer-to property *being morally wrong*. The facts about the naturalness of classes are supposed to capture the structure of reality: and this is supposed to be a paradigmatically factual subject matter. The application of this idea to non-factualism about the *de re* is not entirely straightforward. Non-factualists about the *de re* will not agree that *all* truths about naturalness are factual truths—*de re* truths about naturalness, such as the truth that a certain particular class is very natural, will share in the non-factual character of all *de re* truths. It is only the purely qualitative questions

about naturalness—such as the question ‘Is there an instant whose unit class is much more natural than those of most other instants?’—whose answers are factual. Because of this, it is far from clear how non-factualists about the *de re* should go about forming opinions about the naturalness of classes on the basis of their other opinions. It is clear enough, on the other hand, that non-factualists about the changeable [contingent] should take only permanent [necessary] properties into account in assessing the naturalness of a class. Since non-factualists about the *de re* who also think that all change [contingency] is *de re* change [contingency] must be non-factualists about the changeable [contingent], such philosophers have principled grounds for rejecting the present section’s argument from temporal [modal] counterpart theory to for temporal [modal] elitism.

Non-factualism about the *de re* is an interesting view to think about. But the main point I want to make about it is the simple one that it is certainly *not* a commitment of temporal or modal counterpart theory in general, or of the particular versions of these theories which the last two chapters have developed. And I am emphasising this because I expect temporal A-theorists to resist my attempts to convince them that they should be temporal counterpart theorists, and I suspect that a lot of this resistance will stem from the mistaken assumption that temporal counterpart theory would have to involve non-factualism about the *de re*, or at least some attitude of metaphysical unseriousness about *de re* facts. It is true that a temporal counterpart theory that was coupled with this kind of non-factualism would not give A-theorists what they want, since this combination of views entails non-factualism about the changeable. But there is no reason for a temporal counterpart theorist to feel any sympathy at all for non-factualism about the *de re*; the kind of facts that change according to temporal counterpart theorists, namely *de re* facts, can be every bit as factual and robust and fundamental as they seem to be.

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